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BARBERSHOP: INTEGRATED MONEY & BUSINESS MANAGEMENT SYSTEM

MUHAMMAD FAUZI NAIM BIN MOHD ASRRI

BACHELOR OF INFORMATION TECHNOLOGY (HONS) IN COMPUTER APPLICATION DEVELOPMENT

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BARBERSHOP: INTEGRATED MONEY & BUSINESS MANAGEMENT SYSTEM

MUHAMMAD FAUZI NAIM BIN MOHD ASRRI
AM2505018945

FYP4112

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Abstract

Small barbershops in Malaysia still run on phone calls, WhatsApp messages and paper. At KI'AD Barbershop in Ayer Hitam, Kedah, a customer who wanted a haircut at a certain time had to ask by message, the barber had to remember who was coming, and payments, staff attendance and monthly salary were kept in separate places. The usual results are double bookings, receipts nobody can match to a payment, and an owner who cannot easily tell how much the shop earned or who came to work late.

This project built BarberShop: Integrated Money & Business Management System as a web application for three kinds of user. A customer registers, chooses services, picks a barber, a date and a free 30-minute slot, and pays either in cash at the shop or by QR with a receipt upload. The owner checks every QR receipt before the booking is counted as paid, and the barber marks the haircut as done. Staff clock in and out by scanning a QR code that changes every minute, and the system turns those scans into late minutes, worked hours and a monthly payroll page next to each barber's sales and ratings. The system is written in PHP and MySQL, uses prepared statements, bcrypt password hashes, CSRF tokens, a login lock after repeated wrong passwords and role-based access, and it is live on shared hosting over HTTPS.

The system was tested with 37 test cases on a local copy of the deployed code, covering performance, security, reliability, compatibility, unit, integration and system tests, and all of them passed. In a user acceptance questionnaire answered by 27 people (23 customers, 3 barbers and the owner), the average score across 297 Likert answers was 4.54 out of 5, 89% agreed they were satisfied overall, and 85% said they would recommend the system. The sample is small and mostly customers. The report also records what was left out: online payment is still a manual QR check, and the expense and profit module and audit log that were proposed in FYP1 were not built.

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INTRODUCTION

1.1 Introduction

This chapter presents the final year project reported here, called BarberShop: Integrated Money & Business Management System. It clarifies why it is necessary to have a booking/management system for a small barbershop and what issues the project is addressing and what it is meant to do and for whom. The case study is KI'AD Barbershop in Ayer Hitam, Kedah and the completed system is hosted at barbershopesmes.site.je.

From the street, a barbershop can appear to be simplistic. Yet, behind the counter the owner has a number of things to track: who was booked at what time with which barber, who paid, how they paid, who showed up on time and who didn't, and who should be paid at the end of the month and how much. In a chat message, notebooks and memory, that information can be easily lost and difficult to locate.

1.2 Project Background

KI'AD Barbershop is a small, owner-operated barbershop shop consisting of a few barbers and one owner. As is common with most shops of its size, it had phone call and WhatsApp bookings. When a customer would ask if a slot was available at a specific time, it would be based on the memory of the first person who read the message. Two customers sometimes called for the same time and the one found out only upon arrival.

The money was dealt with in much the same way. Customers paid in cash, transferred money via QR and sent a screenshot of the receipt in the chat. The owner had to go back through the conversation to check if a payment was actually received. Staff attendance and salary were also recorded on paper, and it was hard to determine how many days a barber had worked in a month, the number of days he missed or how many haircuts he had done.

The customers were also impacted. They could not find out from the price list, nor from the barbers themselves, which of them were free unless they asked, and they received no receipt for their visits, nor did they have a record of their earlier visits. Some just walked in and waited, causing the line to be unpredictable for others.

There are many digital tools that could potentially address some of this. While a general booking system is effective for booking appointments and a large business suite will handle accounting and staff records, a small shop rarely requires the heavy load of either. This project straddles the fence on both sides; small and simple enough for the owner and barbers to run without training or specialized experience, yet with enough sophistication to perform a few important tasks this shop does daily on a budget of cheap shared hosting.

1.3 Problem Statement

The following problems have been identified from the owner's description of the way the shop works in the requirement interview (Section 3.4.1), and by observing how the bookings and payments are carried out.

1.3.1 Booking by Phone and WhatsApp Causes Double Bookings

If a booking is made in a chat, there is nothing to prevent that two customers can be promised the same barber at the same time. The barber's availability is only in someone's thought, in a record book. You can't tell which slots are taken, and customers have to ask, wait for an answer and then ask again. A system that displays free slots and will reject a slot that is already occupied eliminates the question in its entirety.

1.3.2 Cash and QR Payments Are Checked by Hand

For the customer, a QR transfer is simple, but not convenient for the shop. The proof is in the form of a screen shot in a message and is not linked to any particular booking and no one can see afterwards which payments have been checked. The owner also doesn't have an accurate running total of what is collected this month. The payment status should be on the booking itself and then there should be an easy step where the owner says the money has arrived.

1.3.3 Staff Attendance, Salary and Customer Data Are Scattered

Attendance, salary notes and customer details are stored in separate places. A late-arriving barber won't leave any traces, and a monthly pay is calculated from notes and memory. Phone-based customer records. All of this is not secure, so it can be altered or lost without anyone knowing. The owner requires one place where attendance, sales per barber and salary are linked, and each person only gets what they should.

1.4 Project Objectives

Every objective addresses one of the problems above and can be verified with the completed system. One by one the testing chapter comes back to them.

1.4.1 To Develop Online Booking With Barber Choice and No Double Booking

The first goal is a booking system that allows for a customer to select services, select a barber, select a date, select a time, and not allow for a date and time that is already booked. Success is judged on the following tests: A barber should not be able to schedule another appointment at the same time, a customer should not be able to schedule a barber for a closed day or holiday, and if a customer chooses 'any barber', that should be the one not booked.

1.4.2 To Record Cash and QR Payments With Owner Verification and a Sales Dashboard

The second goal is to maintain payment status for all bookings. This means cash bookings don't have to be converted to anything else and QR bookings require a receipt to be uploaded and verified/rejected by the shop, and only verified payments are considered collected in the dashboard. It is met when the status changes adhere to these rules and the totals in the dashboard are the same as in the database.

1.4.3 To Centralise Attendance and Payroll With Secure Role-Based Access

The third goal is to integrate staff attendance, performance and payroll into the same system and safeguard it all. Each time they punch in, the Barbers scan a QR code, the owner will see late minutes, hours worked, sales and ratings next to every salary, and every page will verify the user's role. It is satisfied when users do not have access to pages from other roles, a locked salary and the right attendance is recorded.

1.5 Scope and Target User

This section defines the limits of the project to make it clear what the system is and is not supposed to do.

1.5.1 Project Scope

The project includes the study, design, development, deployment and testing of a web application to support one barbershop. It comes with customer accounts, a list of services, a shopping cart, barber and slot selection, cash and QR payment, receipts, booking status, ratings, staff and owner panels, QR attendance, performance figures, payroll, and shop settings, including opening and closed days, the QR image for payment and contact details.

The system is written in PHP/MySQL and can be accessed with a standard web browser, from a computer or on a phone. It was deployed to shared hosting at barbershophesmes.site.je. Testing covers performance, security, reliability, compatibility, unit, integration, system and acceptance testing. This system is not an accounting package. It doesn't accept payments by card, it's not able to send SMS and it's not designed for a chain of shops.

1.5.2 Product Scope

The product comes in three pieces. The Customer side will be the Home page, Sign Up and Log In, List of Services with search, Cart, Barber selection and Time selection, Payment page, receipt and a My Bookings page to rate the completed services. The second one is the admin panel, which is accessed by the owner and the barbers: dashboard, bookings and

payments, attendance, performance, payroll and owner only pages (services, staff, customers, settings).

The third component is the array of security measures which operate beneath both sides. These include bcrypt password hashing, login lock, CSRF tokens for every form, prepared statements, setting for sessions, attendance check via signed QR code, and checking of uploaded files. These are explained in Chapter 4.

1.5.3 Target User

Customers

People who want a haircut at KI'AD are customers: teenagers, older men. They should view the price list, select a barber and time, pay their money as they like and must have a receipt. The majority will be using a phone, so the customer pages are optimised for a smaller screen.

Barbers (Staff)

The system is used by the barbers throughout the day. They see their own bookings, mark cash as received, mark a haircut as done and clock in and out with the counter QR. They can also view their own attendance and after marked as paid, their own salary. They are unable to open owner's pages.

Owner

The owner is granted the maximum permission level. The owner validates the QR payments, cancels/reschedules bookings, checks the dashboard, the performance page, fixes the forgotten attendance scan, prepares the payroll, edits the services and staff accounts, resets the customer's password and changes the shop's settings.

1.5.4 Changes From the FYP1 Proposal

It is important to note that a number of changes have occurred since the FYP1 Proposal was written.

The FYP1 proposal was: A system that consisted of an expense and profit module, a live queue status page, daily, weekly and monthly financial reports and an audit log will be built using the Laravel framework. The scope changed while the process of development was taking place. It is plain PHP/MySQLi to stay light enough for any regular shared host, and focuses on the aspects that the shop employs daily. Booking, payment verification, attendance, payroll and performance were developed and tested. The expense and profit module, the queuing display and audit log was not designed and is cited as future work in Chapter 6. The goals outlined in Section 1.4 were revised to reflect services provided.

1.6 Conclusion

This chapter outlined a little barbershop that operates on chat messages and paper, and the three issues that arise from it: double bookings, manually audited payments and fragmented attendance and payroll records. Established three objectives, scope and the three types of user. It also clearly indicated what aspects of the FYP1 proposal were abandoned. Chapter 2 presents related work and the technology to develop such a system.

LITERATURE REVIEW

2.1 Introduction

A literature review demonstrates what has been attempted and helps to ensure that the project does not do the same thing or claim new ideas that have already been tried. This system impacts four areas: online appointment booking, small business management software, small business payments and attendance management in a small store, and web app security. The chapter reads sources from each area and then compares three existing products, that a barbershop could use instead of a custom system.

The aim is not to prove that the existing products are bad. There are mature and widely used square appointments, bookly and odoo. So, the question is a little bit narrower: which of them suit a one-shop barbershop that takes QR payments, wants to know who arrived late and doesn't want a monthly bill, and what is left over for a purpose-built system to do?

2.2 Investigation

Simple 4W1H outline is used for the investigation. What is the challenge of running bookings, payments and staff records via chat and paper? Where is a small barbershop in Kedah. The *who* are the customers, barbers and the owner. Manual records lead to double bookings, poor information security and payment clarity – the *why*? The *how* is a web system that stores all of this in one location and has access based on roles.

2.2.1 Investigation Overview

Sources were selected because they relate directly to a design choice in the system. The slot based booking process is supported by work on the appointment booking. Security guidance supports the login, session and upload controls. Malaysian data protection law backs up the decision to keep to a minimum the amount of personal data collected and to publish a concise privacy notice. A comparison is made in Section 2.4, with the help of the product websites.

Liu, van de Ven and Zhang (2016) investigated the appointment slots that should be offered by a service provider when consumers select them. This is their main argument: the slots a business displays DOES make a difference on the fullness of the day. This is readily apparent in a barbershop, where if customers don't know when they are free, the shop will fill the calendar by negotiation. The concept that is being brought into this project is that availability should be calculated by the system and displayed to the customer and NOT by the owner in a chat.

In terms of security, the OWASP Top 10 (2021) identifies broken access control, cryptographic failures, injection and identification and authentication failures as the top four most common risks in Web applications. All of these correspond to real things on this server: only pages for owners can be checked, bcrypt hashing, prepared statements and a login limit. The default algorithm, in this project, behind PHP's password_hash function is bcrypt, a password scheme that includes a built-in salt and an adjustable cost (Provos and Mazières 1999). The 15 minute login lock is recommended in accordance with NIST Special Publication 800-63B (Grassi et al., 2017) which states that there should be a limit on the number of failed authentication attempts.

Personal Data Protection Act 2010 (Act 709) is applicable to personal data in commercial transactions in Malaysia. A barbershop where customers' names and emails are stored is in that. The design implication is rather small, but still important: only gather what is necessary, inform the customer what you hold and why, and ensure that the passwords and uploaded receipts stay out of the sight of those who shouldn't have access to them. This is actually stated clearly on the Terms & Privacy page of the system.

Table 2.1 : Summary of Investigation

Year	Title / Source	Aim	Findings / Relevance
2016	Managing Appointment Booking under Customer Choices (Liu, van de Ven & Zhang)	To study how appointment slots should be offered to customers who choose among them.	Slot choice affects how well a schedule is filled; availability should be computed and shown by the system.
2021	OWASP Top 10: 2021	To list the most common web application security risks.	Access control, cryptography, injection and authentication failures shaped the security framework in Chapter 4.
1999	A Future-Adaptable Password Scheme (Provos & Mazières)	To describe the bcrypt password hashing scheme.	Salted, adjustable-cost hashing; basis of PHP password_hash used for all accounts.
2017	NIST SP 800-63B Digital Identity Guidelines (Grassi et al.)	To give guidance on authenticators and passwords.	Supports limiting failed login attempts and not storing passwords in readable form.
2010	Personal Data Protection Act 2010 (Act 709), Malaysia	To regulate personal data in commercial transactions.	Supports collecting little data, publishing a privacy notice and protecting stored data.
2026	Square Appointments	Cloud booking and payment platform for service businesses.	Good reference for the booking flow; payments and reminders run on its own platform.
2026	Bookly	Booking plugin for WordPress sites.	Shows a simple booking flow; depends on WordPress and paid add-ons for extras.

Year	Title / Source	Aim	Findings / Relevance
2026	Odoo	Open-source business suite with many modules.	Shows the value of connected modules; heavier than a one-shop barbershop needs.

2.3 Related Works

Three of the products currently available on the market have been consulted. They are in three forms: a commercial booking site, a plugin to install booking on a website, and a business suite. Each is examined as to its strengths and weaknesses for this type of shop.

2.3.1 Square Appointments

Square Appointments is a cloud-based solution for any time selling business, like barbershops or salons. Customers can book online, staff have a calendar, and reminders can be sent out via SMS or email. Payments are handled via Square and card handling and receipts are included as part of the platform. It is a solid option for a shop that desires all of this out of the box.

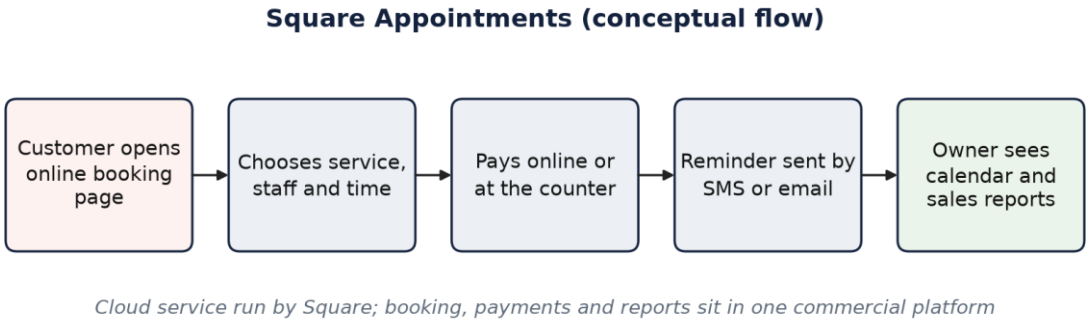


Figure 2.1 : Conceptual overview of Square Appointments

The boundaries of this project in its context is more of a fit than quality. It's a subscription service and the payment functionality is based on Square's payment functionality, not some local QR code paid by a customer and a local owner looking it over. There is no QR attendance scan or payroll page like there is, and the shop's data is stored on the provider's platform, not in a database that is controlled by the shop.

2.3.2 Bookly

Bookly is a booking plugin for WordPress. The visitor selects a service, staff member and time via a form on the website and the booking is saved to the WordPress database. Staff and services are run from the WordPress admin panel, and additional features like online payments or text messages are provided by add-ons.

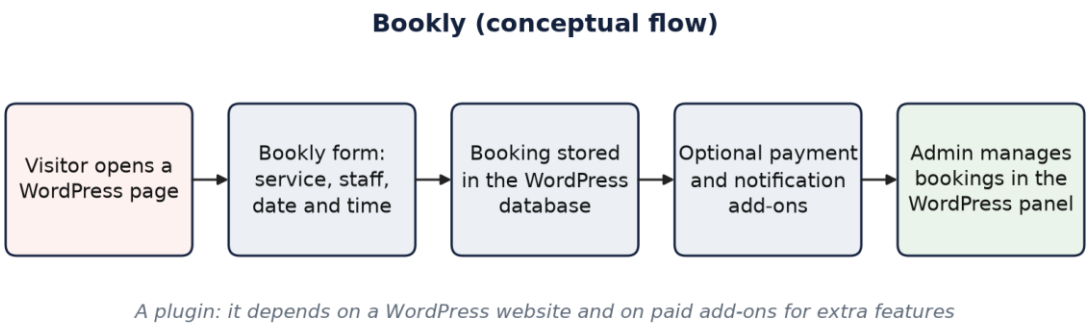


Figure 2.2 : Conceptual overview of Bookly

This is ideal for a store with an existing WordPress site. However, the drawback is that the booking tool isn't the one and only thing the shop requires. Attendance and payroll/performance per barber is beyond its scope, so the owner would still need a tool, or paper, for them. They would also rely on maintaining the shop up to date as well as the plugin itself.

2.3.3 Odoo

Odoo is an open source ERP that's a suite of business applications. Its apps are appointments, point of sale, accounting, inventory, CRM, attendance, and they all share a database, so that a sale in one app will be reflected in the others. It is the product that is most similar to the concept of integration in the project title.

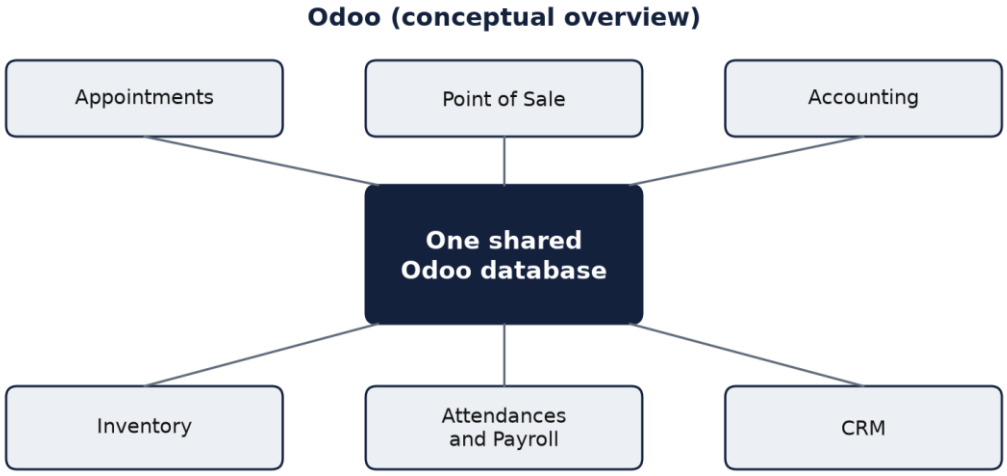


Figure 2.3 : Conceptual overview of Odoo

This is its downfall, as well: the same width. The effort required in setting up and keeping up a server, selecting modules & training personnel is huge contrasted with what the store does on an everyday premise and some applications (such as payroll) are related with paid variants. The project focuses on maintaining Odoo's concept that bookings, payments, attendance and salary should be linked and removes the burden.

2.4 Comparison

Table 2.2 compares the three products with the system developed in this project. Some of the descriptions in the current products are a reflection of their general public description and are not an assessment of every plan or edition. If a property is not a primary feature of a product, the table does not say "impossible" but rather "not a main feature.

Table 2.2 : Comparison of Existing Products

Feature	Square Appointments	Bookly	Odoo	KI'AD system
Online booking with staff choice	Yes	Yes	Yes, through the appointment app	Yes, with named barber or any barber
Free-slot check against double booking	Yes	Yes	Yes	Yes, checked at slot choice and again at payment
Payment handling	Card payments through Square	Add-ons for payment gateways	Through accounting and POS modules	Cash, or QR with receipt verified by the owner
Staff attendance	Not a main feature	Not a main feature	Attendance app	QR scan with late minutes and hours worked
Payroll view	Not a main feature	No	Payroll app (edition dependent)	Monthly salary page with attendance, sales and rating
Barber performance and ratings	Reports	Basic	Reports	Customers, sales, picked-by-name and star rating per barber
Roles	Owner and staff	WordPress roles	Detailed access rights	Customer, barber and owner
Hosting and cost	Monthly subscription	WordPress hosting plus plugin	Server or subscription	Shared hosting, no per-user fee
Suitability for a one-shop barbershop	Good, commercial	Good if a WordPress site exists	Powerful but heavy	Small and focused on this shop

2.5 Discussion

Square Appointments and Bookly are good examples on the booking flow, and it goes in the same sequence that customers are used to: select a service, select a person and a time, then pay. Odoo serves as the reference to link the records. None of the three eliminates the need for the owner to manually verify a QR transfer, and none of them provide attendance and salary coverage as a small shop would do without an additional fee.

This system built here is not attempting to out-feature these products. It's intended to match this specific store. It is based on the payment habit that customers already have, a QR transfer with a screenshot, then moves it from a chat thread to a booking record with a status.

It will take advantage of the phone already carried by every barber used as attendance scanner and thus no hardware is purchased. And it stores everything in a small MySQL database on inexpensive hosting he owns.

This review also uncovered a clear gap. Card gateways and automatic confirmation of payment would eliminate the necessity for QR checking, although this requires a business to be registered with a payment provider and that is beyond the scope of this project. For this reason, online payment is not half completed in this book, but is only stated as future work in Chapter 6.

2.6 Conclusion

The literature and product review are in one direction. Customers should not be asking for things to become available, while security controls need to be in place based on guidance, and a system that combines booking control, payment systems, attendance, and salary control is best for a barbershop without the overhead of a big booking system. The methodology used to develop such a system is described in the next chapter and the requirements came out of the interview with the owner are explained.

METHODOLOGY

3.1 Introduction

The development of the system is described in this chapter. It outlines the methodology, the phases the work was taken through, how the requirements were collected from the owner, what the system needs to do and how well it needs to do it, and the models used to analyse the requirements prior to coding the system.

3.2 Agile Methodology

The project is an Agile project where the system is developed in small bits and presented to the user for corrections rather than being fully specified and delivered at the end of the project (Beck et al., 2001). It was a necessity that led to the choice. The owner is familiar with the business but would not be able to explain all of the screens in advance and some ideas did not come to light until a working page was created. When a customer selects "any barber", for instance, viewing the booking form, what should occur?

Those late discoveries would have been costly if it were a strict waterfall plan. Changes to the booking rules or to a report layout could be made in the next increment without having to redo the rest, with short cycles. In our case, having one developer, the framework is not used rigidly, there are no daily stand-ups but the plan, build, test and review cycle is observed with each feature.

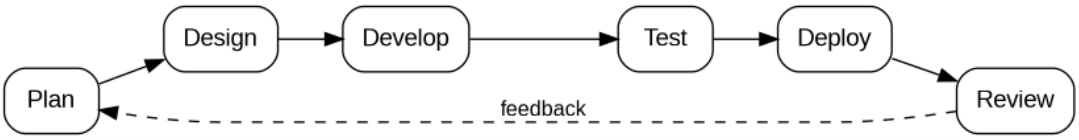


Figure 3.1 : Agile methodology used in the project

3.3 Phases in Agile Methodology

In Agile Methodology, there are three stages: 1. Conception 2. Elaboration 3. Construction.

Six phases were repeated for each increment. They are indicated in the sequence in which they occurred in one cycle, but the project itself circled the cycle a few times.

3.3.1 Phase 1: Planning

The planning process began from the FYP1 project brief and interview with the owner. The result was a set of problems (in Section 1.3), a first set of objectives and a requirement list. At this point I also determined what was "wishful thinking" and what was essential to call

the system useful to me. Booking and checking payments were added to the first group, while concepts like a loyalty scheme were put into a "future work" list.

3.3.2 Phase 2: Design

The requirements were translated to the following models in design phase: use case diagram, flowcharts of the 3 major flows, sketching of the important pages which became the actual interface, data flow diagram, entity relationship diagram, and a security framework. This was done on paper in order to identify gaps early. The flow chart for the payment for example indicated that when a QR payment was rejected, the booked slot has to be released again, became a rule in the booking code.

3.3.3 Phase 3: Development

Development was done in increments. The booking flow was the first, because everything else is dependent on a booking record. The next logical steps were payment and receipts, followed by an admin panel, and finally attendance, payroll and performance, which are built on the concept of bookings and staff accounts. Security measures were implemented on a page-by-page basis and the entire system reviewed at the end, not as one security step.

3.3.4 Phase 4: Testing

Each feature was tried by hand as it was finished. Towards the end test cases from Chapter 5 were developed and executed on the entire system, which includes performance, security, reliability, compatibility, unit, integration and system tests. User acceptance testing was conducted with a Google Form questionnaire and a client walk through was completed with the owner.

3.3.5 Phase 5: Deployment

A shared hosting account was used to deploy the system. Deployment involved generating the database with a clean install script and putting files on the server and then setting config.php to production mode and checking if HTTPS was enabled, and if configuration files were not accessible via a browser. A brief hosting guide was provided in Malay language and left with the project to enable the owner to repeat the steps.

3.3.6 Phase 6: Review

The outcome of each cycle was checked at the end of the cycle to see how it met the requirements list and the feedback received. Items not fitting the time left moved to the future work list. The final review is Chapter 6.

3.4 Requirement

3.4.1 Data Gathering Techniques

To determine what the system should do, two approaches were adopted, an interview with the owner and an observation of the bookings and payments process. They were chosen because they complement each other. An interview with the owner is combined with observation of the actual counter activity. Later, after the system was constructed, a questionnaire was used to assess the acceptance of the system (Chapter 5).

Interview

The interview was conducted with the shop owner; it was designed around the daily routine, such as how appointments are made, how the customers pay, and how staff attendance and salary are monitored and stored. The answers are analysed in section 3.5.1. The interview took place on 5 September 2026 at the owner's shop. The interview questions are in Appendix A.

Observation

I also observed in practice how bookings and payments were made. The booking was done in a chat window, the payment proof was sent as a screen shot and the barber's own list of the day was pretty much what the barber could remember. This made it obvious that it would be best to have the booking and the payment status in a single record.

3.4.2 Functional Requirement

The functional requirements are broken down into user categories. The rules are given in Table 3.1 for customers, Table 3.2 for barbers, Table 3.3 for the owner, and Table 3.4 for rules which the system applies without the involvement of any players.

Table 3.1 : Functional Requirements for Customers

ID	Requirement	Description
FR-C1	Register and log in	A customer creates an account with a username, email and password and logs in with the username or the email.
FR-C2	View services	The customer sees each service with a photo and price and can search by name or price.
FR-C3	Use a cart	The customer adds or removes services and sees the total before choosing a time.
FR-C4	Choose barber, date and time	The customer picks a named barber or "any barber", a date and a free 30-minute slot, and can add a note.
FR-C5	Pay	The customer pays cash at the shop or by QR and uploads a receipt (JPG, PNG, WEBP or PDF, up to 5 MB).
FR-C6	See a receipt	A printable receipt shows the booking, barber, services, total and payment status.

ID	Requirement	Description
FR-C7	See bookings	My Bookings lists past and upcoming bookings with a clear status.
FR-C8	Rate a haircut	After a booking is done, the customer can give 1 to 5 stars and a short comment, once per booking.
FR-C9	Manage password	The customer can change the password, and must do so after logging in with a temporary one.

Table 3.2 : Functional Requirements for Barbers

ID	Requirement	Description
FR-B1	Log in to the staff panel	A barber logs in with the same login page and lands on the STAFF panel.
FR-B2	See own bookings	The barber sees today's schedule and the next seven days, limited to their own bookings.
FR-B3	Mark cash received	For a cash booking, the barber can mark the money as received.
FR-B4	Mark a haircut done	The barber marks a booking as done when the haircut is finished.
FR-B5	Clock in and out	The barber scans the counter QR code with a phone: the first scan is arrival, the next is going home.
FR-B6	See attendance and salary	The barber sees their own attendance and, after the owner marks it paid, their own salary.

Table 3.3 : Functional Requirements for the Owner

ID	Requirement	Description
FR-O1	Read the dashboard	The owner sees bookings today, booked value, money collected this month, payments waiting for a check, a seven-day sales chart, top services and barber figures.
FR-O2	Verify QR payments	The owner views the uploaded receipt and verifies or rejects it. A rejected payment frees the slot.
FR-O3	Manage bookings	The owner can reassign a booking to another barber, mark it done or cancel it.
FR-O4	Manage attendance	The owner sees arrivals, late minutes and hours, fixes forgotten scans and can renew the counter QR link.
FR-O5	See performance	The owner sees customers served, sales, bookings picked by name and star ratings for each barber in a chosen month.
FR-O6	Prepare payroll	The owner enters a salary for each barber, saves a note and marks it paid. A paid salary is locked.
FR-O7	Manage services	The owner adds, edits and deletes services with name, price and photo.
FR-O8	Manage staff	The owner creates and edits staff accounts, sets role, barber status, active status and start time, and resets passwords.
FR-O9	Manage customers	The owner searches customers and gives a temporary password to someone who forgot theirs.

ID	Requirement	Description
FR-O10	Change shop settings	The owner edits opening hours, slot length, closed weekdays, holidays, payment QR, logo, contact details, map, social links and the terms and privacy text.

Table 3.4 : System Rules

ID	Rule	Description
FR-S1	No double booking	The same barber cannot be booked twice for the same time. The check runs when a slot is chosen and again when the booking is saved.
FR-S2	Fair automatic assignment	For "any barber", the system picks the free barber with the fewest bookings that day.
FR-S3	Closed days and past times	Closed weekdays, holidays and times that have already passed cannot be booked.
FR-S4	Login protection	Five wrong passwords for one account, or twenty from one address, block login for 15 minutes.
FR-S5	Attendance token	The attendance QR changes every 60 seconds and an old code is refused.
FR-S6	Role separation	Customer, barber and owner pages are separated. Owner-only actions are refused to barbers on the server.

3.4.3 Non-Functional Requirement

Table 3.5 : Non-Functional Requirements

ID	Requirement	Description
NFR1	Usability	A new customer should be able to finish a booking without help. The booking has four visible steps.
NFR2	Performance	Common pages such as the services page, the dashboard and payroll should load within two seconds.
NFR3	Security	Passwords are hashed, sessions are protected, forms carry CSRF tokens, queries are prepared and access depends on role.
NFR4	Reliability	A booking and its items are saved together or not at all, and records remain after logout, refresh and a restart.
NFR5	Compatibility	The pages work on a desktop browser and are usable on a phone screen.
NFR6	Maintainability	Opening hours, slot length, prices, QR image and contact details are settings, not code, so the owner can change them.
NFR7	Privacy	Collect only username, email, password hash and bookings, and publish a short terms and privacy notice.

3.4.4 System Requirement

The system needs a development setup, a server to run on and the devices people use. The minimum hardware is listed in Table 3.6, and software in Table 3.7. The project was developed locally using XAMPP, and deployed to a shared hosting account.

Table 3.6 : Hardware Requirement Specification

Hardware	Minimum requirement	Purpose
Laptop or PC	Windows 10 or 11, 8 GB RAM, 10 GB free disk	Development, local testing and documentation
Smartphone	Camera and a current mobile browser	Customer booking, barber attendance scan, mobile testing
Counter screen	Any tablet, phone or PC with a browser	Shows the attendance QR code in the shop
Internet connection	Stable broadband or mobile data	Hosting access, libraries loaded from CDNs, live site
Web hosting	PHP 8.1 or newer, MySQL or MariaDB, HTTPS	Runs the live system

Table 3.7 : Software Requirement Specification

Software	Version / tool	Purpose
Operating system	Windows 11	Development environment
XAMPP	PHP 8.2, MariaDB 10.4, Apache, phpMyAdmin	Local server and database for development
Programming languages	PHP, SQL, JavaScript, HTML, CSS	Back end, database, front end
Front-end libraries	Bootstrap 5.0.2, Font Awesome 6.4, qrcode.js, jsQR	Admin layout, icons, QR display and QR scanning
Browser	Google Chrome	Testing and screenshots
Test tools	Python scripts with requests, Playwright	Automated tests and page screenshots
Hosting	Shared PHP and MySQL hosting	Live site at barbershophesmes.site.je

3.5 Analysis

3.5.1 Data Gathering Analysis

Table 3.8 is a summary of the interview. Every topic is connected with the requirement it answers. **[CHECK: Check with interview notes before submission]**

Table 3.8 : Analysis of Interview Questions

Topic	Question asked	Finding	Requirement
Booking	How are appointments taken now?	By phone call and WhatsApp. Times are written on paper or remembered. Double bookings and missed slots happen.	FR-C4, FR-S1, FR-S3
Payment	How do customers pay and how do you check it?	Cash, or a QR transfer with a screenshot sent in the chat. The owner scrolls back to confirm. There is no running total.	FR-C5, FR-O1, FR-O2
Staff	How are attendance and salary tracked?	By notes. Late arrivals and days worked are hard to count at month end.	FR-B5, FR-O4, FR-O6
Customer data	Where are customer details kept?	In phones and notes. Customers' privacy is a concern, and a lost phone would mean lost data.	FR-C1, FR-S6, NFR7
Devices	What do staff and customers use?	Mostly smartphones.	NFR5, FR-B5

3.5.2 Use Case Model

The use cases of the three actors are shown in Figure 3.2. Customer signs up, peruses the services, schedules a Barber, date and time, pays, and follows the booking. The barber is responsible for the day's appointments and for clocking in. The owner has full control of the barber's skills, verification, reassignment, attendance and payroll and the shop administration. The owner's connection to the barber's use cases is not drawn, to keep the diagram readable.

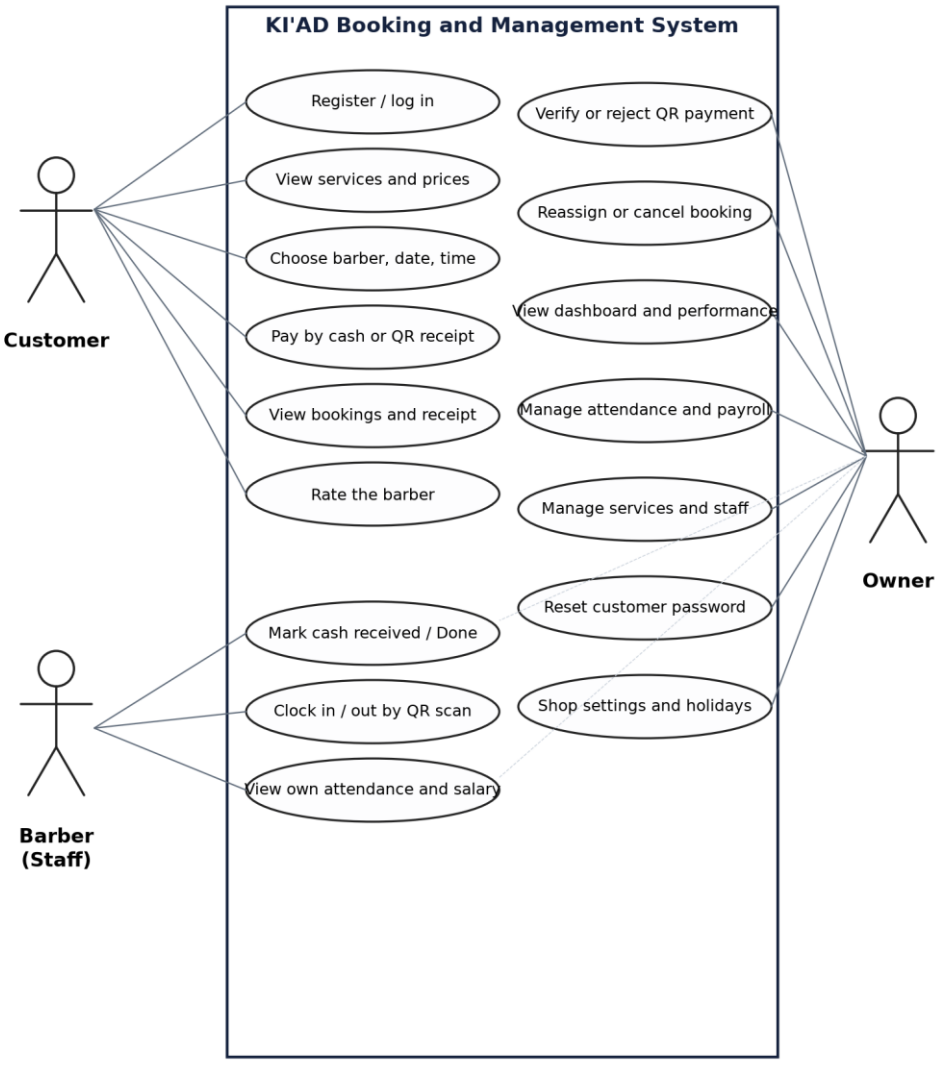


Figure 3.2 : Use case diagram of the KI'AD system

Three use cases are described in more detail in Tables 3.9, 3.10 and 3.11 as these use cases have the main rules of the system.

Table 3.9 : Use Case Description: Book an Appointment

Item	Description
Use case	UC-01 Book an appointment
Actor	Customer
Precondition	The customer is logged in and has at least one service in the cart.
Main flow	1. The customer opens Barber & Time and chooses a barber or "any barber". 2. The customer picks a date; the page greys out times that are taken. 3. The customer chooses a free time and may add a note. 4. The customer chooses cash or QR on the payment page (QR needs a receipt file). 5. The system saves the booking and its items in one transaction and shows the receipt.

Item	Description
Alternative flows	A. The slot is taken while the customer is paying: the system returns to Barber & Time with a message. B. The date is a closed day or a holiday: the system refuses it and explains why. C. QR is chosen without a receipt, or the file is not an allowed type or is over 5 MB: the system shows an error.
Postcondition	A receipt exists with status cash (cash) or pending (QR) and job status booked.

Table 3.10 : Use Case Description: Verify a QR Payment

Item	Description
Use case	UC-02 Verify a QR payment
Actor	Owner
Precondition	A booking exists with payment status pending and an uploaded receipt.
Main flow	1. The owner opens Bookings & payments; pending rows are highlighted. 2. The owner opens the receipt image. 3. The owner presses Verify payment. 4. The status changes to verified (Paid) and the amount counts in Collected this month.
Alternative flows	A. The receipt cannot be confirmed: the owner presses Reject, the status becomes rejected and the slot is freed. B. A barber tries the same action: the server refuses it because it is owner-only.
Postcondition	The booking is marked Paid or Rejected.

Table 3.11 : Use Case Description: Clock In by QR

Item	Description
Use case	UC-03 Clock in by QR
Actor	Barber
Precondition	The counter screen shows the attendance QR and the barber has an active account.
Main flow	1. The barber scans the QR with a phone camera and logs in if asked. 2. The system checks the signed token and that it is recent. 3. If there is no record for today, it saves the arrival time and late minutes against the start time. 4. The page shows Welcome with On time, Early or Late.
Alternative flows	A. The token is old: the page says the QR code has expired and asks for a new scan. B. A record exists and the barber scans again within five minutes: the page says Already clocked in. C. A record exists and more than five minutes have passed: the system saves the leaving time.
Postcondition	An attendance row exists for the day with arrival and, later, leaving time.

3.5.3 Flowchart

There are three flowcharts that display the main processes. Figure 3.3 traces the customer's journey from when they come to the site until they have a booking saved, as well as the loops for if they cannot log in and for if the slot is already booked. A barber or owner deals with the booking and payment as can be seen in Figure 3.4. The attendance scan and checks behind are displayed in figure 3.5.

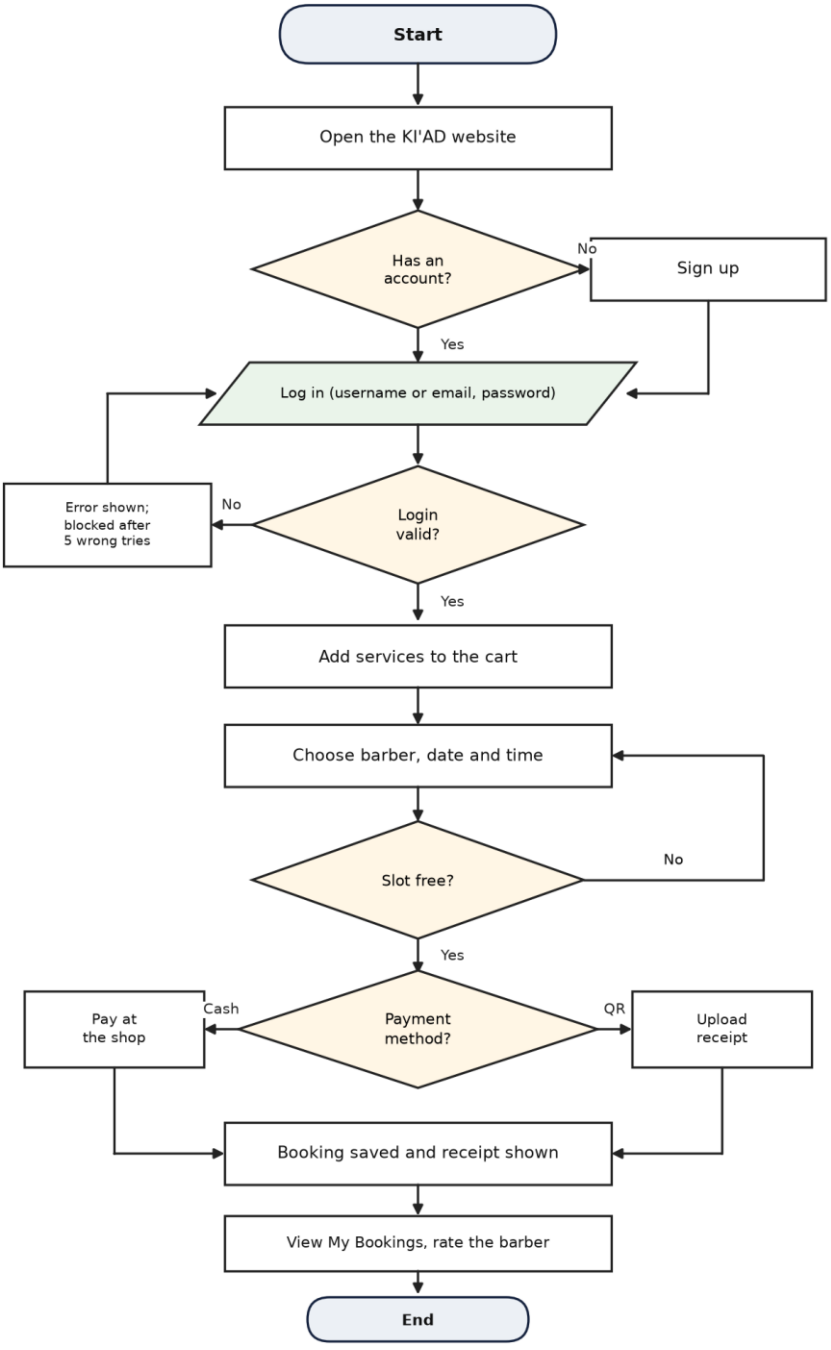


Figure 3.3 : Customer booking flowchart

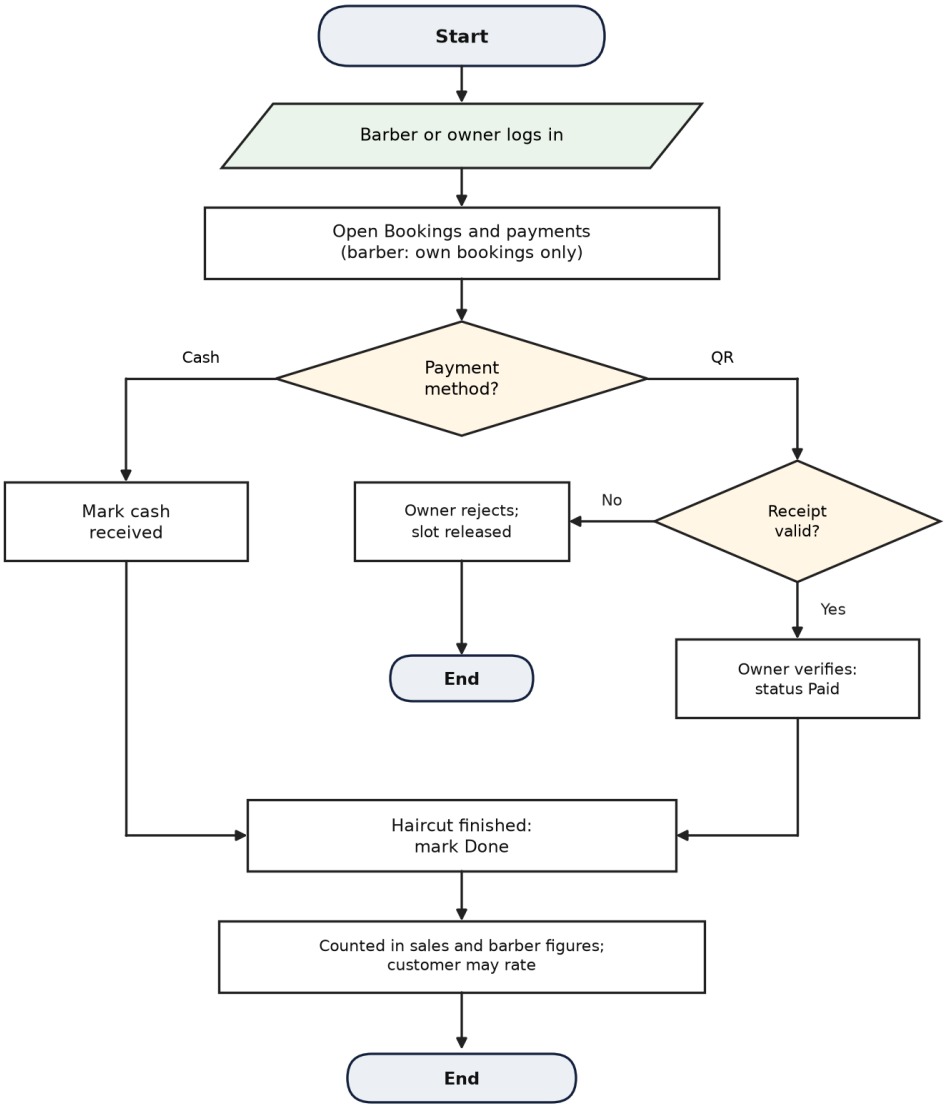


Figure 3.4 : Barber and owner booking and payment flowchart

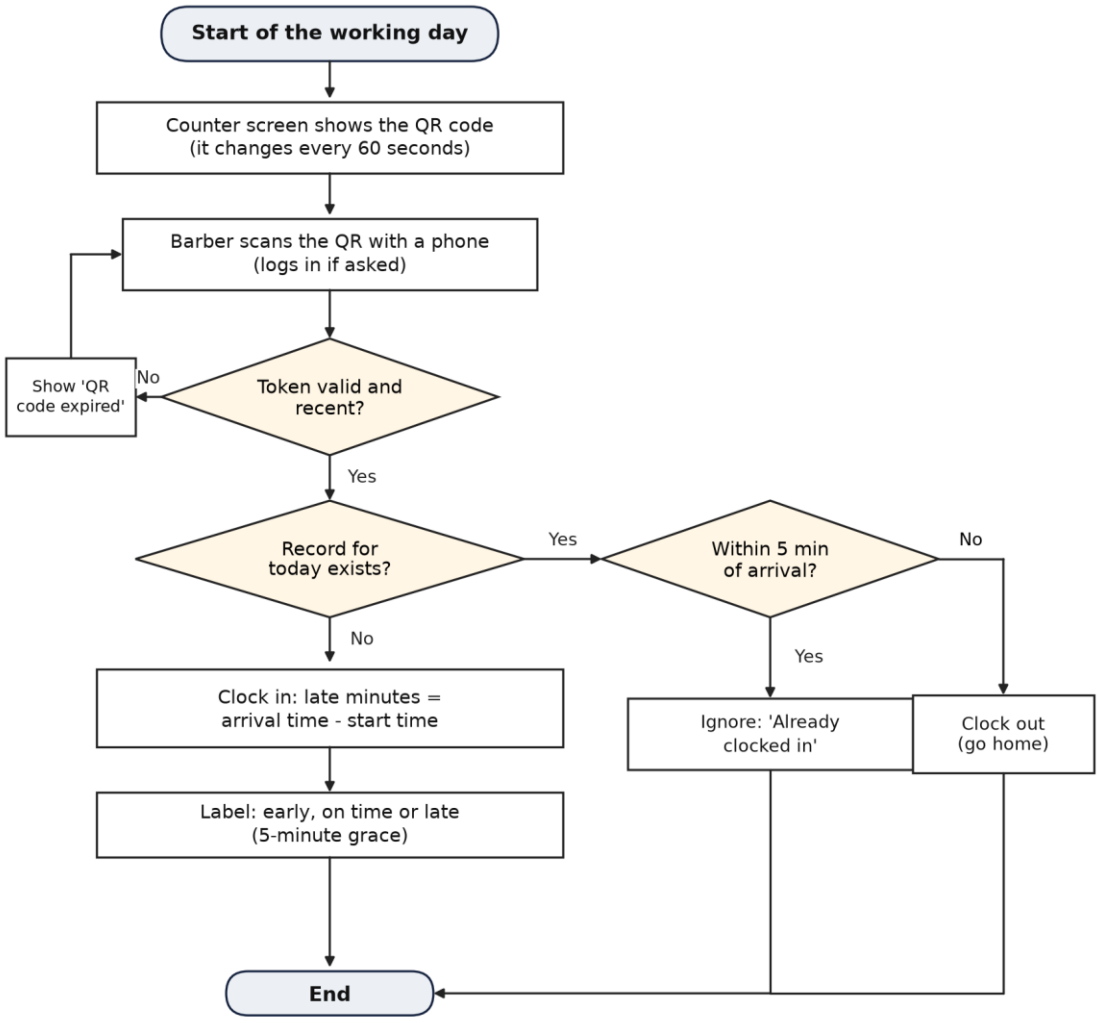


Figure 3.5 : Attendance (QR scan) flowchart

3.6 Conclusion

This chapter selected the Agile methodology, outlined the six phases of Agile and listed the requirements derived from the owner interview and observation. Use cases and flowcharts were used to analyse the requirements. The next chapter makes them a design and explains how the system was constructed.

DESIGN & IMPLEMENTATION

4.1 Introduction

This chapter covers how we got from the requirements in Chapter 3 to a working system. The first half is devoted to design, interface of the main pages, database, data flow, entity relationship diagrams, and security framework. The second half focuses on implementation: the place where the system is executed, the tools that are employed, what the modules look like and how the most important functions operate, and short excerpts of real code.

The finished system has 42 PHP files with about 5,060 lines, four style sheets with about 2,360 lines and one JavaScript file. Customer pages are in the main folder, and all barber/s owner pages are in an admin folder.

4.2 Interface Design

The interface was designed as sketches and then assembled and developed step by step in the actual system. The main pages presented in this section are as they will look in the completed KI'AD system. A dark and gold aesthetic is used for the customer side, harmonizing with the shop's photos and logo. The admin side has a simple and light design with density, as the owner reads it between the customers, and requires to see the numbers and actions quickly. The same menu position and same buttons for the same actions are maintained on each page, allowing a user who has learned the buttons of one page to use the next page.

4.2.1 Customer Interface Design

The customer pages are based on the booking steps. Figure 4.1 is a login/signup page. It also has two tabs: one a shop photo and another a short promise (look sharp, book in seconds) and the opening hours, address and rating are displayed at the bottom so that a new visitor can see that the shop is real before signing up.

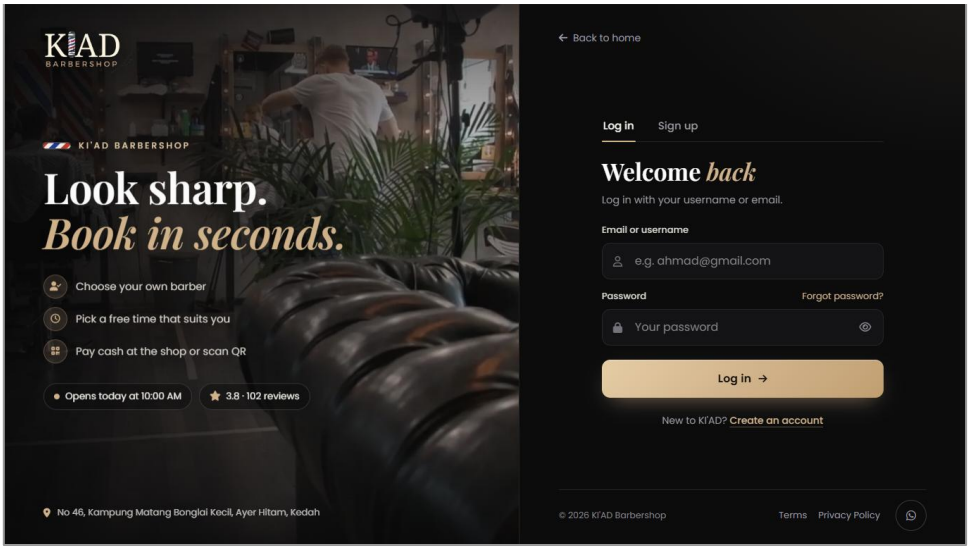


Figure 4.1 : Login and sign-up page of the KI'AD system

The services page (Figure 4.2) is a grid of cards that contain a photo, name and price, and there is a search box above the grid. Each booking page has a four-step progress bar (Services, Cart, Barber & Time, Payment) that allows the customer to always see his or her progress.

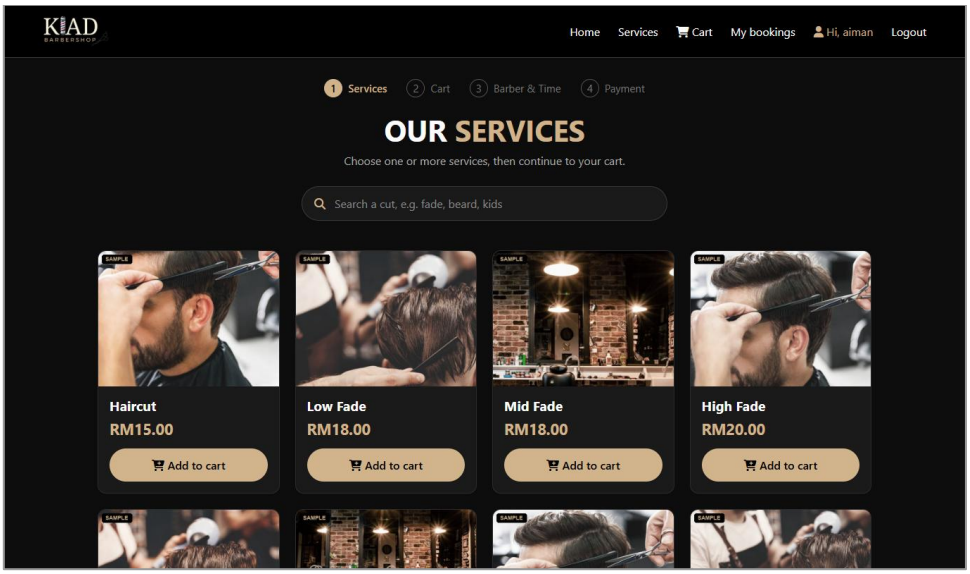


Figure 4.2 : Services page

On the Barber & Time page (Figure 4.3) the three things that a slot relies on are asked in the order that a customer will probably think about them: who, which day and what time. Barbers are shown as buttons, the time list is limited to the free slots and a short line shows how many slots are still free.

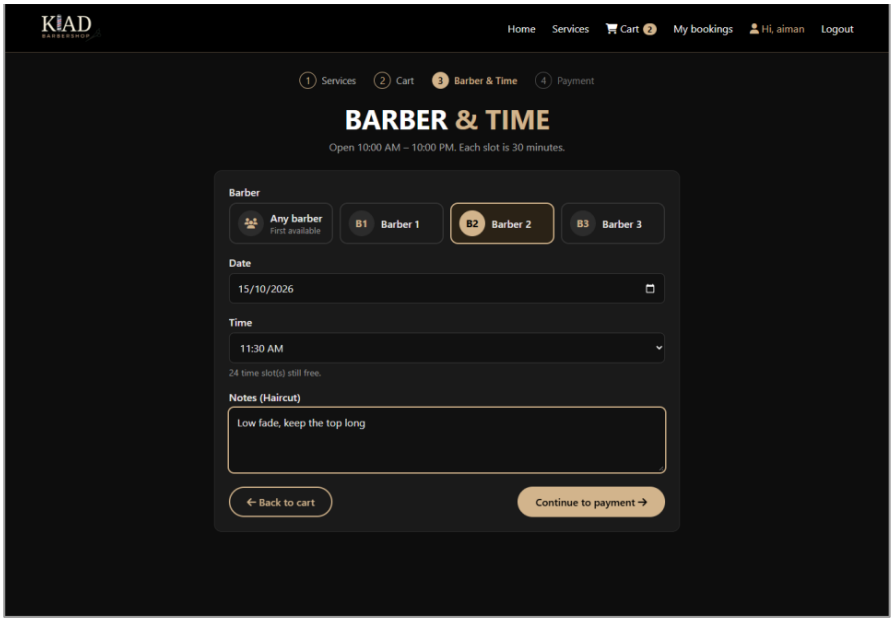


Figure 4.3 : Barber and time page

The payment page (Figure 4.4) shows the booking summary again to ensure that the customer has that information again before paying. The QR code and upload box are only visible if the online payment is selected as the payment method, so that the page is kept short for the customers who pay cash.

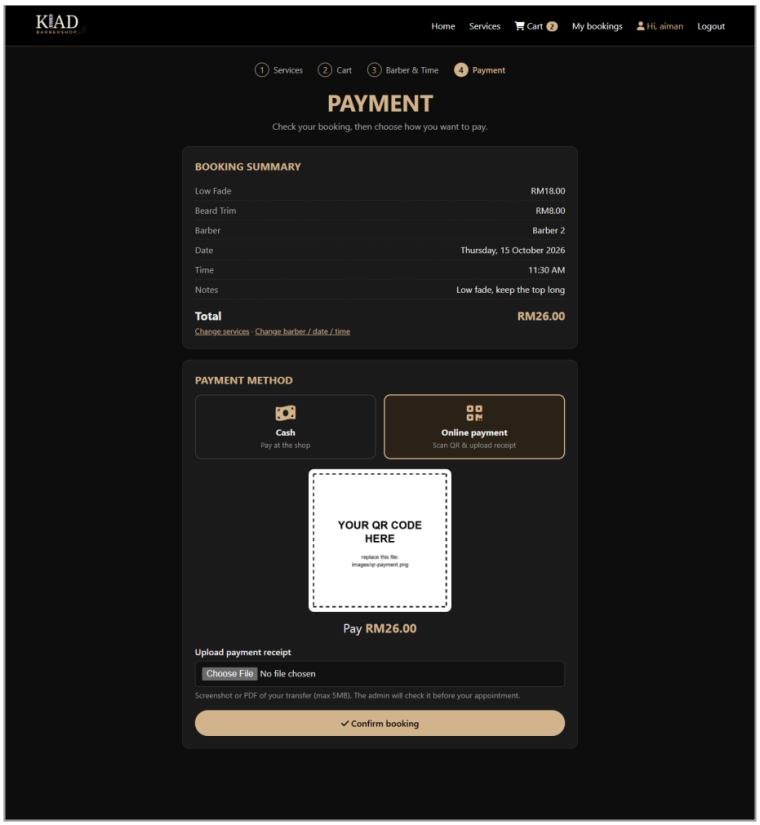


Figure 4.4 : Payment page with online QR payment selected

My Bookings (Figure 4.5) shows all bookings on a card format, including date, services, barber, price, and a colour-coded status indicator – a link to the receipt – and a rating form for completed haircuts.

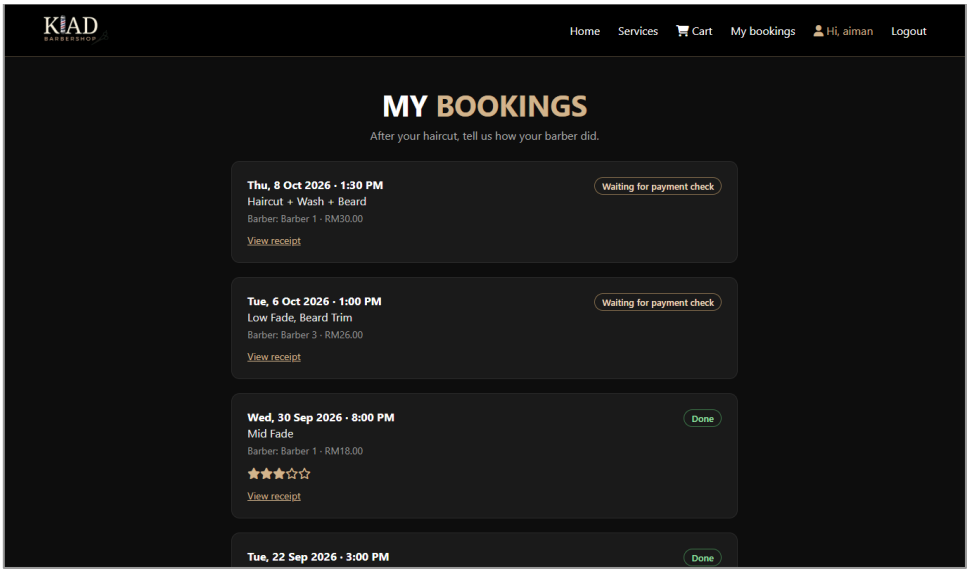


Figure 4.5 : My Bookings page

4.2.2 Barber and Owner Interface Design

The dashboard (Figure 4.6) displays four numbers at the top, the schedule for the current day in the middle and a seven-day sales chart, and three small tables at the bottom. The bookings page (Figure 4.7) is a table that has one row for each booking and the actions that are allowed at the end of each row. Owner's attention highlight will be yellow. In barbers, they will have the same format but they will be missing some of the buttons and the figures will be his/her own.

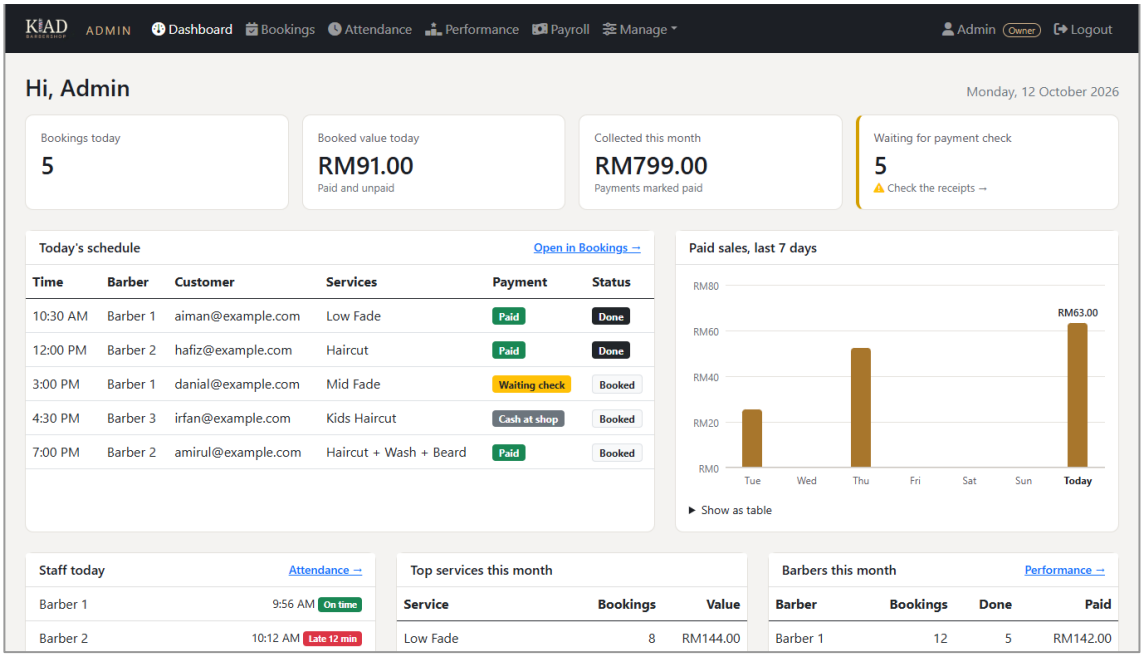


Figure 4.6 : Owner dashboard

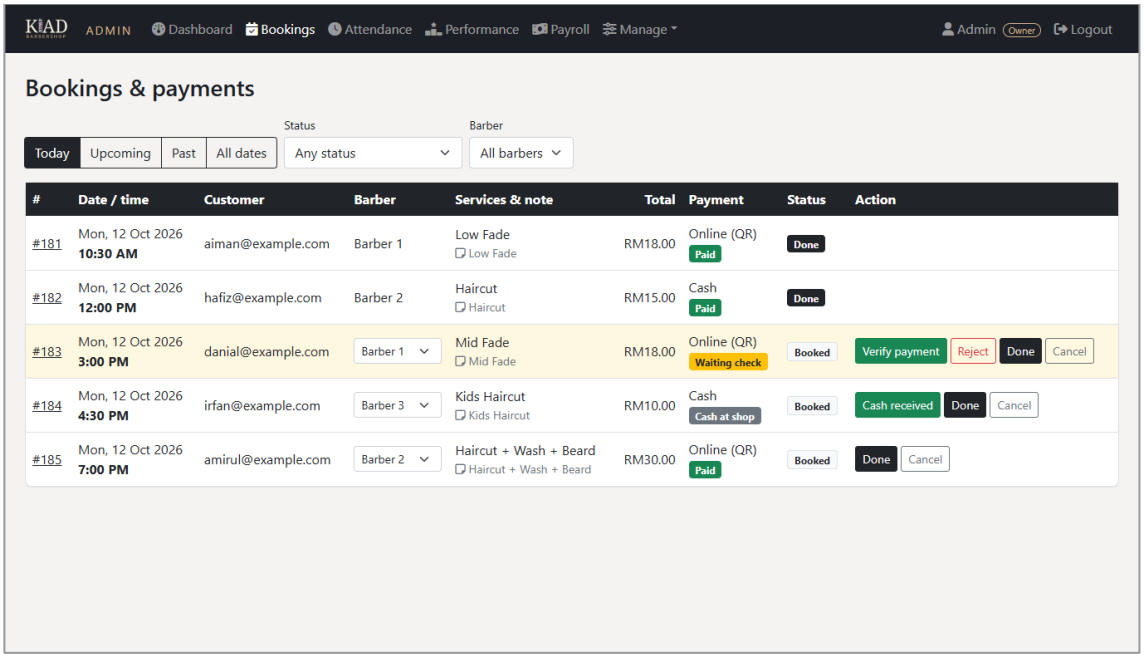


Figure 4.7 : Bookings and payments page

The counter QR link is at the top of the attendance page (Figure 4.8) followed by the arrivals of the day and a monthly summary. Each barber's days worked, lateness, customers, sales and rating are listed next to a barber's salary box on a payroll page (Figure 4.9) so the owner can decide upon the salary based on the facts in front of him.

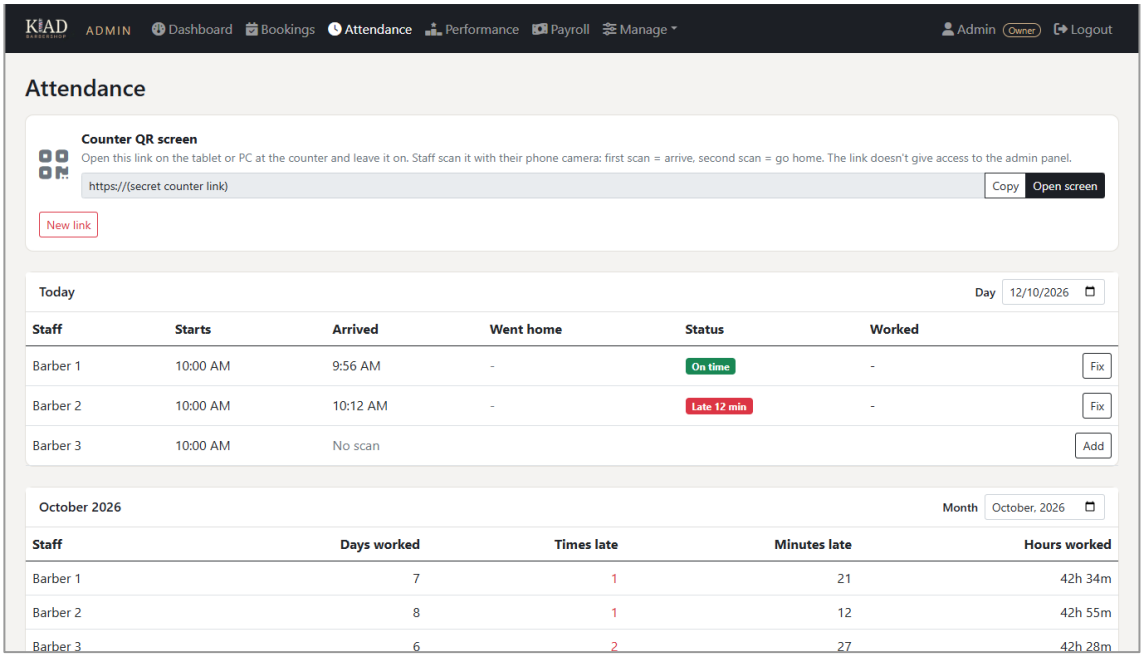


Figure 4.8 : Attendance page

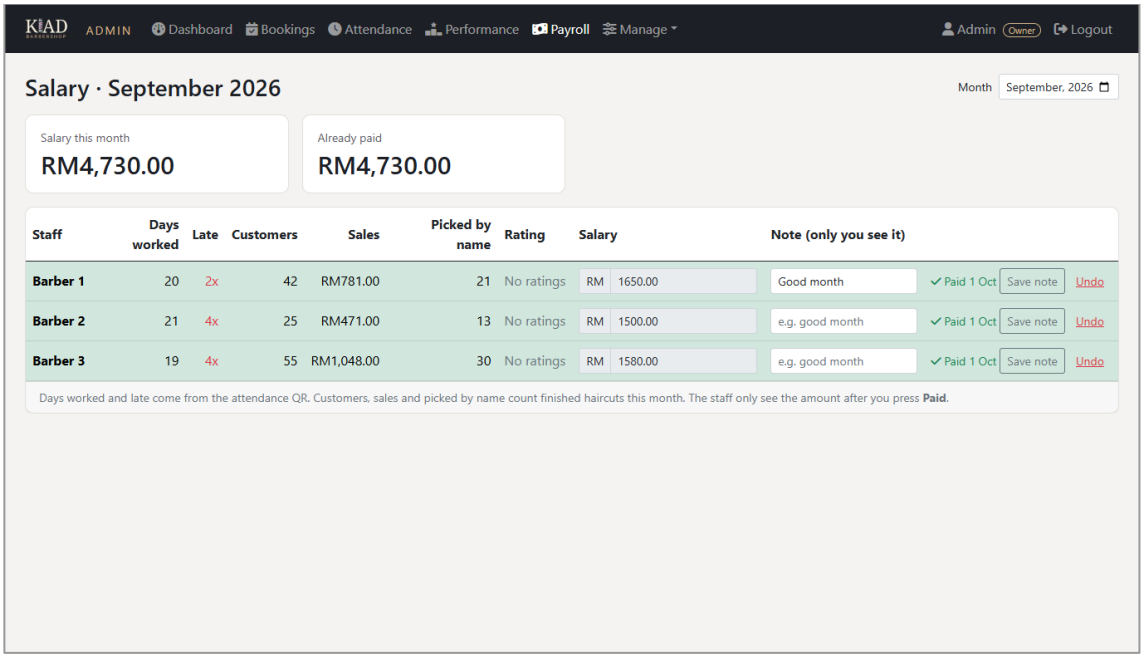


Figure 4.9 : Payroll page (September 2026, demo data)

4.3 Database Design

The information is saved in a MySQL database containing 11 tables. There are nine of these with business data and two small ones with settings and failed logon data. Each fact remains in one place: A receipt is one row in the receipt table, and services are one row in receipt_item, so an old receipt is never rewritten when the price of a service is changed in the list of services.

4.3.1 Data Dictionary

Each table is described in the tables 4.1-4.9. Types are displayed as they're declared in the install script.

Table 4.1 : Data Dictionary of Table "customer"

Field	Type	Key	Description
cust_id	int	PK	Customer number, auto increment
username	varchar(255)	Unique	Name used to log in
password	varchar(255)		bcrypt hash; older plain-text rows are converted at next login
email	varchar(255)	Unique	Email, also accepted for login
must_change_password	tinyint(1)		1 when the owner gave a temporary password

Table 4.2 : Data Dictionary of Table "staff"

Field	Type	Key	Description
staff_id	int	PK	Staff number
name	varchar(100)		Name shown to customers when they choose a barber
username	varchar(50)	Unique	Login name
password_hash	varchar(255)		bcrypt hash
role	varchar(10)		owner or staff
is_barber	tinyint(1)		1 if customers can choose this person
active	tinyint(1)		0 locks the account out at once
created_at	datetime		Date the account was made
start_time	varchar(5)		Own start time (HH:MM); empty means shop opening time

Table 4.3 : Data Dictionary of Table "service"

Field	Type	Key	Description
service_id	int	PK	Service number
name	varchar(100)		Service name
price	decimal(8,2)		Price in Ringgit
image	varchar(255)		Path of the photo

Table 4.4 : Data Dictionary of Table "receipt"

Field	Type	Key	Description
receipt_ID	int	PK	Booking and receipt number
cust_id	int	FK	Customer who booked
name	varchar(255)		username/email copied at booking time
date, time	date, time		Appointment day and start time
notes	text		Customer's note, for example the style wanted
total	decimal(8,2)		Sum of the item prices
payment_method	varchar(10)		cash or online
payment_proof	varchar(255)		Path of the uploaded receipt (QR only)
payment_status	varchar(10)		cash, pending, verified or rejected
barber_id	int	FK	Barber assigned
job_status	varchar(10)		booked, done or cancelled
chosen_barber	tinyint(1)		1 if the customer picked the barber by name
created_at	datetime		Time the booking was made

Table 4.5 : Data Dictionary of Table "receipt_item"

Field	Type	Key	Description
item_id	int	PK	Row number
receipt_ID	int	FK	Booking the item belongs to
service_name	varchar(100)		Service name copied at booking time
price	decimal(8,2)		Price copied at booking time

Table 4.6 : Data Dictionary of Table "review"

Field	Type	Key	Description
review_id	int	PK	Review number
receipt_ID	int	Unique, FK	Booking being rated; one review per booking
cust_id	int	FK	Customer who rated
barber_id	int	FK	Barber who was rated
stars	tinyint(1)		1 to 5
comment	varchar(500)		Optional comment
created_at	datetime		Time of the review

Table 4.7 : Data Dictionary of Table "attendance"

Field	Type	Key	Description
attendance_id	int	PK	Row number
staff_id	int	FK	Staff member
work_date	date	Unique with staff_id	Day of work; one row per person per day
clock_in	datetime		Arrival time
clock_out	datetime		Time of leaving, empty until the second scan
late_minutes	int		Minutes after start time (negative if early)
edited_by_owner	tinyint(1)		1 if the owner fixed this row by hand

Table 4.8 : Data Dictionary of Table "payroll"

Field	Type	Key	Description
payroll_id	int	PK	Row number
staff_id	int	FK	Staff member
month	char(7)	Unique with staff_id	Month as YYYY-MM
amount	decimal(10,2)		Salary in Ringgit
note	varchar(255)		Private note, shown only to the owner
paid_at	datetime		Time marked paid; empty while only saved

Table 4.9 : Data Dictionary of Tables "holiday", "setting" and "login_attempt"

Table / Field	Type	Key	Description
holiday.holiday_date	date	PK	A day the shop is closed
holiday.note	varchar(100)		Reason, shown to customers
setting.name	varchar(50)	PK	Setting name
setting.value	text		Setting value
login_attempt.attempt_id	int	PK	Row number
login_attempt.ip	varchar(45)	Indexed	Address of the failed attempt
login_attempt.username	varchar(255)	Indexed	Name that was tried
login_attempt.attempted_at	datetime		Time of the attempt

The setting table is a basic name/value repository. The settings and the initial values of the system are provided in table 4.10. The advantage they have, is that they exist in the database and the owner can alter them from the Settings page without editing the code.

Table 4.10 : Main Settings and the Values Used

Setting	Value used	Meaning
open_time, close_time	10:00, 22:00	Opening hours; slots are generated between them
slot_minutes	30	Length of one slot in minutes
late_grace_minutes	5	Minutes after start time before arrival counts as late
closed_days	(none)	Weekdays on which the shop is closed
qr_image, logo_image	images/qr-payment.png, images/logo.png	Payment QR and shop logo
shop_address, shop_phone, whatsapp_number	Ayer Hitam address and phone	Contact details on the site
attendance_secret, kiosk_key	(generated)	Random secrets for the attendance QR and the counter screen link
policy_terms, policy_privacy	Short plain-language text	Content of the Terms and Privacy page

4.3.2 Data Flow Diagram (DFD)

The context diagram is shown in Figure 4.10. The system is in the middle and communicates with three other systems outside of the box. Customers provide registration information, requests for services, receipts, ratings; receive services, free slots, receipts and booking status. Logins, cash and done updates and QR scans are sent by Barbers. The owner receives the dashboard, performance and attendance numbers, and sends out verification decisions, service, staff information, salary information and settings.

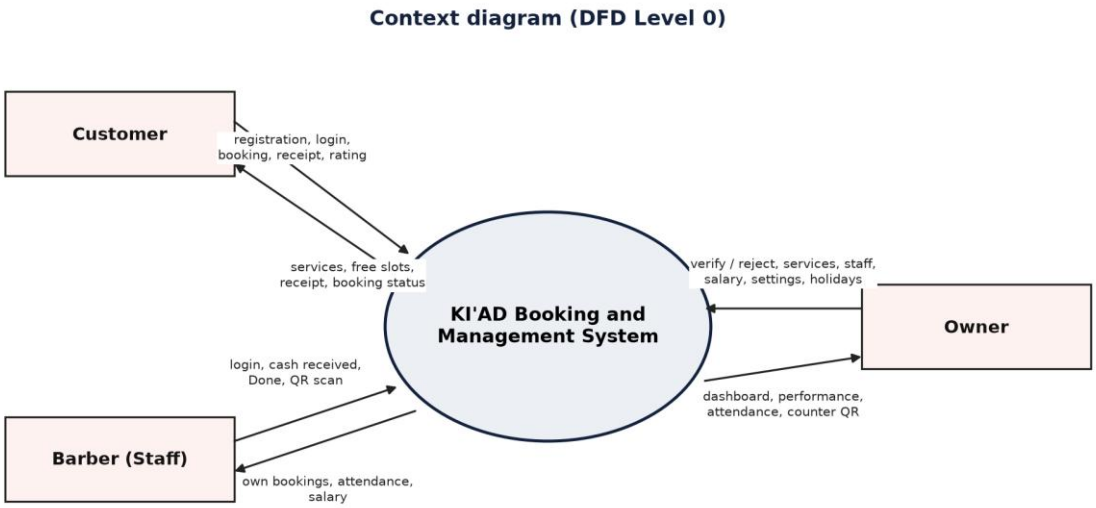


Figure 4.10 : Context diagram (DFD Level 0) of the KI'AD system

Figure 4.11 has divided the system into six processes and seven data stores. Authenticate the customer/staff stores. Booking reads and writes bookings. Payment processing updates bookings and ratings. Attendance writes the attendance store, payroll reads attendance and bookings to generate the monthly view and shop administration maintains services, staff and settings.

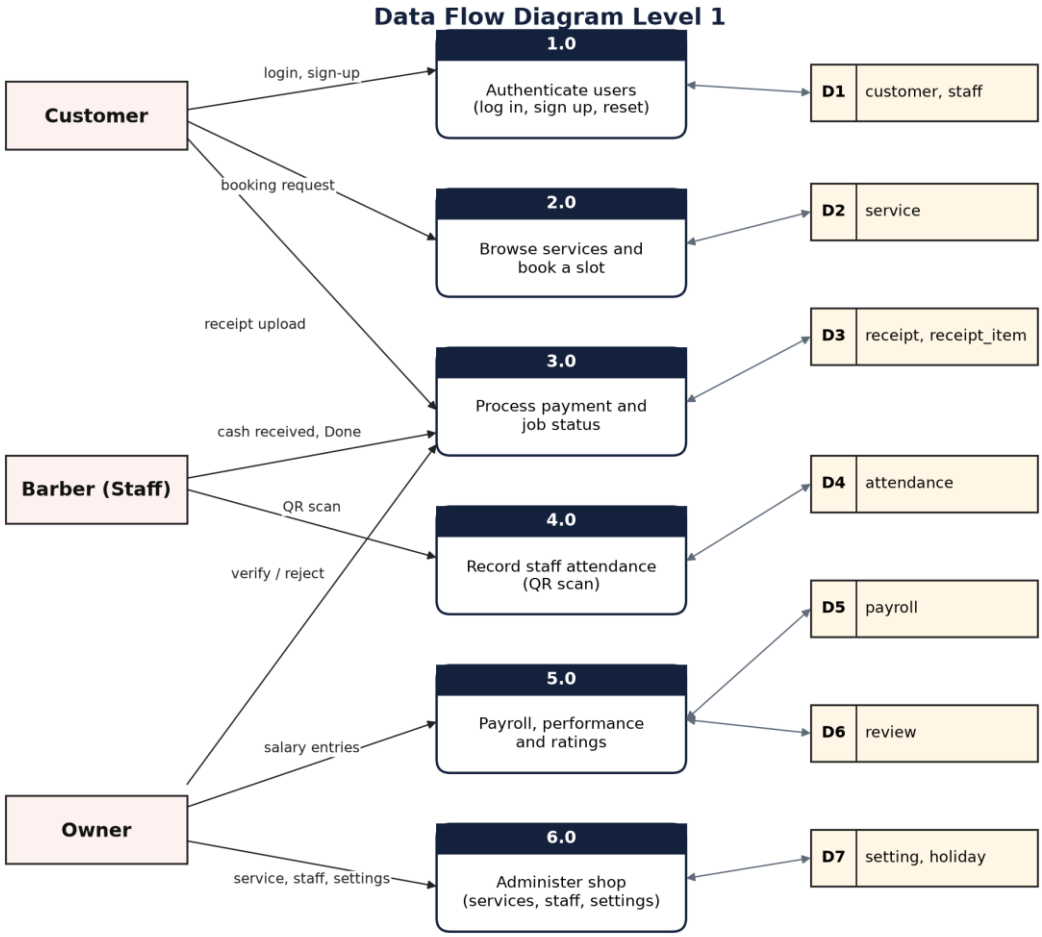


Figure 4.11 : Data flow diagram Level 1

4.3.3 Entity Relationship Diagram (ERD)

The tables and their relationship are shown in figure 4.12. Many customers place many bookings, each of which contains one or more items and may have one review; many barbers serve many bookings. Staff accounts are associated with attendance and payroll rows. The service name and price is copied to receipt_item when the service is booked, and this relation between receipt item and service list is drawn as dashed. The tables of the holiday, setting and login_attempt are isolated.

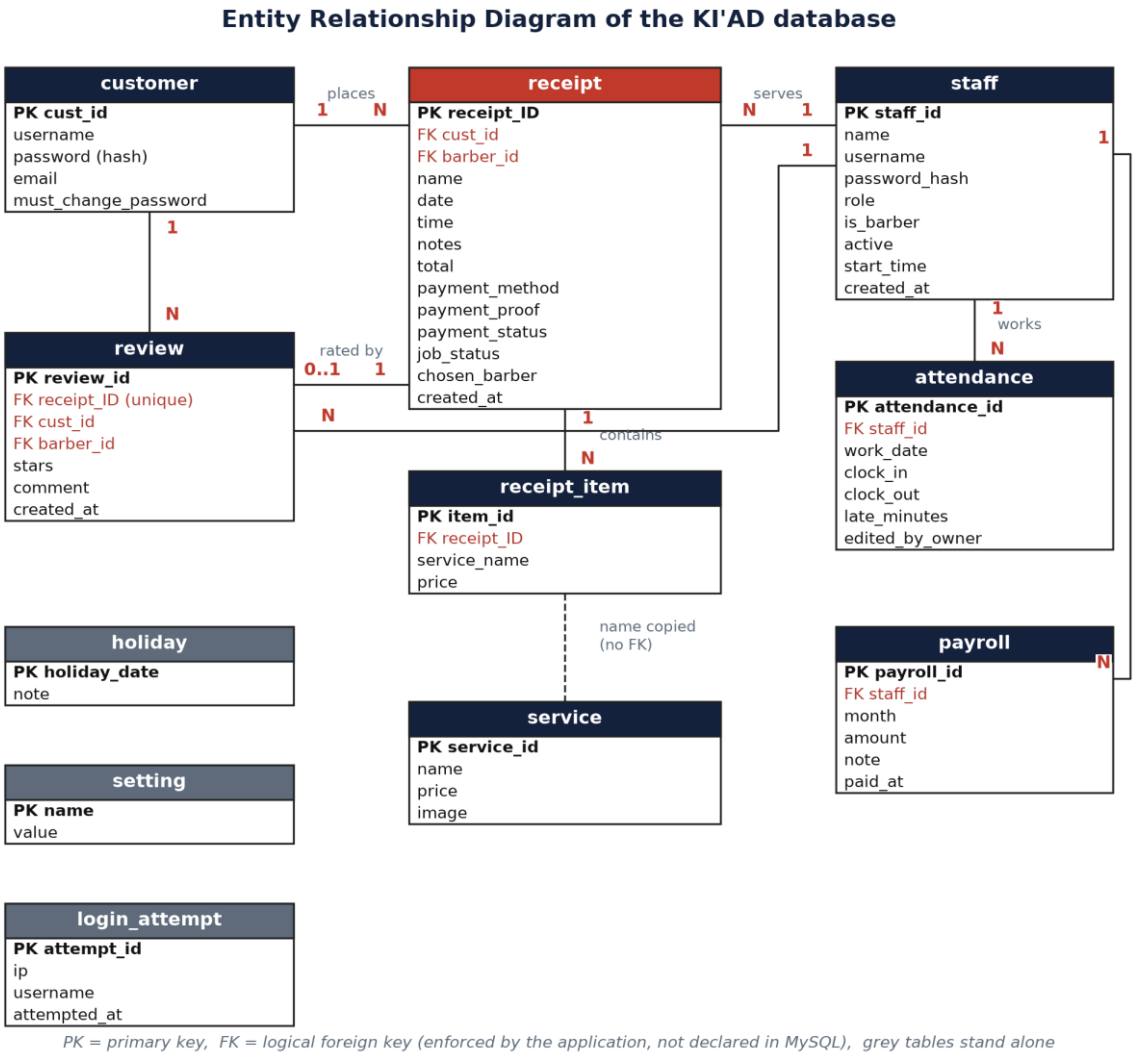


Figure 4.12 : Entity relationship diagram of the KI'AD database

It's important to say straight out that one thing is true. The install script does not create foreign keys, but only keys and indexes, so the relationships of Figure 4.12 are not enforced by MySQL but are maintained by application code. The code is always valid, and it always deletes in a controlled order—doing so in the real constraints is a reasonable hardening measure for the future.

4.3.4 Relationship Matrix

Table 4.11 : Relationship Matrix of the KI'AD Database

Table	customer	staff	receipt	receipt_item	review	attendance	payroll
customer	-	-	1 : N	-	1 : N	-	-
staff	-	-	1 : N (barber)	-	1 : N (barber)	1 : N	1 : N
receipt	N : 1	N : 1	-	1 : N	1 : 0..1	-	-
receipt_item	-	-	N : 1	-	-	-	-

Table	customer	staff	receipt	receipt_item	review	attendance	payroll
review	N : 1	N : 1	0..1 : 1	-	-	-	-
attendance	-	N : 1	-	-	-	-	-
payroll	-	N : 1	-	-	-	-	-

4.4 Security System Framework

Security was built in stages to protect the whole structure from being compromised when a single point is found to be weak. There are 7 layers in between the browser and the data, as shown in figure 4.13. Each of them is described below and the code that performs the function for the section.

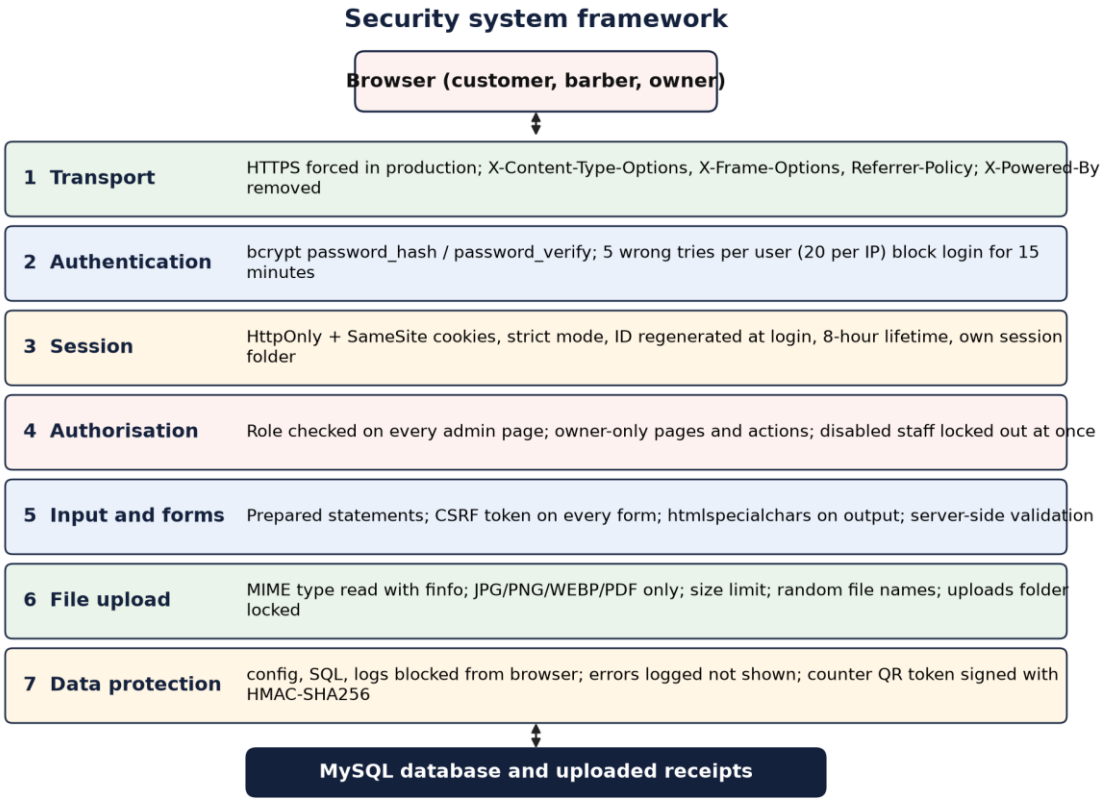


Figure 4.13 : Security system framework of the KI'AD system

4.4.1 Transport and Browser Safety

In production mode, all requests sent via plain HTTP are automatically switched to HTTPS, so that passwords or receipts are not sent in plain text. On each page, three headers are set to prevent the browser from guessing filetypes, to prevent the page from being framed by other websites and to restrict what is passed to other websites via the Referrer-Policy. The X-Powered-By header is stripped, so as not to advertise the PHP version.

4.4.2 User Authentication and Access Control

Customers and Staff login on the same account. Customer passwords and staff passwords are only stored as bcrypt hashes with PHP's password_hash and verified with PHP's password_verify. If you enter five incorrect passwords for the same user id or 20 from the same email address, your login is denied for 15 minutes. If the login fails, it is logged into the login_attempt table and removed from the table when login is successful.

For each admin page, the access control will be checked on the server. The common _auth.php file must be used with a staff session, fetches the account from the database on each use, and logs out the user if the account is disabled. Owner-only pages are defined by require_owner(), and actions within shared pages, e.g. checking payment, recheck the role. A prof who enters an owner URL is taken to the dashboard and a message displayed.

4.4.3 Session Management

The session cookie is HttpOnly, SameSite=Lax and also Secure (on the live site). Strict mode is enabled and the server refuses to accept session IDs not generated by it, the session ID is generated on login and so is unique to each login to prevent session fixation. A login lasts 8 hours of no activity, and sessions are stored separately from the other sites on the shared host so that they are not removed by other sites.

4.4.4 Data Protection and Input Validation

There are bound parameters for every query made with user input – this is the key protection against SQL injection. To prevent cross-site scripting attacks via names or notes, output is escaped with htmlspecialchars prior it is displayed. Each form includes a random CSRF token to compare with the one in the session, using the hash_equals function, which means that it is not possible for another site to submit a form in the user's name. Also server checked for inputs, like username, email, price, date, time, and password – length and rules of form are checked.

4.4.5 Secure File Upload

Customers upload receipts of payment and owner uploads service photos, logos, QR images. The file name or the type sent by the browser is not trusted by the code. It reads the actual MIME type using finfo, only accepts JPG, PNG, WEBP and, for receipts, PDF, limits size (5MB for receipts, 3MB for images) and stores file under a random name. The receipts folder has its own .htaccess to prevent listing and running of files.

4.4.6 Attendance QR Token

A link with a token is stored in the QR code on the counter screen. The token consists of a time window number, followed by an HMAC-SHA256 signature created with a secret

which is only stored in the database. Every 60 seconds a new token appears and the server accepts the current window and the two preceding windows, making a screen shot of the code that is not useful 60 seconds later. The counter screen is opened via a secret link; the shop tablet doesn't require an owner logon and the owner can change the link anytime.

4.4.7 Data Integrity and Privacy

A booking and its items are written within a single database transaction. Should anything go wrong, you will have no half-saved bookings as everything will be rolled back and the uploaded receipt will be deleted. If the owner wants to change a paid salary, he or she has to cancel the payment first. The configuration, SQL and log files are protected from the browser by .htaccess, and in production, PHP errors are not displayed to visitors, but written to a log file. The only data to be stored in the system is a username, an e-mail address, a password hash, booking data and the Terms and Privacy page informs the customer of this.

4.4.8 Flow of the System

The three users are compared in Figure 4.14. The owner adds the customer's needs, charges them for it and then checks the QR payment, after which the customer rates the haircut. Attendance scans are uploaded to the owner's payroll and performance pages by barbers.

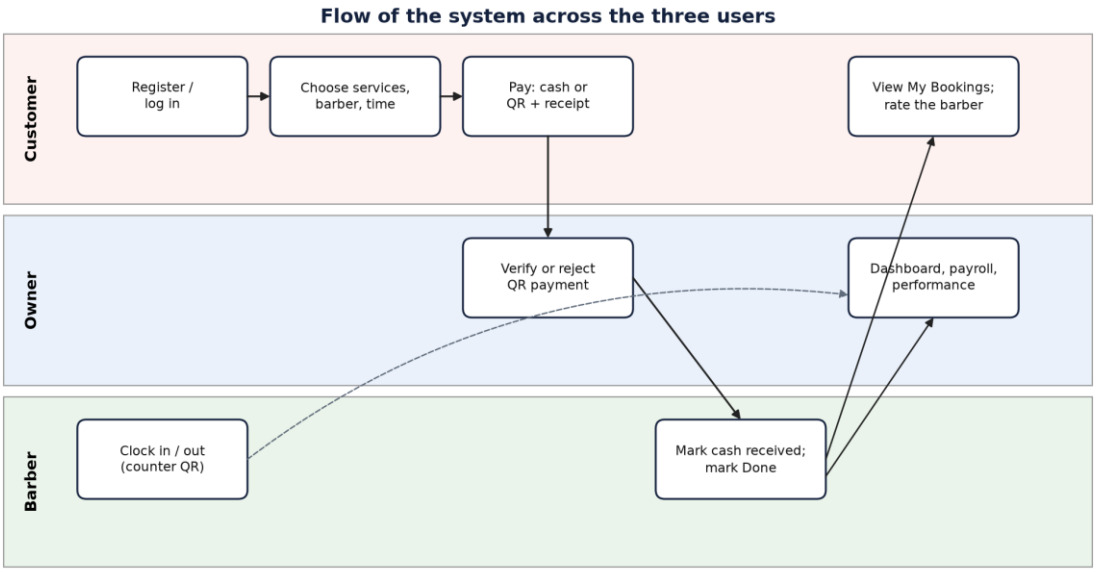


Figure 4.14 : Flow of the KI'AD system across the three users

4.5 Implementation Introduction

It began with the main models of the design being stable. The code is intentionally simple. No framework, each page is a PHP file, a handful of shared files and the logic that multiple pages require, like whether slots are available, exists in booking_lib.php. This allows

easy reading as the project may be passed on to another person some day in the future, and the shop owner might want to get on another project.

4.6 Execution Platform

4.6.1 Development Platform

The system is developed on a Windows 11 laptop having XAMPP (Apache, PHP 8.2, MariaDB 10.4 and phpMyAdmin). Having a local workstation allowed the database to be reset and re-seeded as desired. A script was used for tests that required numerous bookings, and a set of 170 bookings was created using it; these are referred to as demo bookings in this report, and were used for screenshot and performance tests, and were labelled as such where they appeared as demo data.

4.6.2 Hosting Platform

The parts are connected as in Figure 4.15 in a shared host with PHP and MySQL at barbershophesmes.site.je. HTTPS is used to access the site. The web server runs the customer pages and the admin pages, both of which include the shared PHP files that set up the session, the CSRF helpers and the database connection. Data is then stored in MySQL and uploaded receipts are stored in a protected folder on the same server.

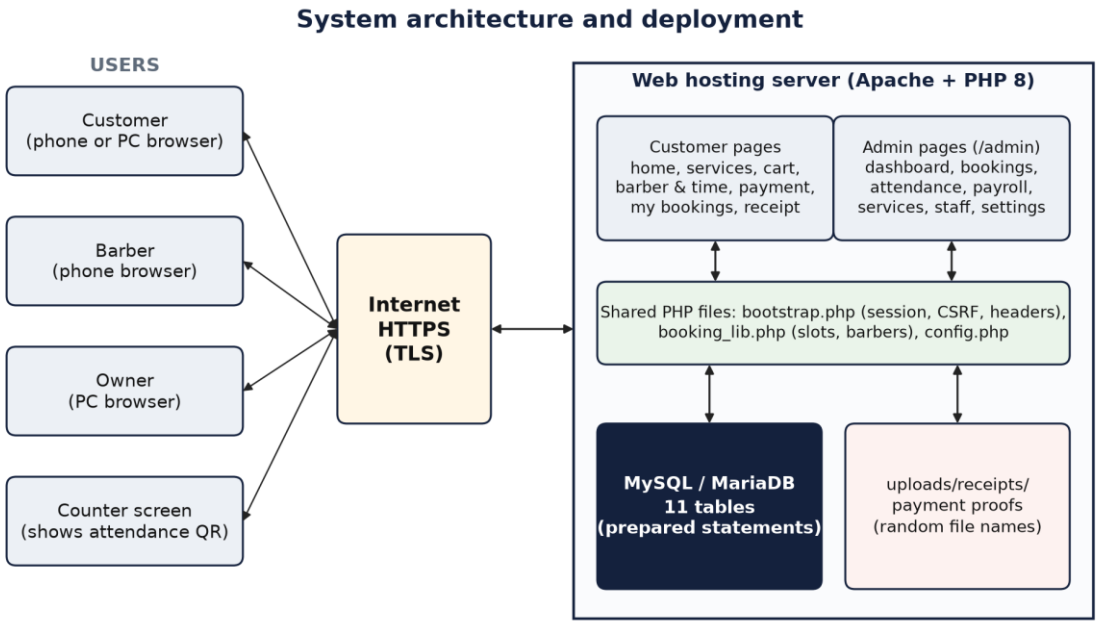


Figure 4.15 : System architecture and deployment

Pages that are visible to customers load Font Awesome and Google Fonts, admin pages load Bootstrap 5 and Font Awesome, and the QR display and scanning uses the public CDN libraries, Font Awesome and qrcode.js and jsQR. The home page also contains a map of the

address of the shop, and links to WhatsApp using the URL wa.me, which is the way customers contact the shop to cancel or request a password reset.

4.7 Implementation Tools

Table 4.12 : Implementation Tools

Tool	Use in the project
PHP 8.2 with MySQLi	All server-side logic, using prepared statements for every query with user input
MySQL / MariaDB	Storage for the 11 tables described in Section 4.3
JavaScript	Greying out unavailable times on the booking page (the page asks slots.php for a JSON list), the login and sign-up tabs, image preview and compression in Settings, QR display and QR scanning
HTML and CSS	Page structure and responsive layout; the customer side has its own style sheets, the admin side uses Bootstrap 5
Bootstrap 5.0.2, Font Awesome 6.4	Admin layout, tables, badges and icons
qrcode.js, jsQR	Drawing the counter QR code and reading it from the phone camera
XAMPP and phpMyAdmin	Local server, database import and inspection
Google Chrome with Playwright	Page screenshots and mobile-viewport checks
Python (requests)	Automated test scripts for the cases in Chapter 5

4.8 System Interface and Modules

This section is a walkthrough of completed pages. Only mentioned are pages which have already been displayed in Section 4.2. The names (e.g. aiman, barber 1) are test accounts in a locally installed copy of the code with demo data, and the dates are when the screenshots were taken.

4.8.1 Customer Interface

Home page, sign-up and login

The home page displays the shop name, a short story, the address including a map, a contact number and a Make an Appointment button. If real customers have left positive ratings, a Review section will be displayed and will include the latest of these. Make an Appointment will take you to Log in page containing two tabs: Log in and Sign up. This same page sends customers to the right place after they log in, as well as barbers and the owner.

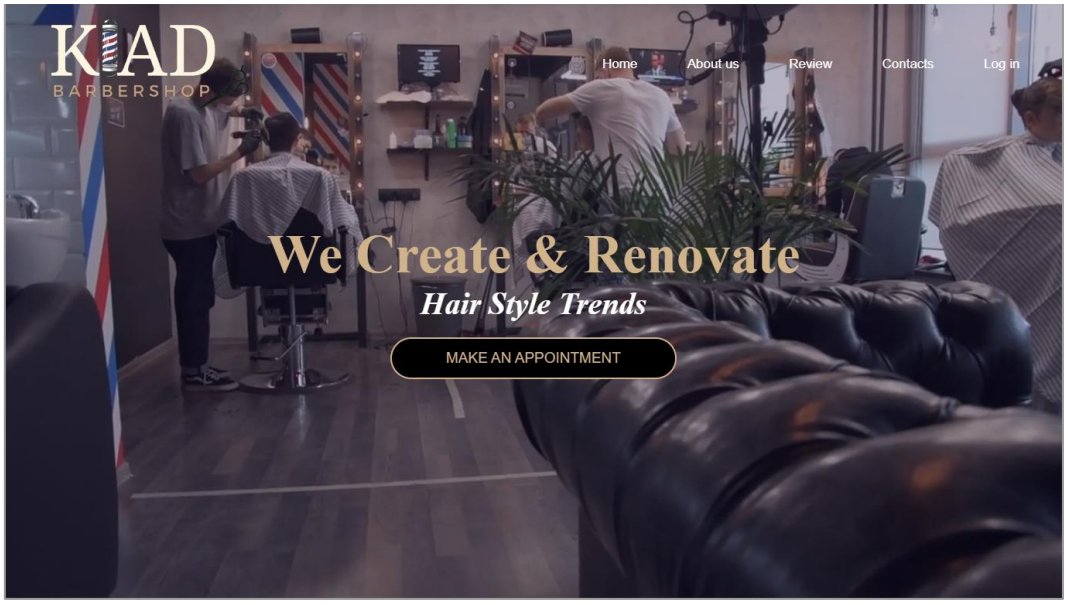


Figure 4.16 : Home page of the KI'AD site

The screenshot of the login page (Figure 4.1) is shown in Section 4.2.

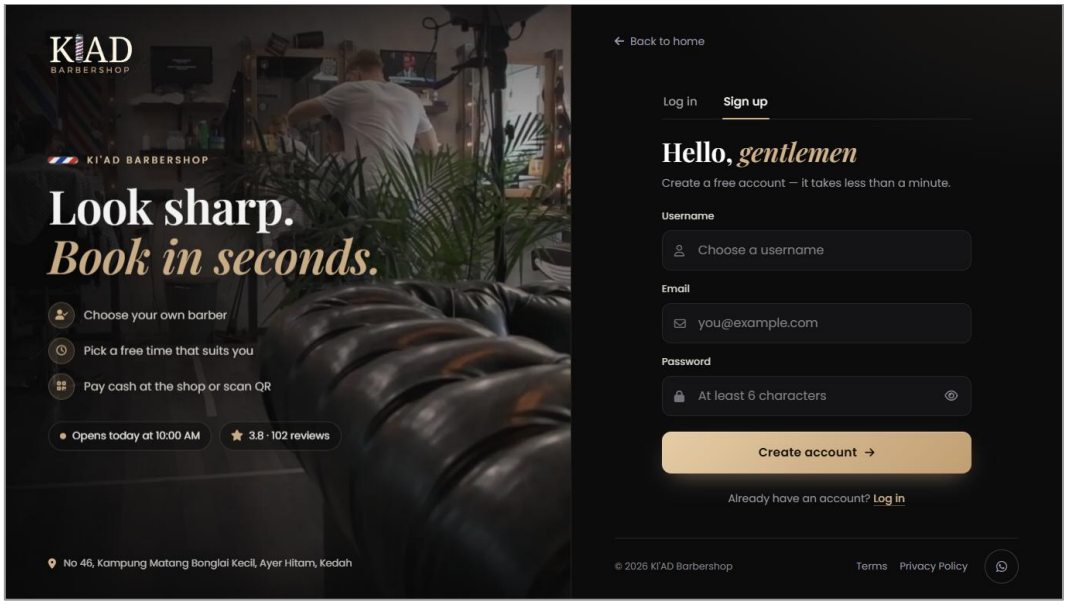


Figure 4.17 : Sign-up tab

There is no minimum requirement for signing up, only a username, email address and password of at least six characters. The username and the e-mail must not be in use. If a customer forgets his/her password, he/she will be redirected to the Forgot password page (Figure 4.18) in which he/she will be taken to WhatsApp with a message that is already pre-typed to the shop and the owner will give a temporary password from the Customers page. The customer will need to change it with their new password the next time he logs in.

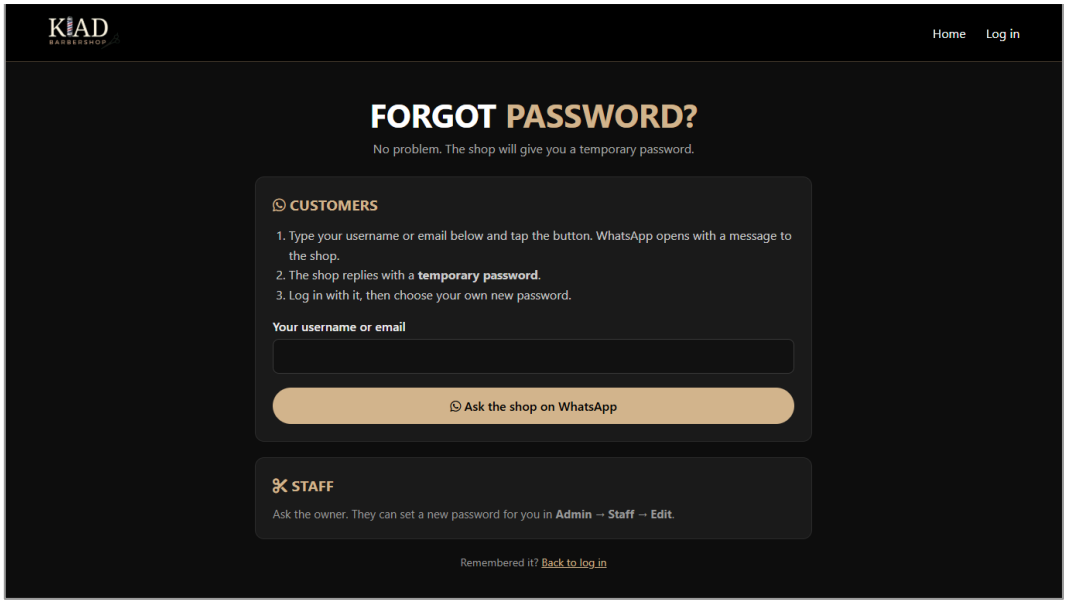


Figure 4.18 : Forgot password page

Services and cart

The services page has a photo and price on all 9 services. The search box is used to filter the cards as the customer enters, and accepts words as they enter, either in the name or the price, such as "fade" or "rm15". Items added to the cart are displayed and can be deleted.

We will look at the services page in section 4.2, with a screenshot of it displayed in figure 4.2.

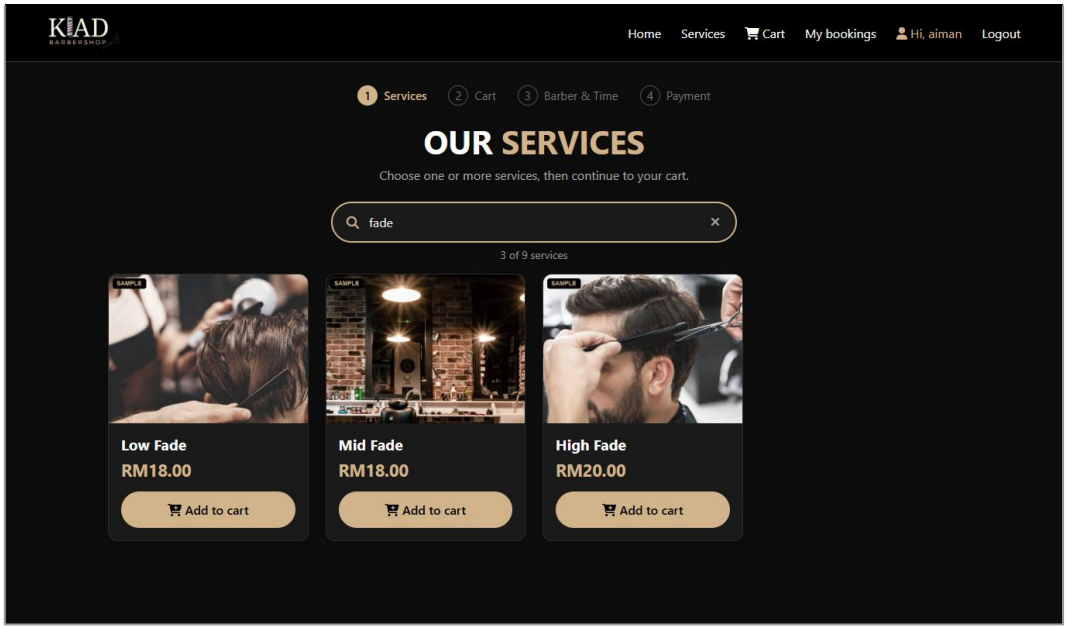


Figure 4.19 : Search on the services page ("fade")

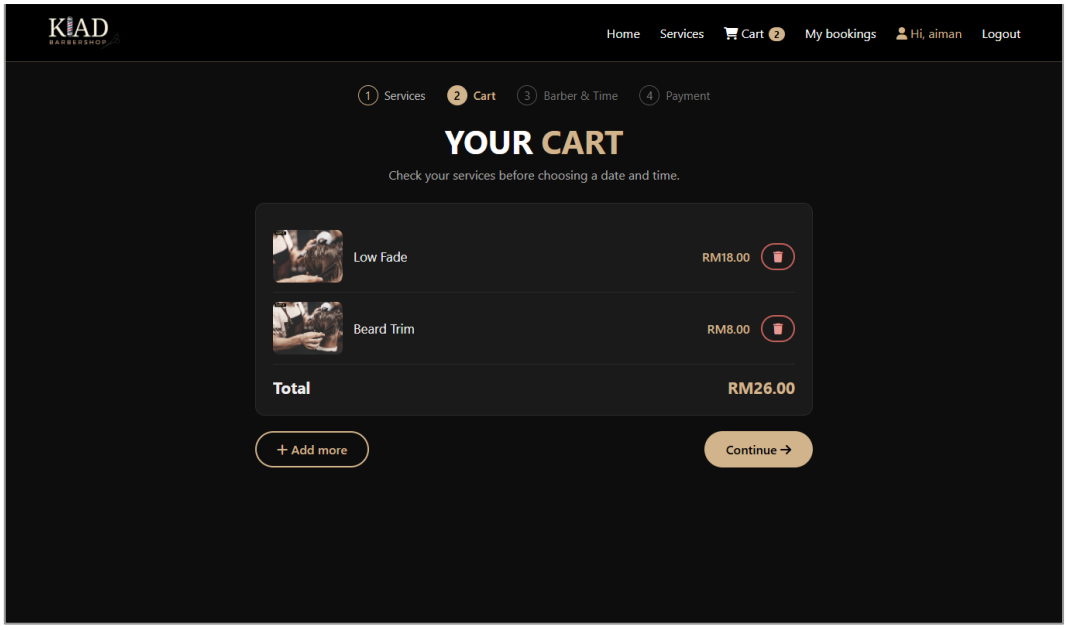


Figure 4.20 : Cart page

Barber and time

This is the page where you won't have to worry for double booking. The customer selects Any barber or a particular barber, a date and a time. If the barber or the date changed, the page requests unavailable times from slots.php and disables the unavailable times in the list and displays the number of slots that are still available. Times in the past, closed days and holidays are blocked too. It is explained in Section 4.9 what the code behind it does.

The screen of the Barber and time page (Figure 4.3) is displayed in Section 4.2.

Payment and receipt

On the payment page, the booking information is repeated and there are two methods. To fill in the booking, cash will be used. Online payment displays QR code and file box of the shop. The customer enters the payment in the banking app, uploads a receipt, and the booking is saved pending the owner's verification. Once saved the customer is directed to a receipt that can be printed.

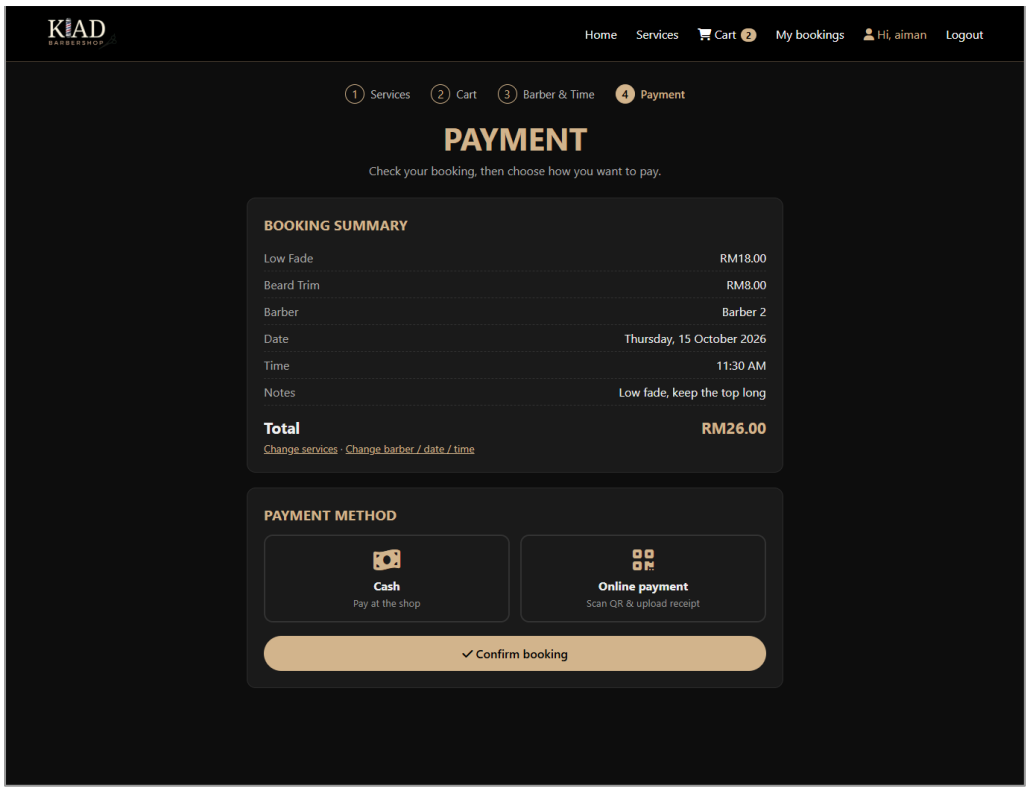


Figure 4.21 : Payment page with cash selected

The screenshot of the payment page with QR (Figure 4.4) is shown in Section 4.2.

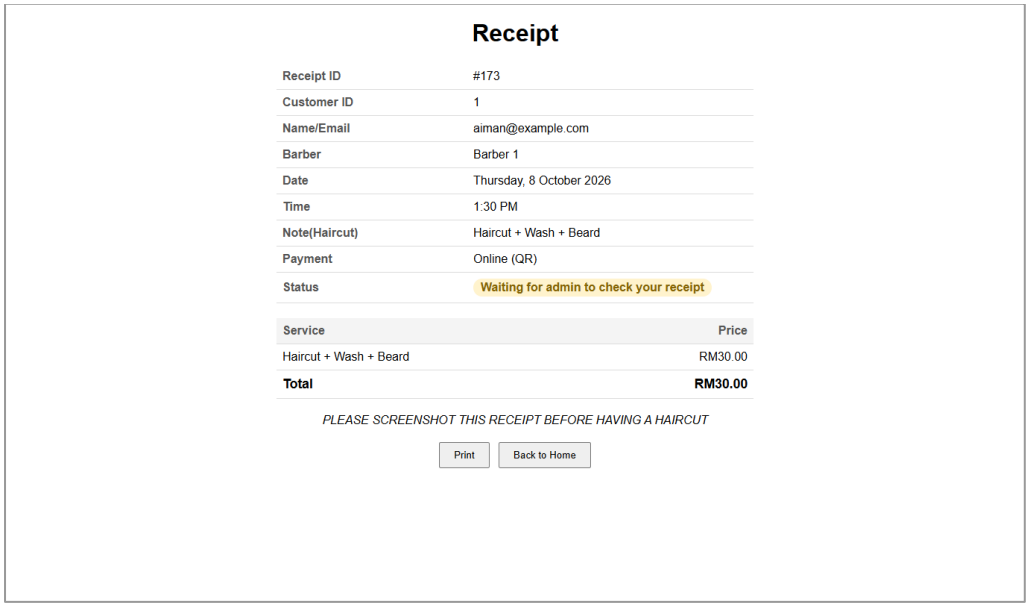


Figure 4.22 : Receipt page

My Bookings, account and terms

In My Bookings, you will see all the bookings with one of the following status: Booked, Waiting for payment check, Payment rejected, Done or Cancelled. A booked booking has a star rating form and a booked and rated booking has a star rating that is displayed. My Account displays the customer's username and email address and allows a customer to change the

password. The Terms and Privacy page is where the rules of booking are displayed and the text on this page of the shop can be edited in Settings, what the shop does with the customer data is displayed.

The screenshot of My Bookings (Figure 4.5) is shown in Section 4.2.

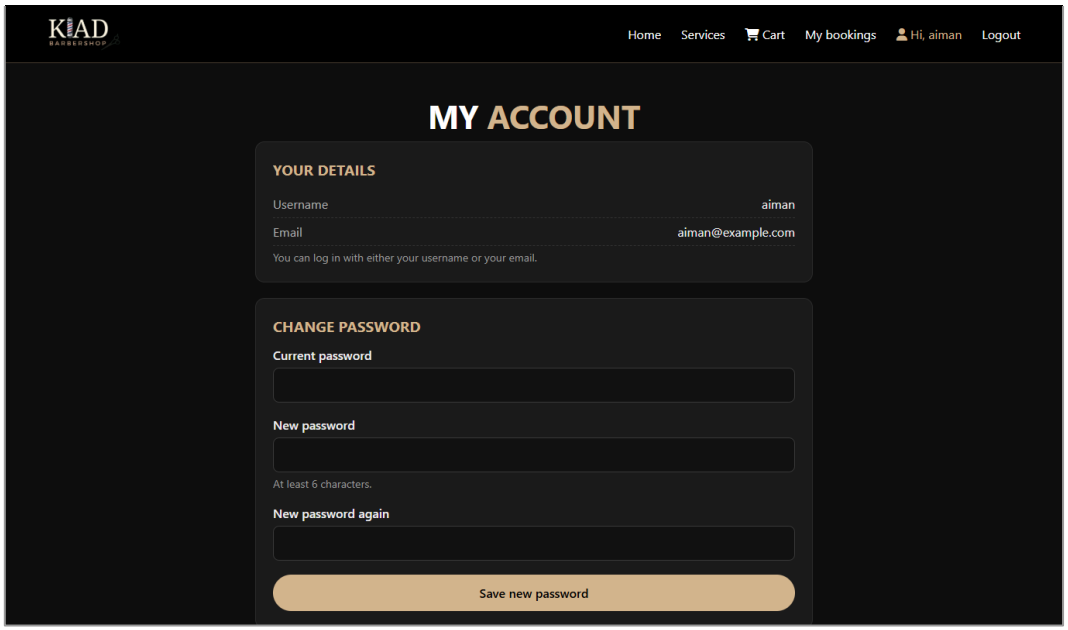


Figure 4.23 : My Account page

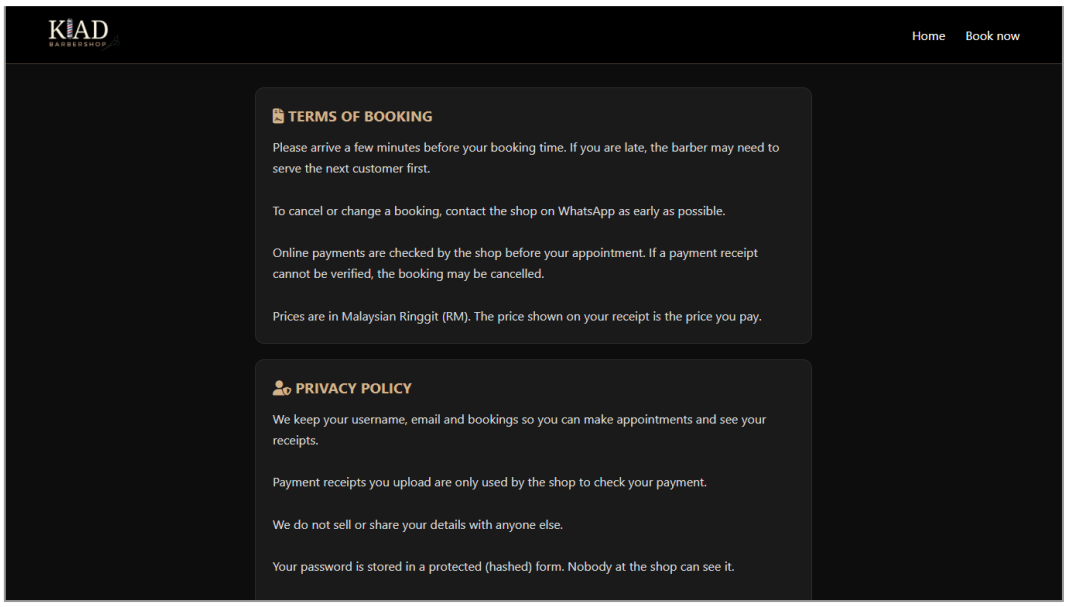


Figure 4.24 : Terms and Privacy page

Mobile view

For most customers it will be on a phone, so the pages were verified at 390 pixels wide. The layout reverts to one column, the menu collapses and the booking steps expand to two lines. The home, services and barber pages look like this at that width, as shown in figure 4.25.

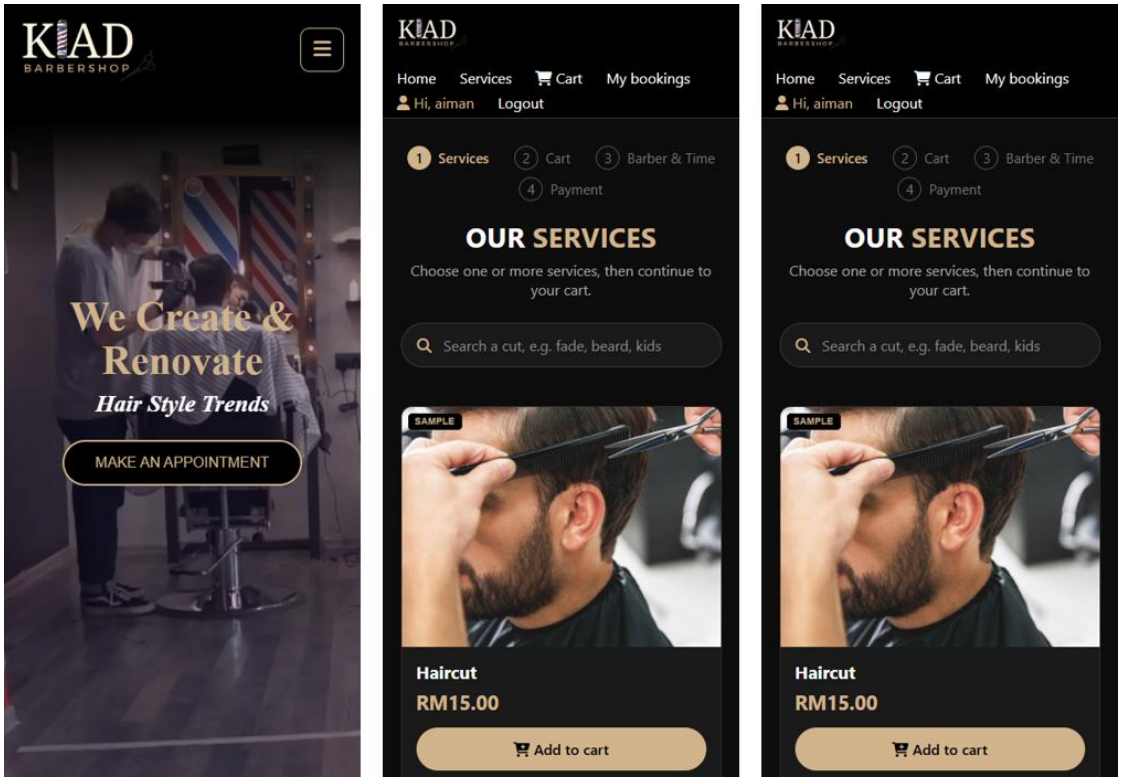


Figure 4.25 : Home, services and barber and time pages on a phone-sized screen

4.8.2 Barber and Owner Interface

Dashboard

The owner's initial page is the Dashboard. It indicates the number of bookings today, the total value booked today, the amount of money collected this month and the number of payments awaiting check. Below are today's schedule, a chart of paid sales for the last seven days, which staff have clocked in, the top services of the month and the figures per barber. The number is only incremented when the payment is verified, not when it is booked, so the number should not be inflated by a booking that you haven't paid for.

In Section 4.2, you will see a screen shot of the owner dashboard (Figure 4.6).

When a barber looks at the same view with their own numbers, they get bookings today and sales this month and today; their average rating; their schedule for today and the next 7 days. The page also reminds the barber to scan the counter QR as he has logged in prior to clocking in.

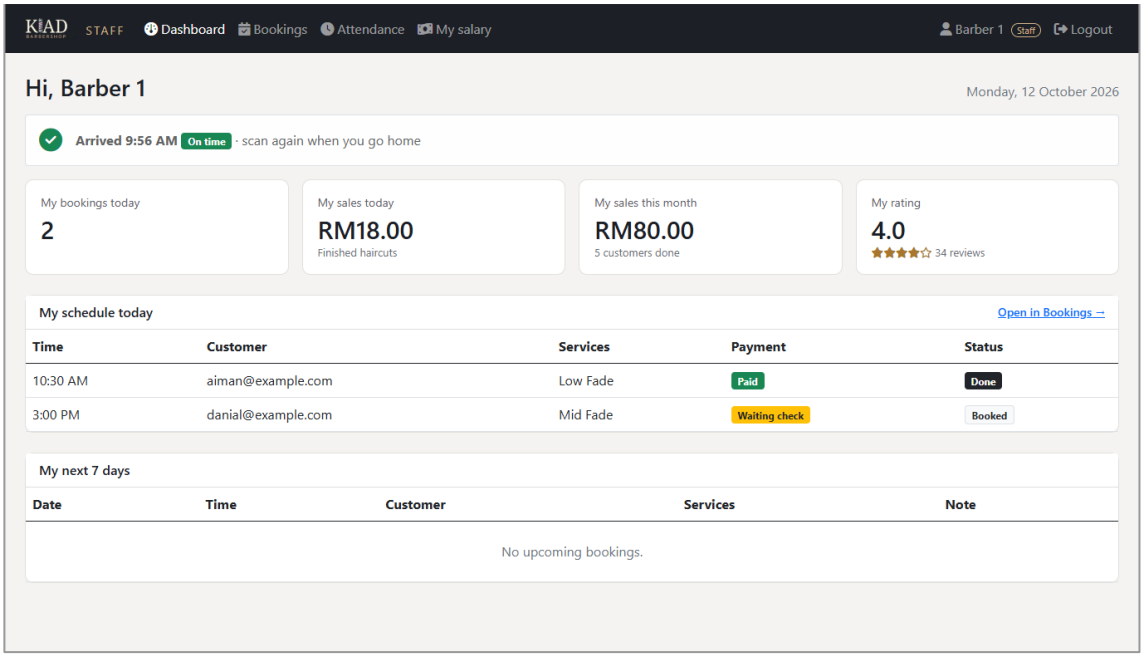


Figure 4.26 : Barber dashboard

Bookings and payments

This is where most of the owner's working time is spent. You can filter bookings by day (Today, Upcoming, Past, All dates), by status and by barber. Highlighted rows are those waiting for a payment check. The receipt is opened by the owner, they can verify or reject a payment, re-assign it to a different barber from the drop-down or cancel it. Only bookings made by the barber are available to him, and he can't verify, reject, reassign or cancel.

The screenshot of the bookings page (Figure 4.7) is shown in Section 4.2.

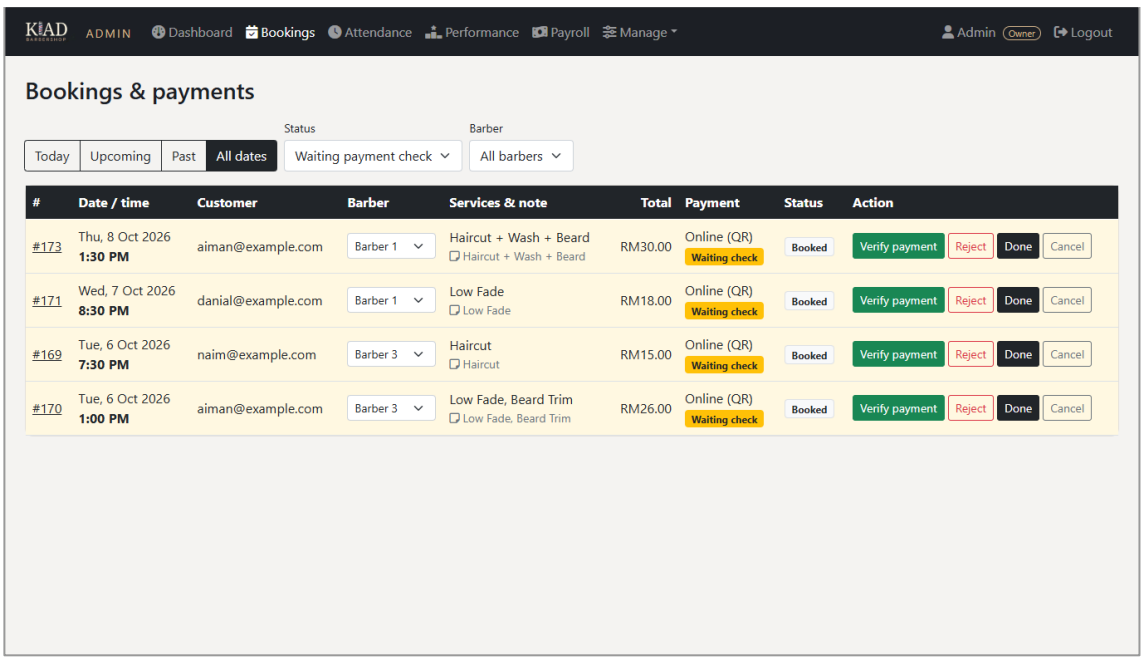


Figure 4.27 : Bookings waiting for a payment check

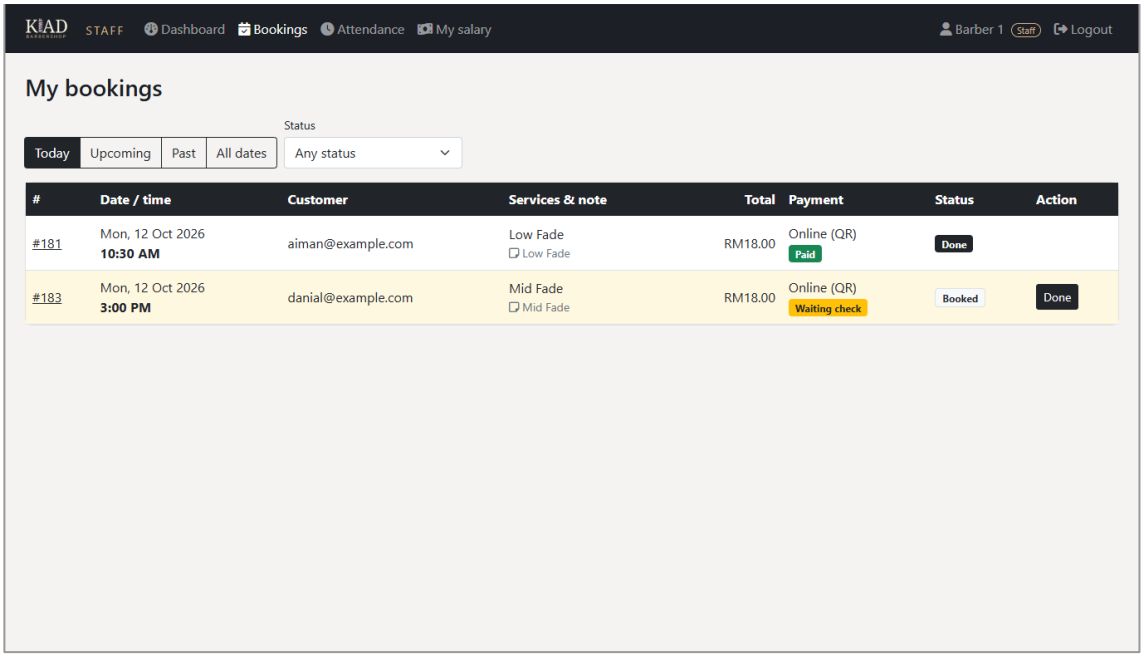


Figure 4.28 : Barber's view of bookings

Attendance

There is a Counter QR screen section in the attendance page. It has a hidden connection to open when the tablet or an extra phone is placed on the counter. The page behind that link (Figure 4.29) shows the date, a QR code that refreshes every minute and the staff who have already clocked in. Below the link are today's records with Fix buttons for forgotten scans and a monthly summary of days worked, times late, minutes late and hours worked.

The screenshot of the attendance page (Figure 4.8) is shown in Section 4.2.

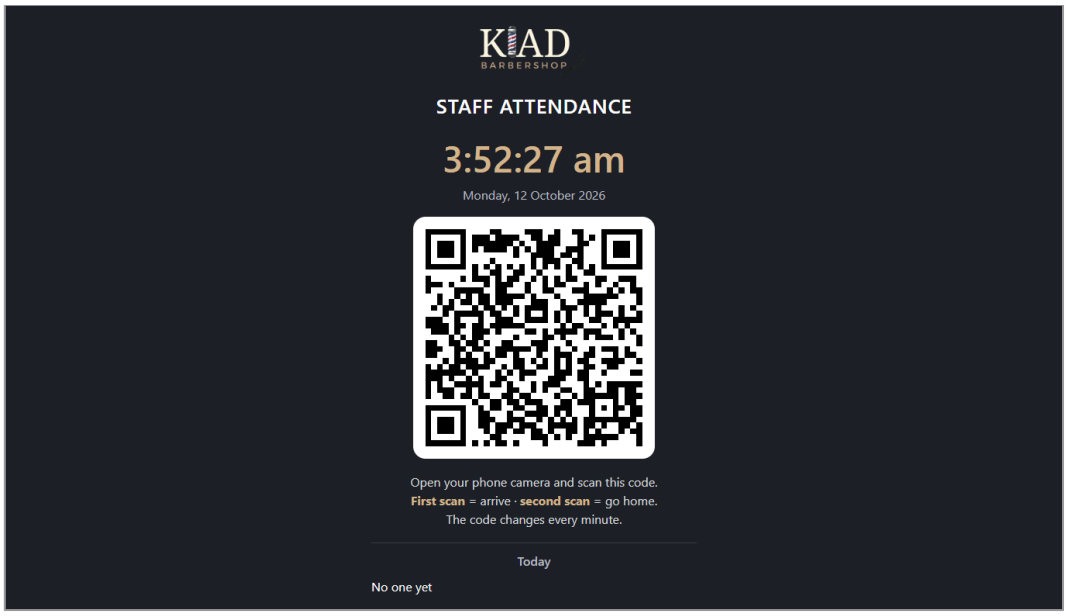


Figure 4.29 : Counter screen showing the attendance QR

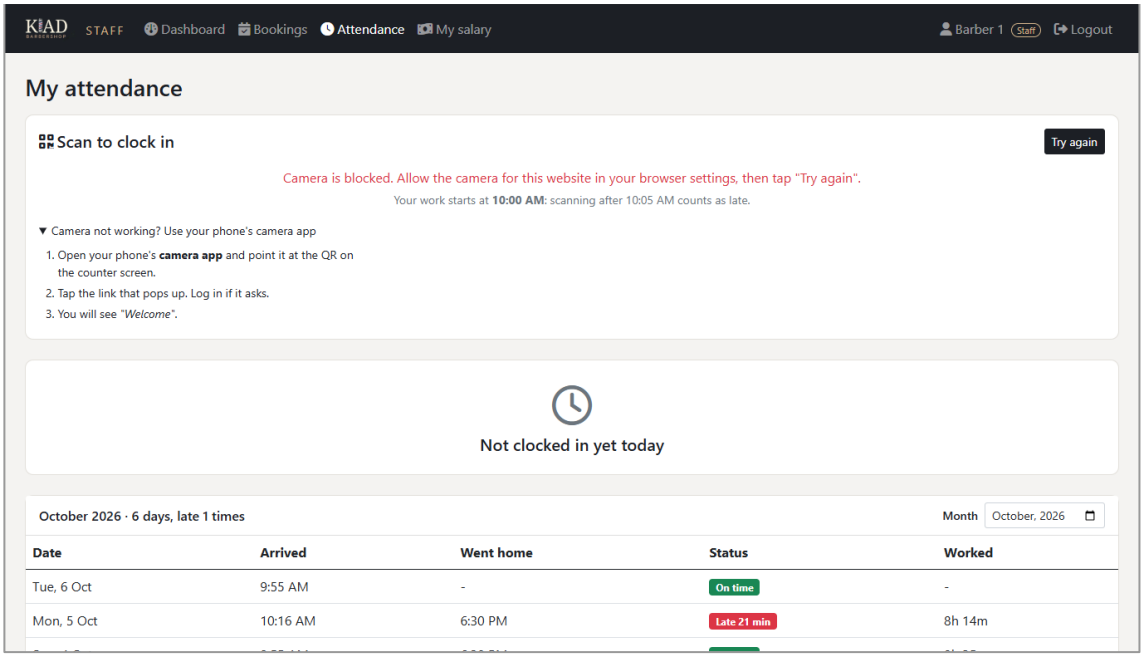


Figure 4.30 : My attendance page (barber)

Performance and payroll

The performance page is used to compare different barbers for a selected month by number of customers served, sales generated, number of bookings picked by name, days worked, number of times late, and star ratings based on the newest customer reviews. The payroll page displays a list of the days worked, lateness, customers, sales and rating for the same month with a salary box and a note which is only visible to the owner. The amount can be saved and locked by Save, and can only be changed again by Undo. Only months that have been marked paid appear on a barber's My salary page.

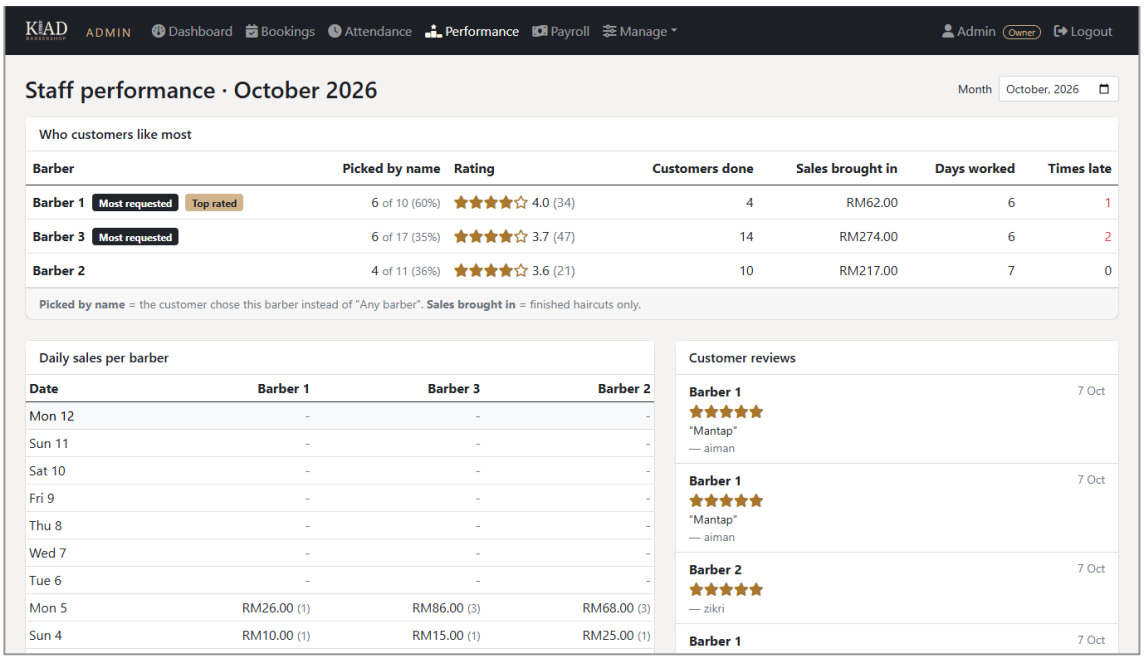


Figure 4.31 : Staff performance page

In Section 4.2, you will see the screen shot of the payroll page (Figure 4.9).

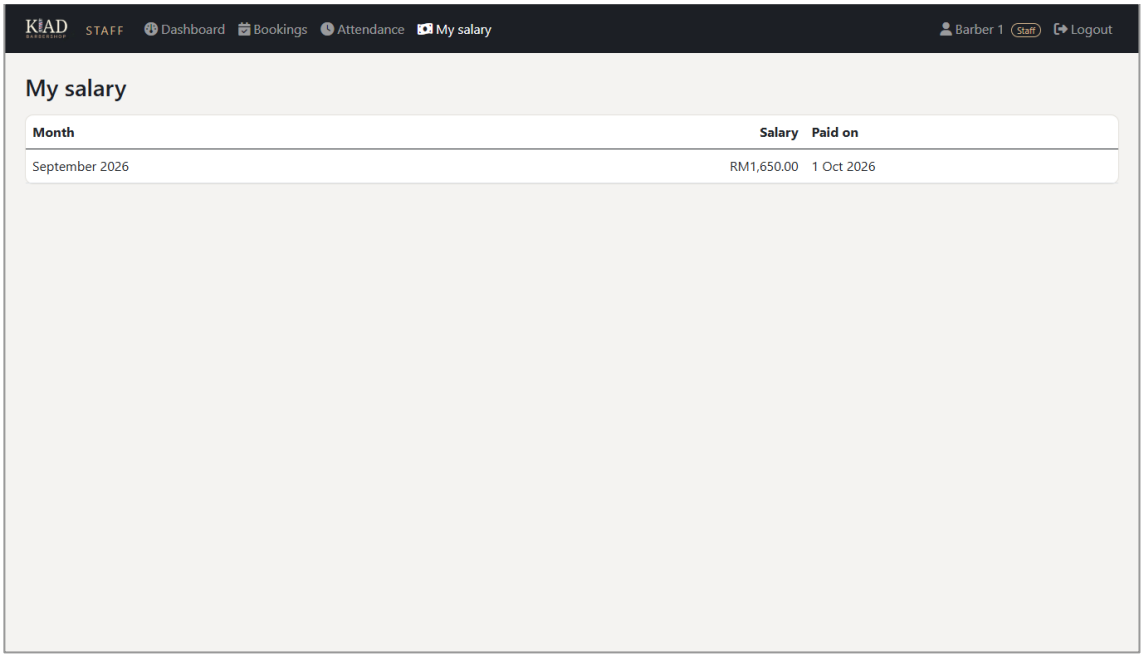


Figure 4.32 : My salary page (barber)

Services, staff, customers and settings

There are four pages under the Manage menu that are available for the owner. Services adds, edits and deletes services having a picture and a cost. Staff creates accounts and sets role, an active flag and optional start time - and must not be able to lock themselves out. When a customer forgets his/her password, he/she is able to search this account and will be able to set a temporary password. Settings determines opening hours, slot length, closed weekdays and closed holidays, the payment QR image, the logo, the login background, the contact information, map, social links and the content of the terms and privacy page.

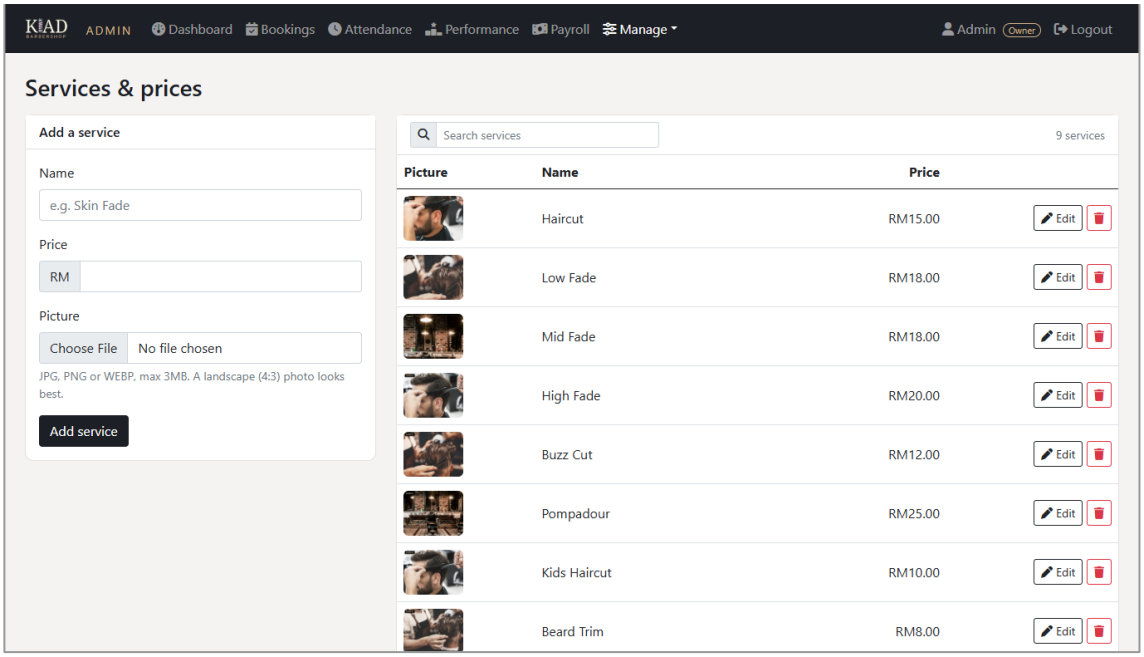


Figure 4.33 : Services and prices page

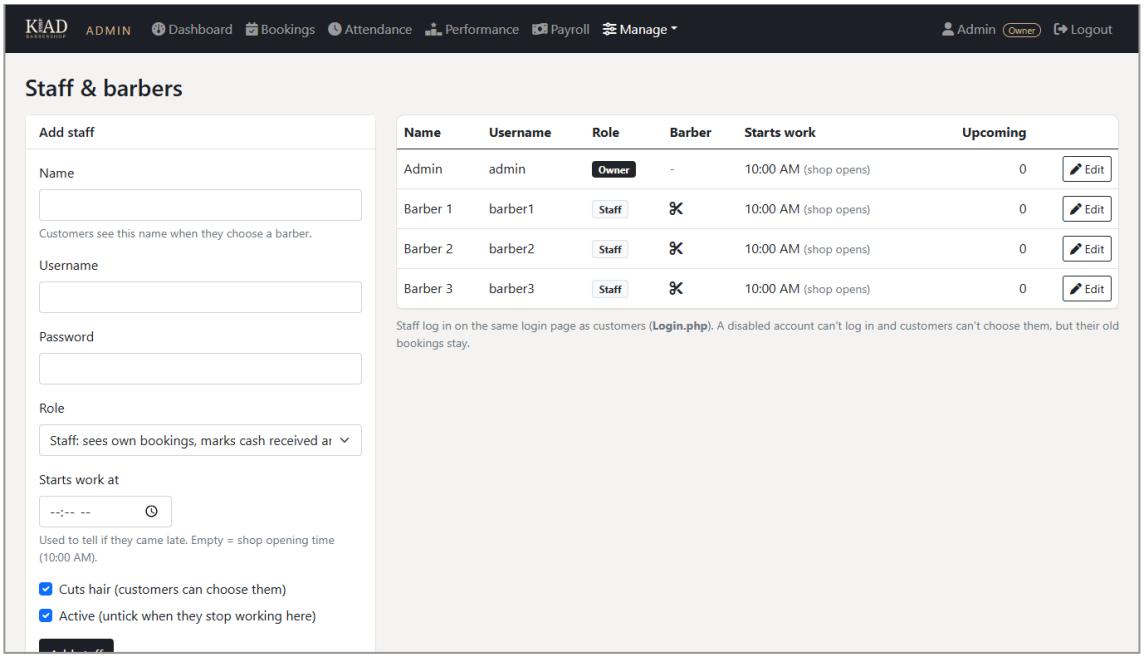


Figure 4.34 : Staff and barbers page

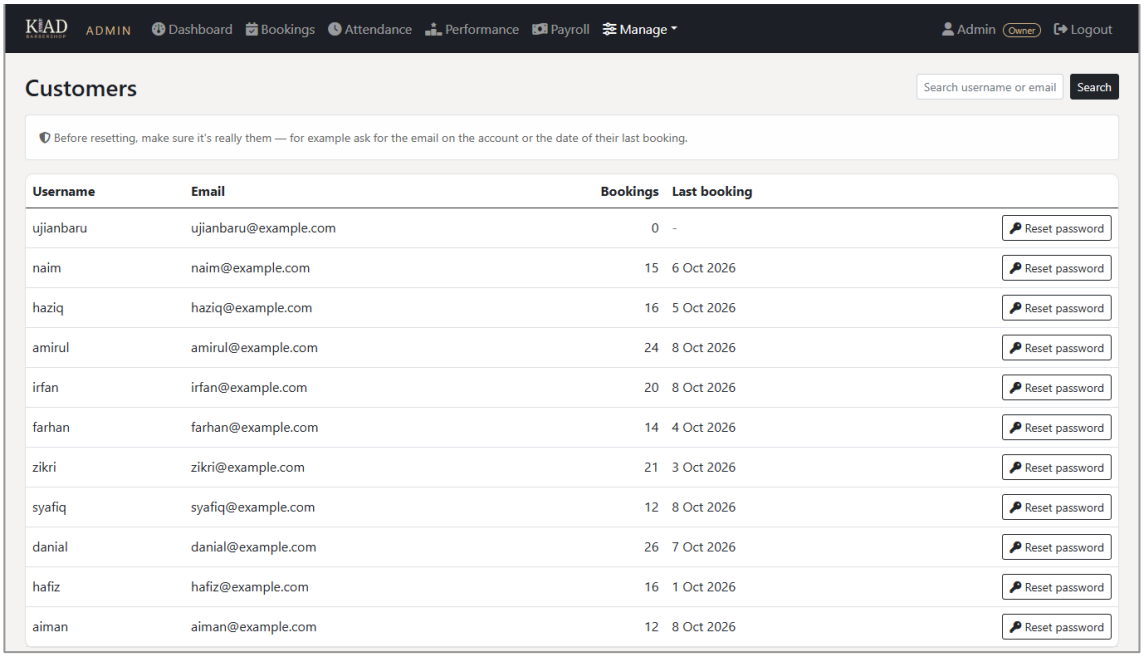


Figure 4.35 : Customers page

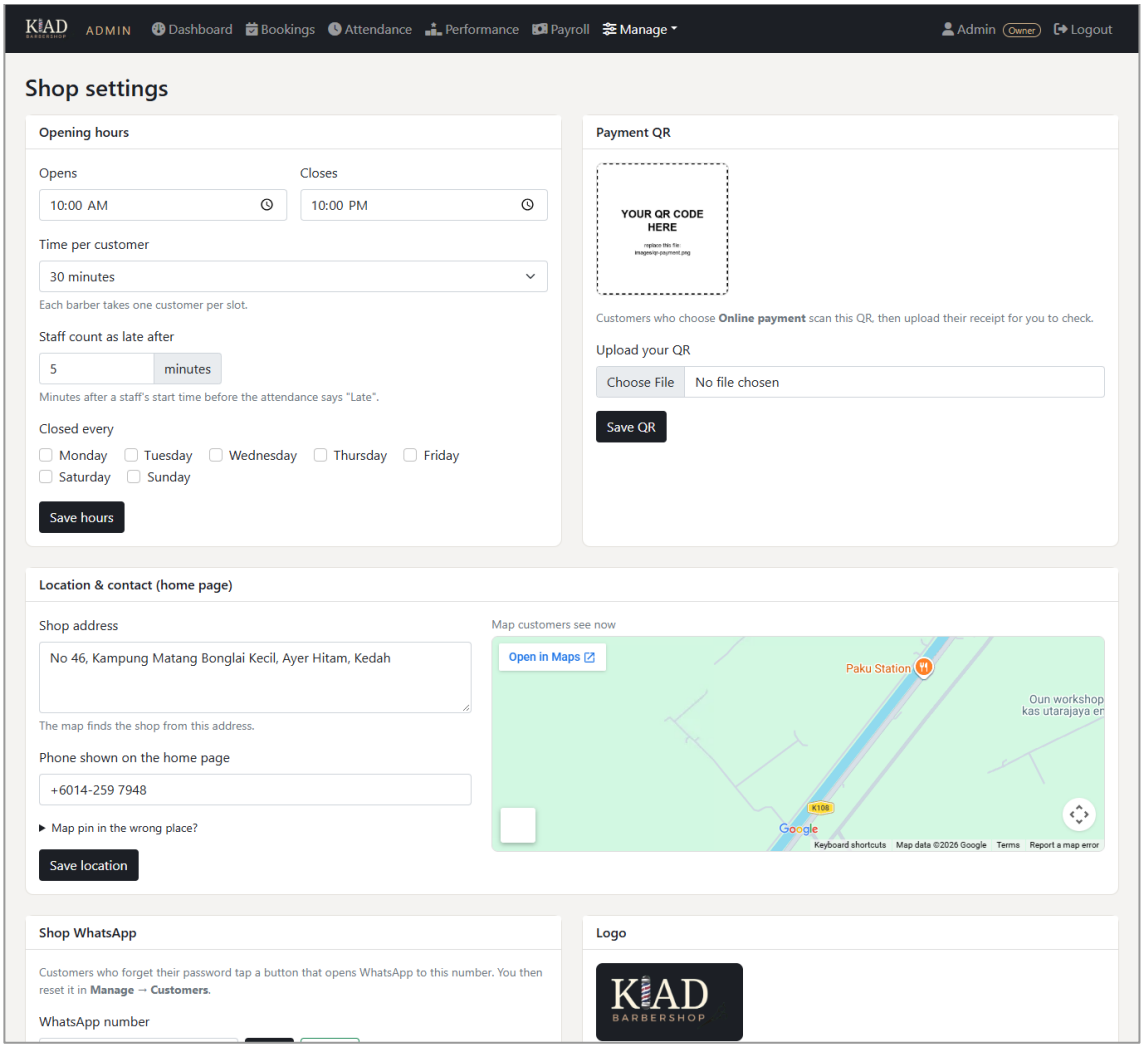


Figure 4.36 : Shop settings page (upper part)

4.9 Significant Functions and Security Algorithms

This section describes the functions which perform main rules of the system. These extracts are from the actual code. Some comments cut down in length and a few lines merged to fit the space.

4.9.1 Core Functional Algorithms

Slot availability

A slot is a set time interval between the two opening and closing times, with step_minutes (default 30) defining the interval, so there are 24 slots in a day from 10:00 to 22:00. Listing 4.1 decides which of them cannot be booked. Closed weekday and holidays - all slots are closed. If any of the above are true, then each of the other slots is blocked if it has already passed today, if the chosen barber has already booked at that time, or, for Any barber, if all of the active barbers are booked at that time. Slots.php is responsible for feeding the page and so is the booking code, which checks immediately before saving.

Listing 4.1 : Function that finds unavailable times (booking_lib.php)

```
function unavailable_times($conn, $date, $barber) {
    if (closed_reason($conn, $date) !== '') {
        return slot_list($conn);            // closed day: every slot is blocked
    }
    $bookings = day_bookings($conn, $date);
    $times = [];
    foreach (slot_list($conn) as $slot) {
        if ($date === date('Y-m-d') && $slot <= date('H:i')) {
            $times[] = $slot;               // time has already passed today
        } elseif ($barber === 'any') {
            if (count(free_barbers($conn, $date, $slot, $bookings)) === 0) {
                $times[] = $slot;           // nobody is free at this time
            }
        } elseif (!barber_is_free($conn, $date, $slot, $barber, $bookings)) {
            $times[] = $slot;               // the chosen barber is busy
        }
    }
    return $times;
}
```

Automatic barber assignment

If the customer selects Any barber, free_barbers() returns the list of barbers not busy at the time of the call, sorted by the number of bookings each has made already today, least to most, and then by staff number, also least to most (Listing 4.2). The first one is to be assigned. This spreads the work without the owner having to do it by hand. Before barber's time, bookings don't have a barber id, each of them still takes one chair at this time.

Listing 4.2 : Function that returns free barbers, fewest bookings first (booking_lib.php)

```
function free_barbers($conn, $date, $time, $bookings = null) {
    $bookings = $bookings ?? day_bookings($conn, $date);
    $busy = []; $load = []; $unassigned = 0;
    foreach ($bookings as $b) {
        if ($b['barber_id']) {
            $load[$b['barber_id']] = ($load[$b['barber_id']] ?? 0) + 1;
        }
        if ($b['t'] === $time) {
            if ($b['barber_id']) { $busy[] = $b['barber_id']; } else { $unassigned++; }
        }
```

```
    }
    $free = [];
    foreach (active_barbers($conn) as $barber) {
        if (!in_array($barber['staff_id'], $busy)) { $free[] = (int)$barber['staff_id']; }
    }
    // fewest bookings that day first
    usort($free, function ($a, $b) use ($load) {
        return (($load[$a] ?? 0) <=> ($load[$b] ?? 0)) ?: $a <=> $b;
    });
    return array_slice($free, 0, max(0, count($free) - $unassigned));
}
```

Saving a booking

Somebody else may have reserved the free slot between when the customer views it and when the customer clicks Confirm booking. The payment page thus checks availability again and if the slot is not available, it gives the customer message and navigates him back to Barber & Time. If the slot is free then the booking and the booking items are stored in a single transaction (Listing 4.3). If it fails for any reason, the transaction is rolled back, and the uploaded receipt-file is removed, which means that a failed booking is nothing.

Listing 4.3 : Saving a booking and its items in one transaction (payment.php)

```
mysqli_begin_transaction($conn);
try {
    $stmt = mysqli_prepare($conn, "INSERT INTO receipt (cust_id, name, date, time, notes, total,
        payment_method, payment_proof, payment_status, barber_id, chosen_barber)
        VALUES (?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?)");
    mysqli_stmt_bind_param($stmt, 'issssdsssii', $cust_id, $name, $booking['date'],
$booking['time'],
        $booking['notes'], $total, $method, $proof, $status, $barber_id, $chosen);
    mysqli_stmt_execute($stmt);
    $receiptid = mysqli_insert_id($conn);


    $stmt = mysqli_prepare($conn, "INSERT INTO receipt_item (receipt_ID, service_name, price)
VALUES (?, ?, ?)");
    foreach ($items as $item) {
        mysqli_stmt_bind_param($stmt, 'isd', $receiptid, $item['name'], $item['price']);
        mysqli_stmt_execute($stmt);
    }
    mysqli_commit($conn);
} catch (mysqli_sql_exception $e) {
    mysqli_rollback($conn);
    if ($proof) { unlink(__DIR__ . '/' . $proof); }   // do not keep the receipt of a failed
booking
    $error = "Booking failed. Please try again.";
}
```

Payment and job status

Payment status and job status are each independent fields on the receipt with each having its own permissible transitions (as given in Listing 4.4). All cash bookings begin as cash and get verified upon receipt of cash. QR bookings are initially listed as pending and will be approved or denied by the owner. If the customer provides a better receipt, the receipt can be verified later if it has been rejected. Only the Booking owner can cancel the booking and it can only be made done while it is still booked. When a payment is rejected and when a booking is cancelled, a chair is no longer held.

Listing 4.4 : Allowed status changes on the bookings page (admin/bookings.php)

```
if ($action === 'verify' && $is_owner && in_array($r['payment_status'], ['pending', 'rejected']))
{
```

```
    $sql = "UPDATE receipt SET payment_status = 'verified' WHERE receipt_ID = ?";
} elseif ($action === 'reject' && $is_owner && $r['payment_status'] === 'pending') {
    $sql = "UPDATE receipt SET payment_status = 'rejected' WHERE receipt_ID = ?";
} elseif ($action === 'cash_paid' && $r['payment_status'] === 'cash') {
    $sql = "UPDATE receipt SET payment_status = 'verified' WHERE receipt_ID = ?";
} elseif ($action === 'done' && $r['job_status'] === 'booked') {
    $sql = "UPDATE receipt SET job_status = 'done' WHERE receipt_ID = ?";
} elseif ($action === 'cancel' && $is_owner && $r['job_status'] === 'booked') {
    $sql = "UPDATE receipt SET job_status = 'cancelled' WHERE receipt_ID = ?";
}
```

Attendance and lateness

The attendance row is created on the first scan of a day. Lateness is the number of complete minutes after the barber's starting time, and is negative if the barber is early (Listing 4.5). The start time is determined by the barber's start_time (if set by the owner) or the shop opening time (if not). The grace time on the label presented to the barber is set to 5 minutes, therefore if a child arrives 2 minutes after the set time it will still show as On time, but the months Times late will be affected by the lateness. INSERT IGNORE and the unique key on staff and day will not allow two scans for the same second to create 2 records or blow up the page.

Listing 4.5 : Recording arrival and labelling lateness (admin/scan.php, admin/_base.php)

```
// First scan of the day = arrive
$start = start_time_for($conn, $staffRow);        // own start time, or shop opening time
$late = (int)floor((time() - strtotime("$today $start")) / 60);


$stmt = mysqli_prepare($conn, "INSERT IGNORE INTO attendance
        (staff_id, work_date, clock_in, late_minutes) VALUES (?, ?, ?, ?)");
mysqli_stmt_bind_param($stmt, 'issi', $me['staff_id'], $today, $now, $late);
mysqli_stmt_execute($stmt);


// "On time" / "Early n min" / "Late n min" use the grace setting (5 minutes)
function arrival_label($conn, $late_minutes) {
    $grace = (int)setting($conn, 'late_grace_minutes');
    if ($late_minutes > $grace) { return ['danger', 'Late ' . $late_minutes . ' min']; }
    if ($late_minutes < 0)      { return ['success', 'Early ' . abs($late_minutes) . ' min']; }
    return ['success', 'On time'];
}
```

Payroll lock and performance figures

The unique combination of staff and month is the ON DUPLICATE KEY UPDATE clause of the payroll query (Listing 4.6). The amount is only updated if paid_at is empty, and if a salary is already paid, a second click on Paid won't update the paid date. The performance page shows only the non-cancelled, non-rejected bookings, and only counts sales from bookings that are done (Listing 4.7).

Listing 4.6 : Saving and locking a salary (admin/payroll.php)

```
// Save the amount; "paid" also records today's date. A paid salary is locked.
$paid = $action === 'paid' ? date('Y-m-d H:i:s') : null;
$stmt = mysqli_prepare($conn, "INSERT INTO payroll (staff_id, month, amount, note, paid_at)
    VALUES (?, ?, ?, ?, ?)
    ON DUPLICATE KEY UPDATE amount = IF(paid_at IS NULL, VALUES(amount), amount),
                            note = VALUES(note),
                            paid_at = COALESCE(paid_at, VALUES(paid_at))");
mysqli_stmt_bind_param($stmt, 'isdss', $staff_id, $month, $amount, $note, $paid);
```

Listing 4.7 : Query behind the performance figures (admin/_stats.php)

```
SELECT barber_id,
       COUNT(*) AS bookings,
       SUM(job_status = 'done') AS customers,
       COALESCE(SUM(CASE WHEN job_status = 'done' THEN total END), 0) AS sales,
       SUM(chosen_barber) AS chosen
FROM receipt
WHERE barber_id IS NOT NULL AND date BETWEEN '$from' AND '$to'
  AND job_status <> 'cancelled' AND (payment_status IS NULL OR payment_status <> 'rejected')
GROUP BY barber_id
```

4.9.2 Security Mechanisms

Login protection

The check that occurs prior to the verification of a password is shown in Listing 4.8. After 5 failed logins, or 20 failed addresses, within 15 minutes the login form denies the login attempt with a message and will not even attempt to look up the password. If the login is successful, the counts for this name are deleted and the session id is re-generated.

Listing 4.8 : Counting failed logins in 15 minutes (Login.php)

```
function login_blocked($conn, $ip, $user) {
    $stmt = mysqli_prepare($conn, "SELECT SUM(username = ?) AS for_user, SUM(ip = ?) AS for_ip
        FROM login_attempt WHERE attempted_at > NOW() - INTERVAL 15 MINUTE
        AND (username = ? OR ip = ?)");
    mysqli_stmt_bind_param($stmt, 'ssss', $user, $ip, $user, $ip);
    mysqli_stmt_execute($stmt);
    $row = mysqli_fetch_assoc(mysqli_stmt_get_result($stmt));
    return $row['for_user'] >= 5 || $row['for_ip'] >= 20;
}

// on success:  clear_login_attempts(); session_regenerate_id(true);
// on failure:  INSERT INTO login_attempt (ip, username) VALUES (?, ?)
```

CSRF protection

Random token is assigned to each session at the first drawing of a form. The token is stored in a hidden field and check_csrf() checks if the token in the hidden field is equal to the token stored in the session using hash_equals on each and every POST request (Listing 4.9). If a request is made with the incorrect token, then it returns an HTTP 400. This is tested in chapter 5.

Listing 4.9 : CSRF token helpers (bootstrap.php)

```
function csrf_field() {
    if (empty($_SESSION['csrf'])) {
        $_SESSION['csrf'] = bin2hex(random_bytes(16));
    }
    return '<input type="hidden" name="csrf" value="' . $_SESSION['csrf'] . '">';
}

function check_csrf() {
    if ($_SERVER['REQUEST_METHOD'] === 'POST'
        && !hash_equals($_SESSION['csrf'] ?? '', $_POST['csrf'] ?? '')) {
        http_response_code(400);
        exit('This page has expired. Go back, refresh the page and try again.');
    }
}
```

Signed attendance token

The attendance token consists of a time window number and the first 20 hex characters of an HMAC-SHA256 signature of the number (Listing 4.10). The signature of the window in the link is recalculated by the server, and the server will not accept a forged window without the secret, and only accepts the current window and the two previous ones. To prevent timing differences, the comparison is done using hash_equals.

Listing 4.10 : Creating and checking the attendance token (admin/_base.php)

```
define('QR_WINDOW_SECONDS', 60);
define('QR_VALID_WINDOWS', 3);


function attendance_token($conn, $window = null) {
    $window = $window ?? intdiv(time(), QR_WINDOW_SECONDS);
    $sig = substr(hash_hmac('sha256', (string)$window, secret_setting($conn,
'attendance_secret')), 0, 20);
    return $window . '-' . $sig;
}


function attendance_token_ok($conn, $token) {
    if (!preg_match('/^(\d+)-[a-f0-9]{20}$/', $token, $m)) { return false; }
    $now = intdiv(time(), QR_WINDOW_SECONDS);
    $window = (int)$m[1];
    return $window <= $now && $window > $now - QR_VALID_WINDOWS
        && hash_equals(attendance_token($conn, $window), $token);
}
```

Receipt upload validation

The checks on an uploaded receipt are listed in Listing 4.11. The type is determined by the content of the file being uploaded via finfo, the size limit set is 5 MB and the name of the file that is stored is 32 random hexadecimal characters with an extension based on the file type, which cannot be chosen by the customer. A disadvantage is that the link to the receipt can be opened by anyone with the exact link. Nothing per user to check file access, names are long and random, and the folder can't be listed.

Listing 4.11 : Validating and storing an uploaded receipt (payment.php)

```
$file = $_FILES['proof'] ?? null;
$allowed = ['image/jpeg' => 'jpg', 'image/png' => 'png', 'image/webp' => 'webp',
'application/pdf' => 'pdf'];


if (!$file || $file['error'] === UPLOAD_ERR_NO_FILE) {
    $error = "Please upload your payment receipt.";
} elseif ($file['error'] !== UPLOAD_ERR_OK || $file['size'] > 5 * 1024 * 1024) {
    $error = "Upload failed. The receipt must be smaller than 5MB.";
} else {
    $type = (new finfo(FILEINFO_MIME_TYPE))->file($file['tmp_name']);   // real type, not the
browser's claim
    if (!isset($allowed[$type])) {
        $error = "Receipt must be a JPG, PNG, WEBP or PDF file.";
    } else {
        $proof = 'uploads/receipts/' . bin2hex(random_bytes(16)) . '.' . $allowed[$type];
        move_uploaded_file($file['tmp_name'], __DIR__ . '/' . $proof);
    }
}
```

4.10 Conclusion

We covered the design and implementation: wireframes, 11 tables for the database, data flow and entity relationship models, a layered security system and the platform on which it is built and each module as it would be viewed by a user. It also revealed the code that's essential to the rules that matter most, and this includes slot availability, the booking transaction, payment status changes, attendance, payroll locking and security checks. The next chapter puts the system to the test, to see if it performs the functions it was designed to serve.

FINDING

5.1 Introduction

In this chapter, the testing of the system and the results of the testing are reported. The tests include speed and safety, data retention, function as standalone and combined system, and acceptability by those who tested it including the owner. All results in the tables have been generated by testing it against the system, not by estimation.

5.2 Testing

The tests were designed to be repeatable scripts as much as possible. Real forms were filled and submitted by Python scripts that logged in as various users, read responses and the database by using requests library. Screenshots were taken and checked using Playwright with chrome for viewport at phone sized. A test is only successful if the behaviour conforms to the expected outcome that was scripted in advance.

The scripts were against a local copy of the deployed code to test bookings and test accounts without affecting the live shop's records. One exception to that is the check to ensure that the config.php file is not opened, which was performed on the live site. The environment is listed in Table 5.1.

Table 5.1 : Test Environments

Item	Detail
Server	PHP 8.2.12 built-in web server on the local machine, serving a copy of the deployed files
Database	MariaDB 10.4.32 loaded from the clean install script, then filled with demo data
Demo data	11 customer accounts, the owner and three barbers, and about 170 bookings spread over four weeks, with payments, ratings and attendance
Accounts used	Owner, Barber 1 to Barber 3 and several customer accounts
Client	Google Chrome 1366 px wide for desktop and a 390 x 844 viewport for the phone check
Tools	Python 3 with requests, Playwright, mysql command-line client for checking stored data
Live site	https://barbershophesmes.site.je (used only for the config.php access check)

5.3 Non-Functional Testing

5.3.1 Performance Testing

Performance was assessed by calling the same page 5 times consecutively in a logged-in session and recording the time to get the entire response. 2 seconds was the requirement.

These are local numbers, but it does indicate that the code itself is not a problem and a real phone on mobile data will be slower.

Table 5.2 : Performance Testing Result Table

Test ID	Test Description	Expected Result	Actual Result	Status
NFT-P01	Open the customer services page and measure load time	Loads within two seconds	Average 19 ms over 5 loads (slowest 30 ms)	Pass
NFT-P02	Open the owner dashboard with sales, schedule and chart data	Loads within two seconds	Average 104 ms over 5 loads (slowest 405 ms)	Pass
NFT-P03	Open the monthly payroll page	Loads within two seconds	Average 22 ms over 5 loads (slowest 31 ms)	Pass

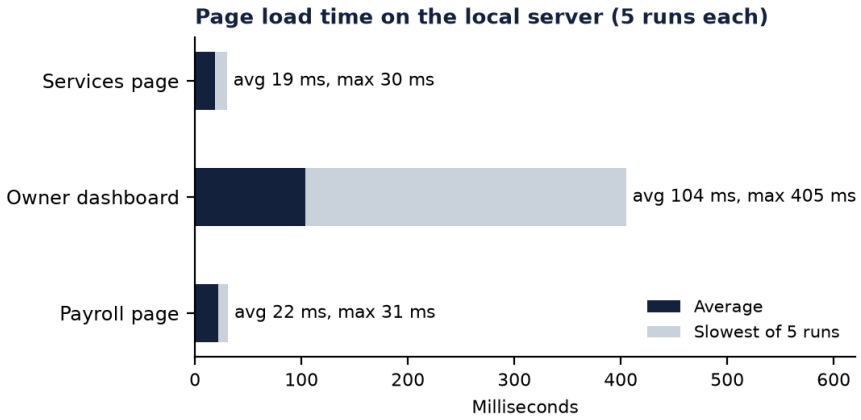


Figure 5.1 : Page load times on the local server

Observation:
The slowest page is the dashboard, where it generates the cards, chart and three tables from multiple queries. All pages are well under the two-second mark with its slowest single run being 405 ms.

5.3.2 Security Testing

Security tests attempted to guess passwords, access pages that don't belong to their role, forge a form, enter SQL and read configuration files, etc. These are given in Table 5.3.

Table 5.3 : Security Testing Result Table

Test ID	Test Description	Expected Result	Actual Result	Status
NFT-S01	Log in with a wrong password	Login rejected with an error message	"Wrong username/email or password." shown	Pass
NFT-S02	Customer opens an owner page by URL (payroll)	Access blocked	HTTP 302 redirect to Login.php	Pass
NFT-S03	Barber opens the owner-only Settings page	Access denied for the staff role	Redirected with "Only the owner can open that page"	Pass

Test ID	Test Description	Expected Result	Actual Result	Status
NFT-S04	Inspect stored passwords in the database	Hashes, not plain text	Staff and customer passwords start with $2y$ (bcrypt)	Pass
NFT-S05	Enter six wrong passwords for one account	Login temporarily blocked	"Too many wrong passwords. Please wait 15 minutes"	Pass
NFT-S06	Submit the login form with a forged CSRF token	Request rejected	HTTP 400 returned	Pass
NFT-S07	Type a SQL injection string as the username	Login rejected and no data exposed	Rejected; prepared statements used	Pass
NFT-S08	Open config.php directly on the live site	Access forbidden	HTTP 403 Forbidden	Pass

Observation:
100% of the eight tests passed. The login lock is on a per account basis and the owner-only checks are applied to the server not just to hide menu items.

One weakness was identified during the creation of the CSRF test. The form's token is compared with the session's token, and, when a request is received for which there is no session, both are empty and equal, so it does not get rejected. If a real session is sent with a wrong token, it is rejected as in NFT-S06. The risk in practice is low, since the first form that a visitor encounters generates the session token, but it would be wise for the check to also reject a request whose session token is missing. This is noted as a recommendation in Chapter 6.

5.3.3 Usability Testing

People should be observed performing tasks to measure the usability. However, since timed task trials were not recorded in this project, evidence is based on the acceptance questionnaire (Section 5.5.2). Weaker than observation: records what people say, not how long it takes to perform a task or where they have hesitation. The number of respondents who support each result is provided to enable readers to understand the strength of the evidence.

Table 5.4 : Usability Testing Result Table

Test ID	Test Description	Expected Result	Actual Result	Status
NFT-U01	A customer finds booking easy	Most respondents agree that choosing a barber, date and time is easy	Q4 mean 4.48 and 85% agreed (n = 27); payment process Q6 mean 4.44	Pass
NFT-U02	Barbers find the staff pages useful and clear	Staff agree the pages help daily work	Q11 from barbers and owner: mean 4.50 (n = 4); too few answers to be conclusive	Pass (small sample)

Test ID	Test Description	Expected Result	Actual Result	Status
NFT-U03	The owner finds the management pages helpful	Owner agrees the pages help run the shop	Owner gave 5 out of 5 on Q11 (n = 1); the client checklist in Table 5.10 also passed	Pass (one respondent)

5.3.4 Reliability Testing

Table 5.5 : Reliability Testing Result Table

Test ID	Test Description	Expected Result	Actual Result	Status
NFT-R01	Save a booking, log out and log in again	The booking is still stored	Booking stored and listed in My Bookings	Pass
NFT-R02	Stop and restart the database and the web server	All records remain accurate	All seeded bookings and payment totals intact after the restart	Pass
NFT-R03	Log in as customer, barber and owner in turn	Each role sees its own page	Home page, STAFF panel and ADMIN panel respectively	Pass

5.3.5 Compatibility Testing

The system was opened at the following width and height: desktop – 1366 width and phone – 390 width and 844 height. It is a mockup phone screen, not a physical phone, only the layout is tested, not the touch behaviour or other browsers.

Table 5.6 : Compatibility Testing Result Table

Test ID	Test Description	Expected Result	Actual Result	Status
NFT-C01	Open the system on a desktop browser	Layout is readable and usable	Chrome at 1366 px: all pages render correctly	Pass
NFT-C02	Open the booking pages at a phone-sized viewport	Layout adapts to the screen	390 px: single column and no sideways scrolling	Pass

5.4 Functional Testing

5.4.1 Unit Testing

Single function is tested with unit tests through the interface, e.g. login or booking. Eight of them are listed in Table 5.7.

Table 5.7 : Unit Testing Result Table

Test ID	Module	Test Scenario	Expected Result	Actual Result	Status
UT01	Login	Enter a valid username and password	Redirected to the right page	Customer to home.php; owner to admin/index.php	Pass
UT02	Login	Enter a wrong password	Login rejected with an error	Error message shown	Pass
UT03	Booking	Choose an available slot and pay cash	Booking saved as booked, cash pending	Receipt created with payment status cash	Pass
UT04	Booking	Choose an already booked slot for the same barber	System prevents double booking	"Selected time slot is not available"	Pass
UT05	Payment	Mark a cash booking paid, then done	Statuses saved	payment_status verified, job_status done	Pass
UT06	Sales	Compare dashboard collected with the database	Totals match	Dashboard RM736.00 equals the sum of verified payments	Pass
UT07	Access control	A barber tries to verify an online payment	Action refused	Payment status unchanged	Pass
UT08	Payment	Choose QR without uploading a receipt	System asks for the receipt	"Please upload your payment receipt."	Pass

5.4.2 Integration Testing

Integration tests follow data across modules, for example a booking made by a customer that must show up on the owner's dashboard.

Table 5.8 : Integration Testing Result Table

Test ID	Modules Involved	Test Scenario	Expected Result	Actual Result	Status
IT01	Booking + barber assignment	Customer books Any barber at a busy time	A free barber is assigned	Booked with a free barber, no clash	Pass
IT02	Booking + payment	Barber marks cash received, then Done	Both statuses update	verified and done saved	Pass
IT03	Payment + dashboard	Owner verifies a QR receipt	Counted in collected sales	Status verified and counted in paid sales	Pass
IT04	Holiday + booking	Owner marks a day closed; customer tries to book it	Customer cannot book that day	"The shop is closed on this date"	Pass
IT05	Role + dashboard	Owner and barber log in	Dashboard depends on role	ADMIN menu for the owner, STAFF menu for the barber	Pass

5.4.3 System Testing

System tests follow whole scenarios from the user's point of view, from registering to being paid.

Table 5.9 : System Testing Result Table

Test Scenario	Test Steps	Expected Result	Actual Result	Status
Registration and login	Sign up, then log in	Account created and login works	Registered with a hashed password; login reached the home page	Pass
Service and price viewing	Open the services page	Name, photo and price shown	Nine services shown, RM8 to RM30	Pass
Appointment booking	Pick barber, date and slot, then pay	Saved, double booking prevented	Saved; a second booking for the same slot refused	Pass
QR payment check	Upload a receipt; owner verifies	pending changes to verified	verified and counted in sales	Pass
Attendance by QR	Barber scans the counter QR	Arrival saved with late minutes	Attendance row created for the day	Pass
Counter screen protection	Open the counter page with a wrong key	Access refused	HTTP 403 and no QR shown	Pass
Payroll	Owner saves a salary and presses Paid	Saved, then locked	RM1500.00 saved; a later change was ignored	Pass
Customer rating	Rate a finished haircut	Saved once per booking	5 stars saved and shown in Performance	Pass

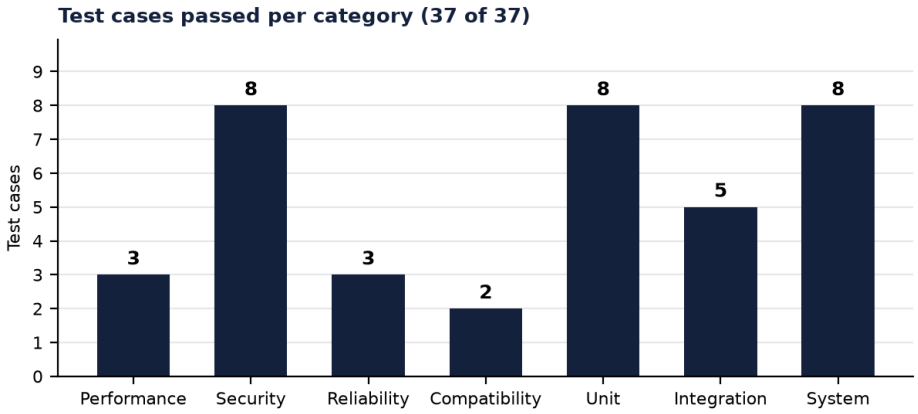


Figure 5.2 : Test cases passed per category

The total number of test cases was 37 and all of them were passed out. This number represents the number of cases that were run on this system and is not meant to be an endorsement of the system having no faults. For the usability tasks, these are not counted as they have not yet been recorded.

5.5 Acceptance Testing

5.5.1 Client Acceptance Testing (CAT)

Client acceptance testing is a process that involves asking the client if the system is doing what the business requires. The owner of the shop, Mr Waqiyuddin, demonstrated the

system function-wise. **[ADD: shop name, date and place of the walkthrough, and the owner's own comments]** The checks and what the system did in each is listed in Table 5.10.

Table 5.10 : Client Acceptance Testing Checklist

Question	System response	Outcome
Can the owner log in with an owner account?	Admin dashboard displayed	Login and role routing work
Can the owner see today's bookings?	Schedule lists time, barber, customer, services and payment	Suitable for daily operation
Can the owner verify a QR payment?	Receipt checked; status moves to Paid	Payment control achieved
Can services and prices be managed?	Service edited with name, price and photo	Price transparency achieved
Can staff attendance be tracked?	Counter QR scan records arrival, lateness and leaving time	Attendance without paper
Can the owner prepare payroll?	Days worked, lateness, sales and rating shown beside the salary	Supports salary decisions
Can sales and barber performance be seen?	Dashboard and Performance show paid sales and ratings	Supports owner decisions
Are owner pages protected from other users?	Customer and barber are blocked from owner-only pages	Security requirement met

Observation:
For each of the eight checks, there was a working function in the system. The system response column can be used to document what the system did during the walkthrough and the outcome column can be used to document the result to the business.

5.5.2 User Acceptance Testing (UAT)

The questions in the user acceptance test were based on a Google Form containing 15 questions (Appendix A). The question 1 is asking the role of the person answering the question. Questions 2 to 10, 12 and 13 are Likert statements (Likert, 1932) from 1 (strongly disagree) to 5 (strongly agree). Only barbers and the owner will be asked the question 11. Question 14 is a yes/no question on whether the respondent would recommend the system, while question 15 is an open box question for comments. No names or email addresses were collected on this form. The responses were gathered on 7 October 2026. CONFIRM: method of reaching respondents and types of questions posed prior to responding.

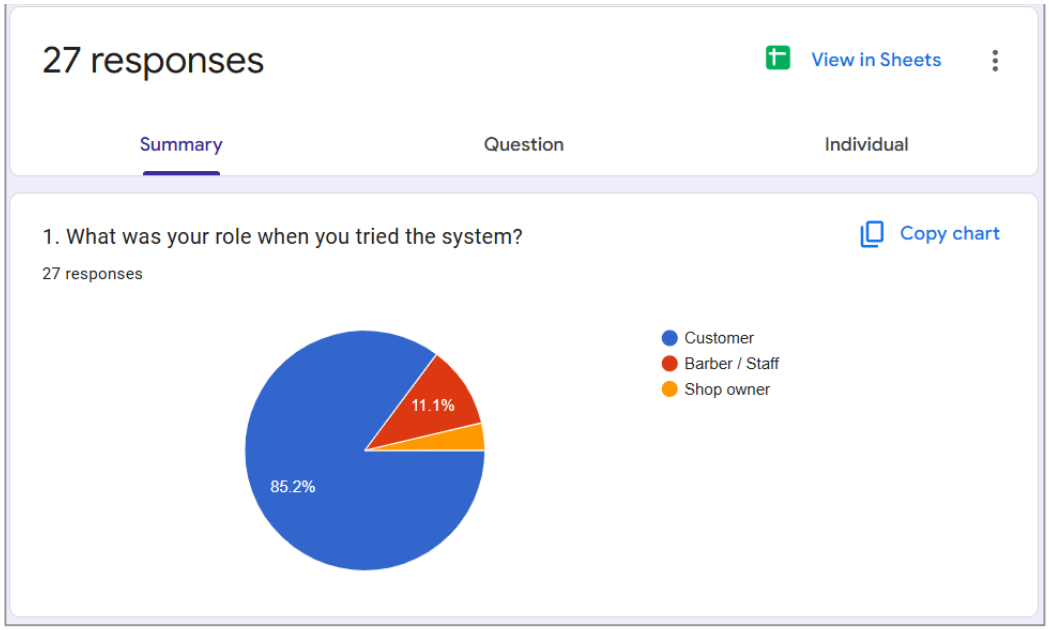


Figure 5.3 : Summary of responses in Google Forms

Twenty-seven people answered: 23 customers, 3 barbers and the shop owner (Figure 5.4). Most answers therefore describe the customer experience, and the sample is too small to represent every customer of the shop. Questions 2 to 13 are reported one by one in the figures below, each with the number of people who gave each score. After that, Table 5.12 and Figures 5.18 and 5.19 bring the results together.

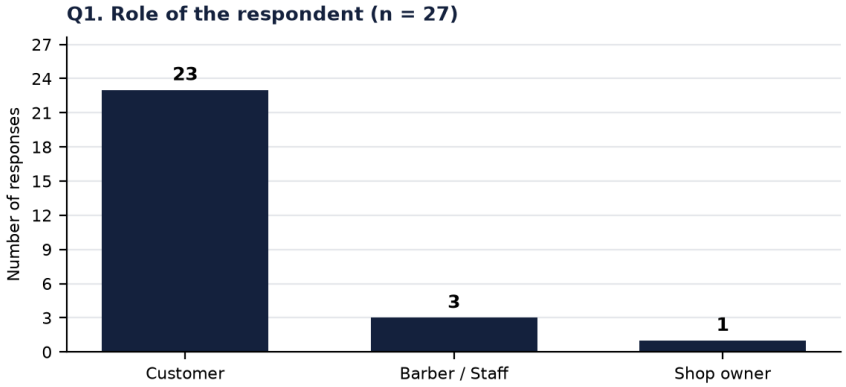


Figure 5.4 : Q1: Role of the respondent

Table 5.11 : UAT Respondents by Role

Role	Respondents	Share
Customer	23	85%
Barber / Staff	3	11%
Shop owner	1	4%
Total	27	100%

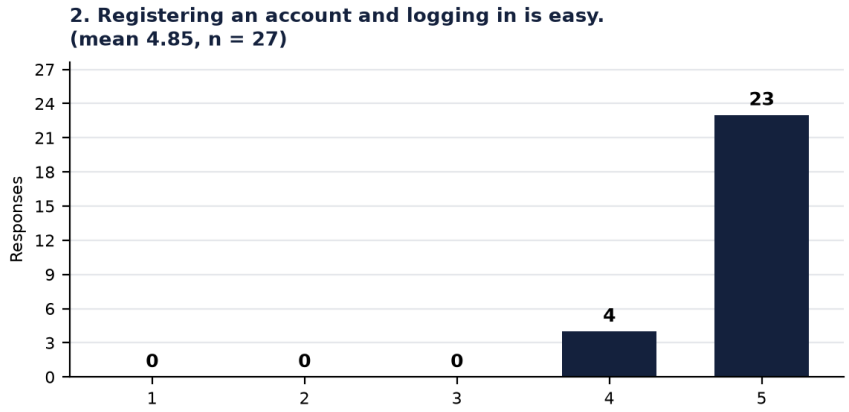


Figure 5.5 : Q2: Registering an account and logging in is easy

Observation:

23 of 27 respondents gave the top score of 5 and 4 gave 4, so 27 of 27 (100%) agreed. The mean score was 4.85.

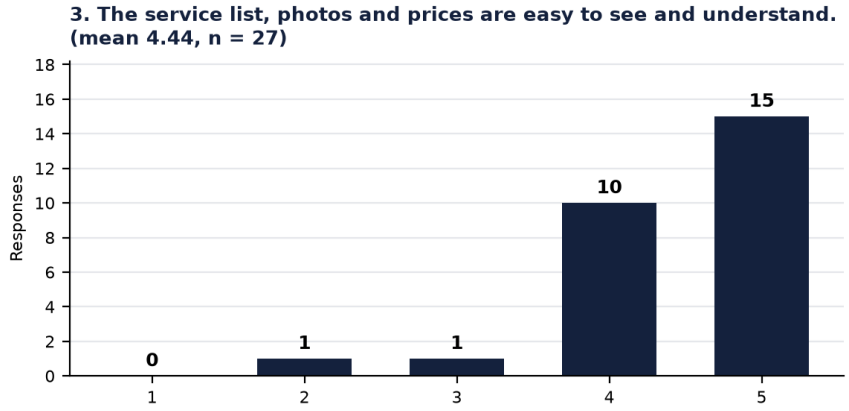


Figure 5.6 : Q3: The service list, photos and prices are easy to understand

Observation:

15 of 27 respondents gave the top score of 5 and 10 gave 4, so 25 of 27 (93%) agreed. 1 was neutral. 1 disagreed (score 2 or lower). The mean score was 4.44.

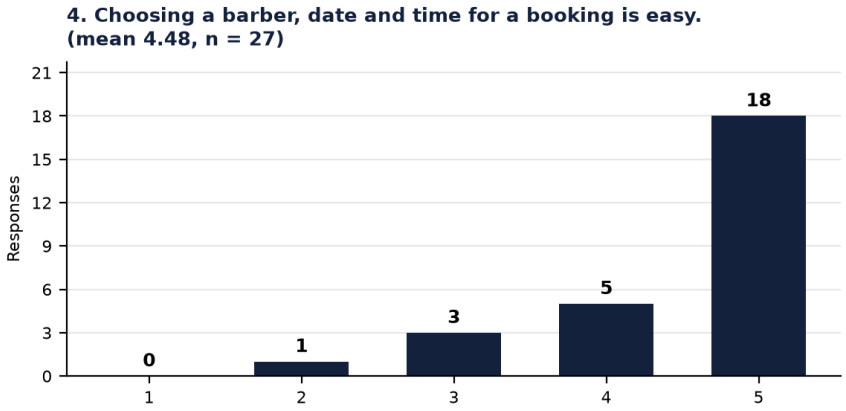


Figure 5.7 : Q4: Choosing a barber, date and time is easy

Observation:

18 of 27 respondents gave the top score of 5 and 5 gave 4, so 23 of 27 (85%) agreed. 3 were neutral. 1 disagreed (score 2 or lower). The mean score was 4.48.

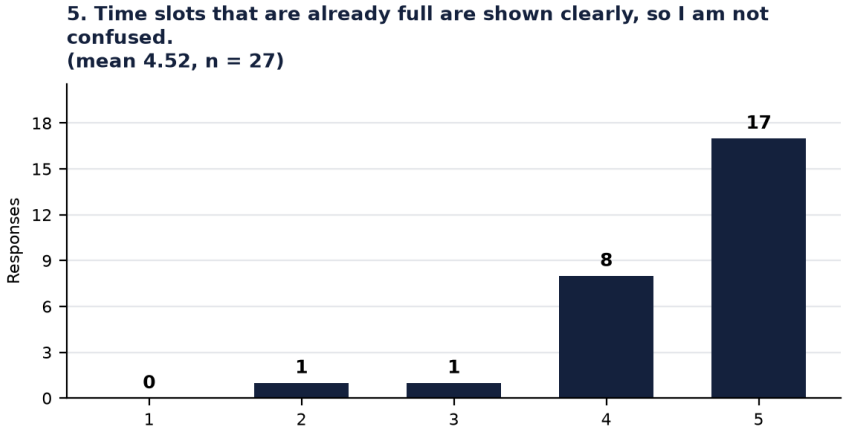


Figure 5.8 : Q5: Full time slots are shown clearly

Observation:

17 of 27 respondents gave the top score of 5 and 8 gave 4, so 25 of 27 (93%) agreed. 1 was neutral. 1 disagreed (score 2 or lower). The mean score was 4.52.

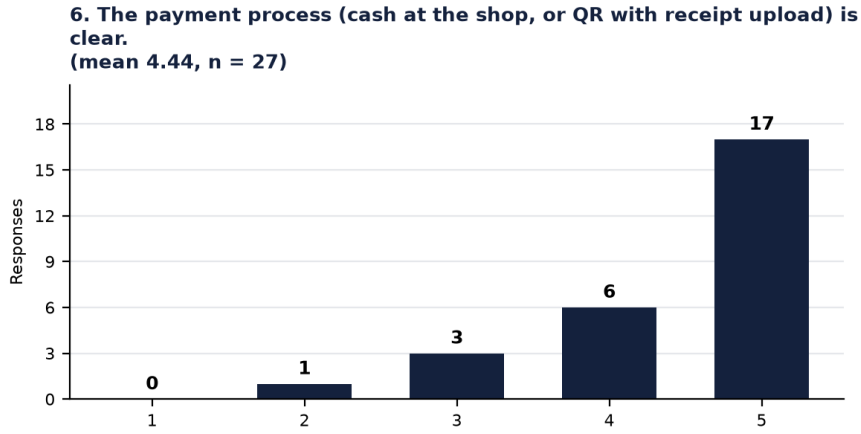


Figure 5.9 : Q6: The payment process is clear

Observation:

17 of 27 respondents gave the top score of 5 and 6 gave 4, so 23 of 27 (85%) agreed. 3 were neutral. 1 disagreed (score 2 or lower). The mean score was 4.44.

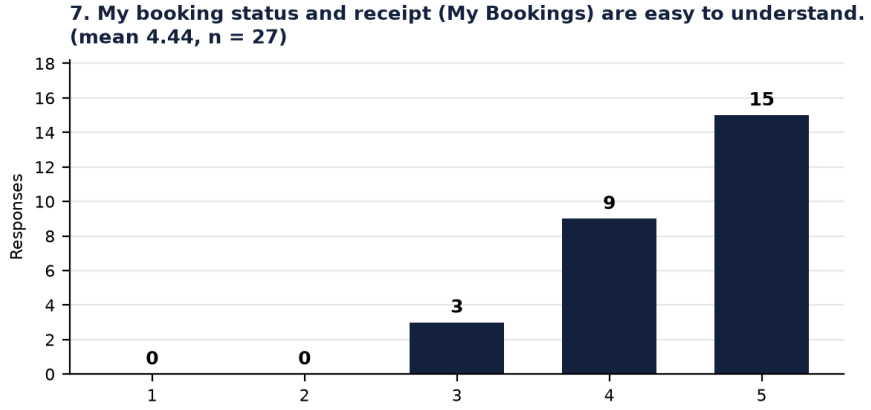


Figure 5.10 : Q7: Booking status and receipt are easy to understand

Observation:

15 of 27 respondents gave the top score of 5 and 9 gave 4, so 24 of 27 (89%) agreed. 3 were neutral. The mean score was 4.44.

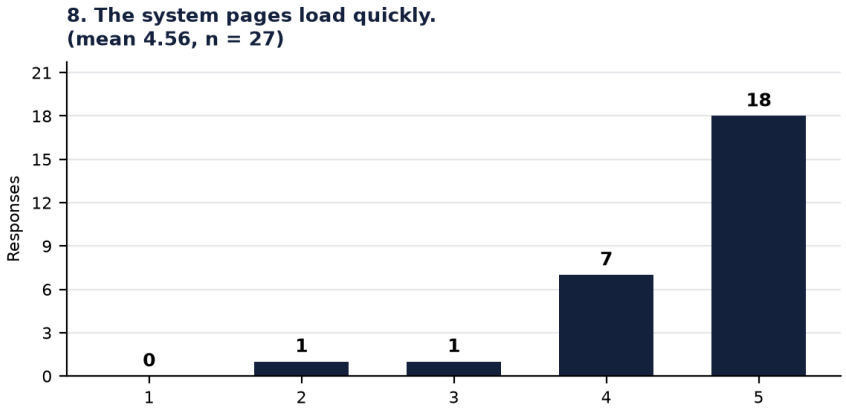


Figure 5.11 : Q8: The pages load quickly

Observation:

18 of 27 respondents gave the top score of 5 and 7 gave 4, so 25 of 27 (93%) agreed. 1 was neutral. 1 disagreed (score 2 or lower). The mean score was 4.56.

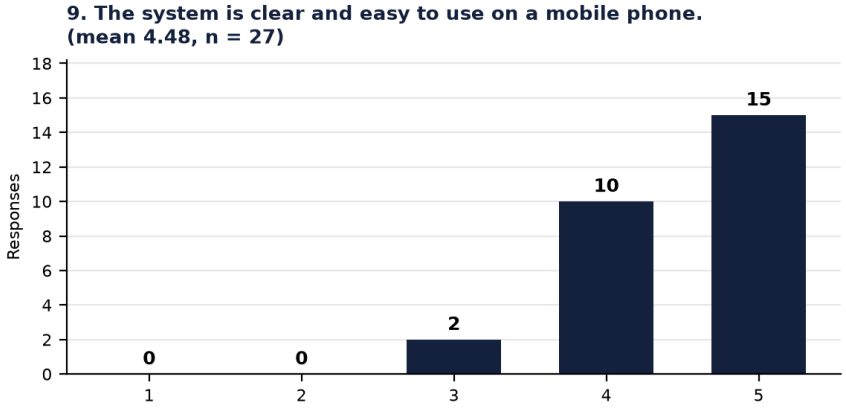


Figure 5.12 : Q9: The system is easy to use on a mobile phone

Observation:

15 of 27 respondents gave the top score of 5 and 10 gave 4, so 25 of 27 (93%) agreed. 2 were neutral. The mean score was 4.48.

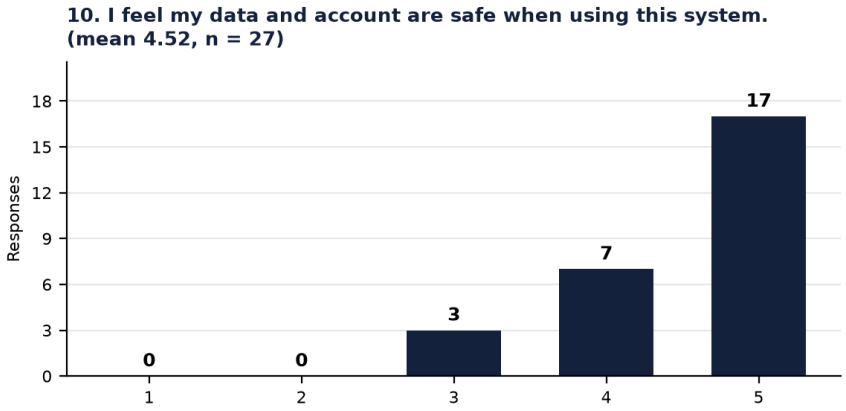


Figure 5.13 : Q10: I feel my data and account are safe

Observation:
17 of 27 respondents gave the top score of 5 and 7 gave 4, so 24 of 27 (89%) agreed. 3 were neutral. The mean score was 4.52.

Question 11 was meant for barbers and the owner only, because the dashboard, attendance, performance and payroll pages are theirs. Customers were asked to choose Not applicable, but many chose a score instead. Only the four answers from the three barbers and the owner are used here (Figure 5.14); the customers' answers to this question are not counted.

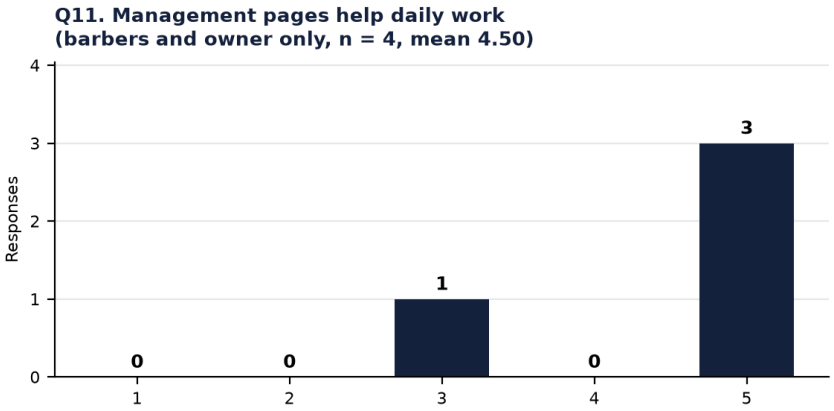


Figure 5.14 : Q11: Management pages help daily work (barbers and owner only)

Observation:
Of the 4 staff and owner answers, 3 were 5 and 1 was neutral, with a mean of 4.50. With only four people, this is an indication and not a result to rely on.

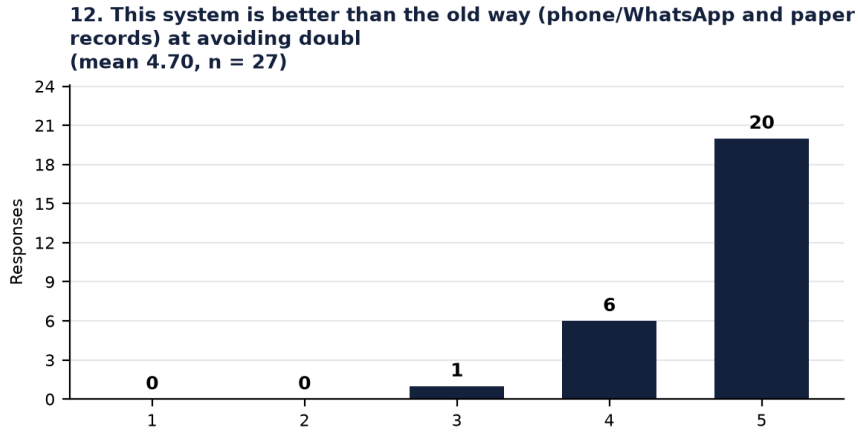


Figure 5.15 : Q12: Better than the old way at avoiding double bookings

Observation:

20 of 27 respondents gave the top score of 5 and 6 gave 4, so 26 of 27 (96%) agreed. 1 was neutral. The mean score was 4.70.

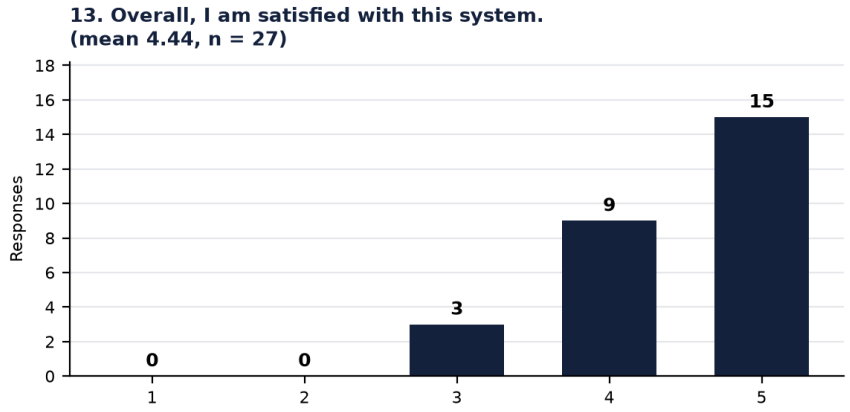


Figure 5.16 : Q13: Overall, I am satisfied with this system

Observation:

15 of 27 respondents gave the top score of 5 and 9 gave 4, so 24 of 27 (89%) agreed. 3 were neutral. The mean score was 4.44.

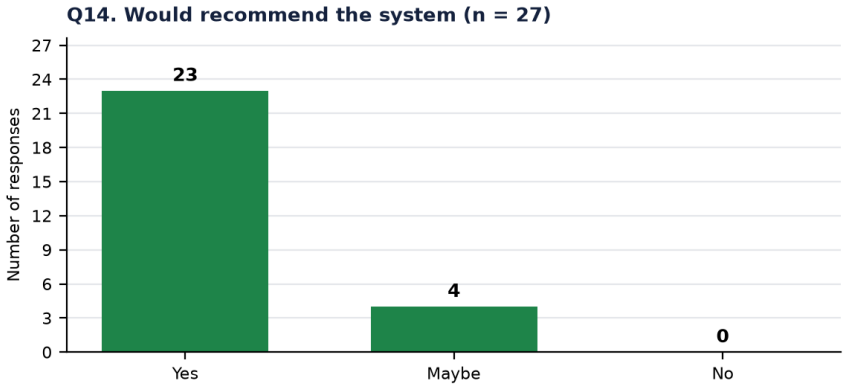


Figure 5.17 : Q14: Would recommend the system

Observation:

23 of 27 respondents (85%) said they would recommend the system, 4 said maybe and none said no.

Table 5.12 : Mean Score and Agreement for Each Statement

Item	Statement	Mean (1-5)	Agree (4-5)
Q2	Registering an account and logging in is easy	4.85	100%
Q3	The service list, photos and prices are easy to understand	4.44	93%
Q4	Choosing a barber, date and time is easy	4.48	85%
Q5	Full time slots are shown clearly	4.52	93%
Q6	The payment process is clear	4.44	85%
Q7	Booking status and receipt are easy to understand	4.44	89%
Q8	The pages load quickly	4.56	93%
Q9	The system is easy to use on a mobile phone	4.48	93%
Q10	I feel my data and account are safe	4.52	89%
Q12	Better than the old way at avoiding double bookings	4.70	96%
Q13	Overall, I am satisfied with this system	4.44	89%
All	All Likert statements together	4.54	91%

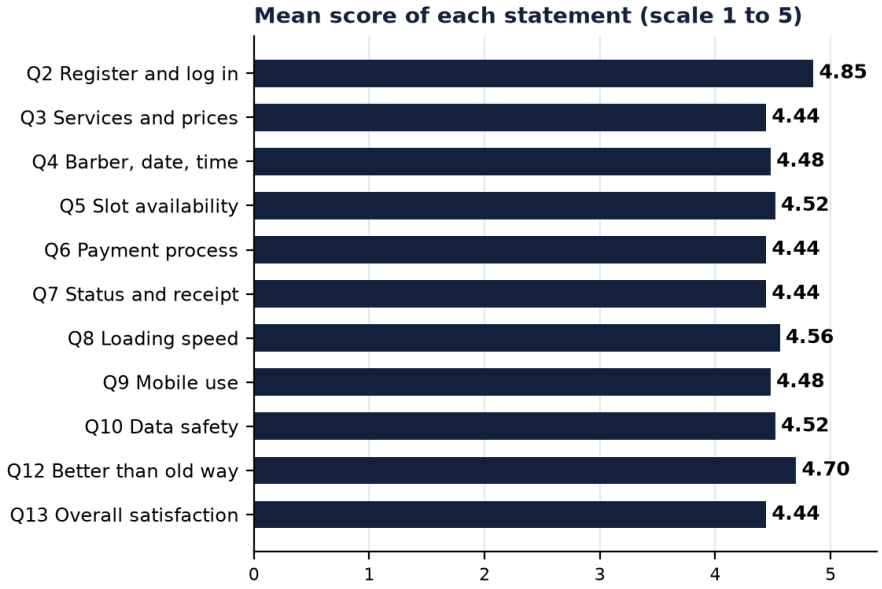


Figure 5.18 : Mean score of each statement

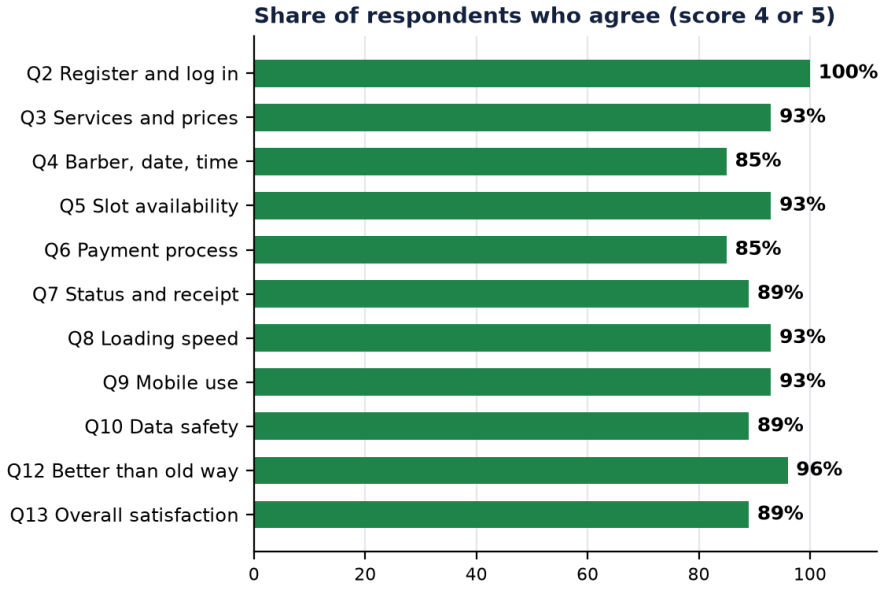


Figure 5.19 : Share of respondents who agree with each statement

Observation:

Across 297 Likert answers, the mean score was 4.54 out of 5 and 91% of answers were 4 or 5. The highest-rated statement was registering and logging in (mean 4.85, every respondent agreed), and the strongest comparison was with the old way of booking (mean 4.70, 96% agreement). The statements with the lowest mean, 4.44, were the services list, the payment process, the status and receipt page and overall satisfaction. They are not low scores, but they point to the places where small improvements, such as clearer wording on the payment page, would help most. No statement received a mean below 4.4.

Question 15 asked for one thing to improve or add. Fifteen of the 27 respondents wrote something. Table 5.13 groups the answers. Most were short remarks of approval and none was a concrete suggestion, so the future work in Chapter 6 reflects my own assessment of the gaps and not requests from users.

Table 5.13 : Open Comments from Question 15

Type of comment	Count	Examples
Short approval	5	"good", "best", "nice this sistem", "so far so good", "win"
Neutral	2	"ok", "ok la"
About a feature	1	"dasboard"
Unclear or filler	7	"base", "yes", "rip", "tt"
No comment	12	(left blank)

5.6 Conclusion

The tests give a consistent picture. The 37 test cases on performance, security, reliability, compatibility, unit, integration and system behaviour all passed, with one security weakness noted and carried forward. Users rated the system 4.54 out of 5 on average and nearly nine in ten were satisfied overall. The caveats are real: the sample is 27 people, most of them customers, usability tasks have not been recorded and the open comments gave little guidance. The next chapter closes the report by checking the results against the objectives.

CONCLUSION

6.1 Introduction

This final chapter looks back at the project. It sets out the schedule and the risks, checks each objective against the evidence from Chapter 5, states the limits of what was built and lists what should be done next.

6.2 Project Timeline

The work was planned around seven activities. Table 6.1 gives the estimated effort for each one. The estimates are planning figures and not time sheets.

Table 6.1 : Project Schedule Timetable

Task	Estimated hours	Main output
Background research and learning new skills	70	Understanding the shop's business flow, PHP and MySQL security
Requirements gathering	30	Owner interview, observation, questionnaire and requirement list
Product design	40	Use case, flowcharts, wireframes, DFD, ERD and security framework
Product development	100	Working booking, payment, admin, attendance and payroll modules
Product evaluation	30	Test results and user feedback
Final report preparation	25	This report
Presentation rehearsal	25	Slides, poster and demonstration
Total	320	Overall estimated effort

6.2.1 Work Breakdown Structure (WBS)

Figure 6.1 divides the project into six groups of work. Planning, design, development, testing and deployment follow the Agile phases, and documentation runs alongside them.

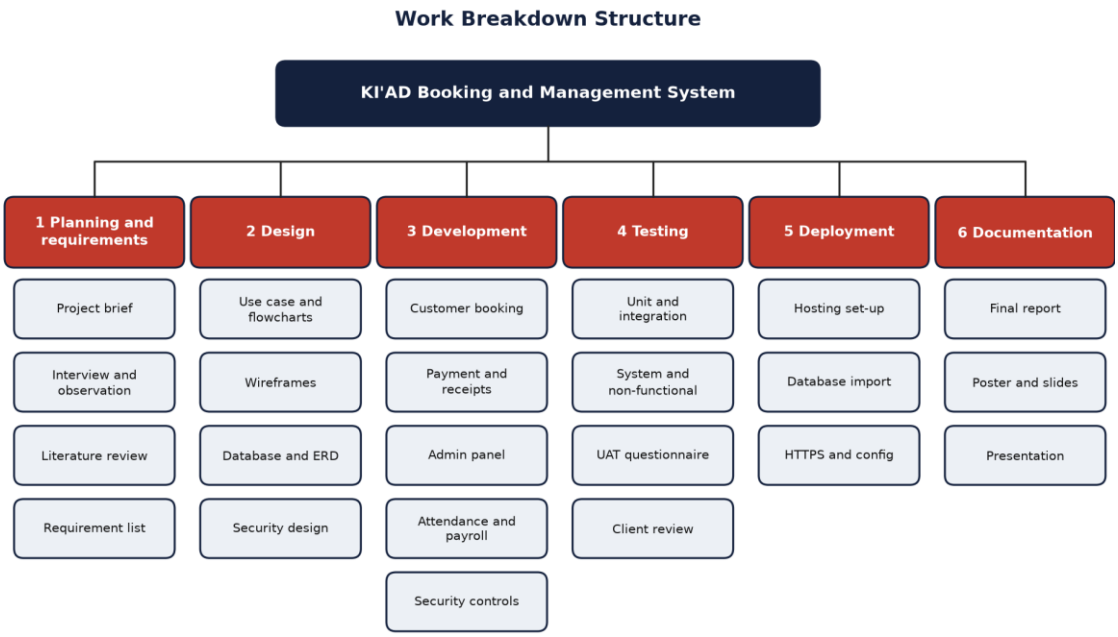


Figure 6.1 : Work breakdown structure of the project

6.2.2 Gantt Chart

Figure 6.2 shows how the activities overlap over 14 weeks. Development ran in parallel with design and testing, because each increment was built, tried and revised before the next one started. **[CONFIRM: check the week numbers against the real project schedule]**

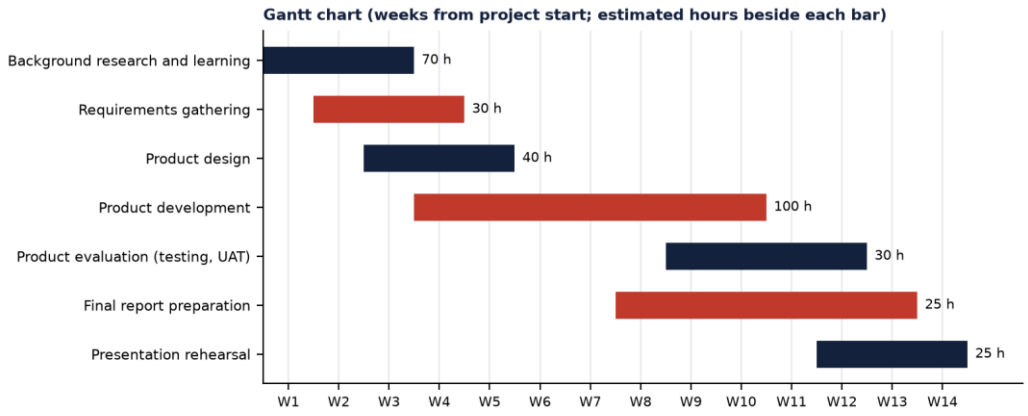


Figure 6.2 : Gantt chart of the project

6.3 Risk Management

Table 6.2 : Risk Management Analysis

Risk	Impact	Mitigation
Requirements change	Redesign and rework of finished pages	Agile increments, early review of each feature with the owner

Risk	Impact	Mitigation
Database error or data loss	Lost bookings, wrong figures, delay in testing	Clean install script, local backups before changes, testing database changes locally first
Security weakness	Exposure of customer data, receipts or payroll	Layered controls (Section 4.4), a dedicated set of security tests
Limited user feedback	System may not match what users need	Owner walkthrough and a questionnaire after development
Time limitation	Some planned features unfinished	Core modules first; expense module, queue display and audit log moved to future work
Deployment problems on shared hosting	System not running online or uploads failing	Tested locally first, hosting steps written down, folder permissions and config checked after upload
Hosting limits (resources, uptime)	Slow or unavailable site at busy times	Keep pages light, avoid heavy queries, plan to move to a paid plan if the shop grows

6.4 Achievement of Project Objectives

6.4.1 Objective 1: To Develop Online Booking With Barber Choice and No Double Booking

This objective is met. A customer can choose services, a named barber or Any barber, a date and a free slot, and the system refuses slots that are taken, in the past or on closed days. Test UT04 showed that the same barber could not be booked twice for the same time, IT01 showed that Any barber receives a free barber, and IT04 showed that a holiday cannot be booked. In the questionnaire, choosing a barber, date and time scored a mean of 4.48 and the clear display of full slots 4.52. The comparison with the old way of working, which was the most direct measure of the problem, scored 4.70 with 96% agreement.

6.4.2 Objective 2: To Record Cash and QR Payments With Owner Verification and a Sales Dashboard

This objective is met. Payment status is kept on every booking. Cash bookings stay as cash until the money is received, QR bookings need an uploaded receipt and wait for the owner, and the dashboard counts only verified payments. Tests UT05, UT06, IT02 and IT03 checked the status changes and confirmed that the dashboard's collected figure equals the sum of verified payments in the database. The payment process scored a mean of 4.44 in the questionnaire. The limit of this objective is that verification is still done by the owner looking at the receipt, which is manual by design.

6.4.3 Objective 3: To Centralise Attendance and Payroll With Secure Role-Based Access

This objective is met. Barbers clock in and out by scanning a rotating QR code, the owner sees late minutes and hours worked beside sales and ratings, and a paid salary is locked. The system tests for attendance, the counter screen and payroll passed, and the eight security tests showed that passwords are hashed, owner pages are closed to other roles and repeated wrong passwords are blocked. In the questionnaire, the feeling of data safety scored 4.52, and the four staff and owner answers about the management pages averaged 4.50. That last figure rests on four people only.

6.5 Constraints and Limitations

The system does what it was built to do, and it also has limits that a reader should know about.

1. **Payments are verified by hand.** There is no payment gateway, so the owner checks every QR receipt. This needs the owner or a trusted barber to be available during the day.
2. **Parts of the FYP1 proposal were not built.** The expense and profit module, the live queue status page, daily, weekly and monthly financial reports and the audit log are not part of the finished system. The dashboard shows collected sales and barber figures, but not expenses or profit.
3. **The database has no foreign key constraints.** The relationships are kept by the application code. A mistake in future code could leave orphaned rows.
4. **Receipt files are reachable by their link.** The file names are long and random and the folder cannot be listed, but anyone who has the exact link can open the file.
5. **The CSRF check accepts a request with no session token.** It rejects wrong tokens, but if the session has no token at all, two empty values compare as equal (Section 5.3.2).
6. **Testing was done on a local copy.** The 37 test cases ran against a local copy of the deployed code, with demo data. Phone testing used an emulated viewport and not a real device, and only Chrome was used.
7. **The user sample is small.** Twenty-seven people answered the questionnaire, mostly customers, and the staff and owner questions rest on four answers. Usability task trials still need to be recorded.
8. **Attendance depends on the phone.** Scanning needs a phone with a working camera, a recent browser and, on the live site, HTTPS. A barber without a phone needs the owner to add the record by hand.

9. **Password reset is manual.** A customer who forgets a password asks the shop through WhatsApp and the owner gives a temporary password. There is no automatic email reset.

6.6 Future Work and Recommendations

The recommendations below follow from the limits above. They are ordered roughly by how much they would improve the system for the shop.

1. **Online payment confirmation.** Connect a payment gateway or a bank API so that a QR or FPX payment confirms itself and the owner no longer checks receipts by eye. This needs a payment provider account.
2. **Expense and profit module.** Let the owner record expenses such as rent, stock and utilities, and show net profit for a day, week or month. This was part of the FYP1 proposal.
3. **Audit log.** Record who verified a payment, changed a price or edited a salary, and when.
4. **Reminders.** Send automatic booking reminders by WhatsApp or SMS to reduce no-shows.
5. **Harden the code.** Add real foreign keys, make check_csrf() refuse an empty session token, and serve receipts through a script that checks who is asking.
6. **Queue display for walk-ins.** Show waiting customers on a screen at the counter.
7. **Two-factor login for the owner**, and regular automated backups of the database and uploads.
8. **Loyalty points and an installable app.** Reward repeat customers and offer the site as a progressive web app on phones.

6.7 Final Summary

The project set out to replace chat messages and paper at a small barbershop with one system that handles bookings, payments, attendance and salary. The result is a working web application, live on HTTPS, that stops double bookings, ties every payment to a booking and an owner decision, records attendance with a QR code that cannot be photographed and reused, and keeps each role to its own pages. It passed 37 tests, and 27 users rated it 4.54 out of 5 on average.

It is not finished business. Payment is still checked by eye, parts of the original proposal were left out, and the evidence from users is small. What it does show is that a small shop can get a useful, secure system from plain PHP and MySQL on ordinary hosting, and that the most valuable features are the unglamorous ones: a free-slot check, a payment status and a lock on a paid salary.

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Appendix A – Questionnaire and Interview

A.1 Interview Record – Shop Owner

Respondent: Mr Waqiyuddin, shop owner. Interviewer: Muhammad Fauzi Naim bin Mohd Asrri. Date: 5 September 2026. Place: the owner's shop.

Table A.1 : Interview Questions

No	Question	Purpose
1	How do customers book an appointment at the moment?	Find out how bookings are taken
2	What problems do you have with that method, for example clashes or missed slots?	Identify booking problems
3	How do customers pay, and how do you check a QR payment?	Understand payment handling
4	How do you record when the barbers arrive and work out their salary?	Understand attendance and payroll
5	Where are customer details kept and who can see them?	Understand data handling and privacy
6	What devices do you and the barbers use during the day?	Decide on the interface and the QR scan
7	What would make an online system worth using for you?	Find priorities
8	What worries you about moving to a computer system?	Find risks and constraints

[ADD: attach the signed interview notes or a photo of them here]

A.2 User Acceptance Testing Questionnaire

The questionnaire is a Google Form titled KI'AD Barbershop Booking System - User Feedback. It does not collect names or email addresses. Statements use a five-point Likert scale (Likert, 1932) from 1 (strongly disagree) to 5 (strongly agree).

Table A.2 : User Acceptance Testing Questionnaire Items

No	Question	Answer type
1	What was your role when you tried the system?	Customer / Barber-Staff / Shop owner
2	Registering an account and logging in is easy.	1-5
3	The service list, photos and prices are easy to see and understand.	1-5
4	Choosing a barber, date and time for a booking is easy.	1-5
5	Time slots that are already full are shown clearly, so I am not confused.	1-5
6	The payment process (cash at the shop, or QR with receipt upload) is clear.	1-5
7	My booking status and receipt (My Bookings) are easy to understand.	1-5

No	Question	Answer type
8	The system pages load quickly.	1-5
9	The system is clear and easy to use on a mobile phone.	1-5
10	I feel my data and account are safe when using this system.	1-5
11	The dashboard, QR attendance, performance and payroll pages help with the daily running of the shop (staff and owner only).	Not applicable, or 1-5
12	This system is better than the old way (phone/WhatsApp and paper records) at avoiding double bookings.	1-5
13	Overall, I am satisfied with this system.	1-5
14	Would you recommend this system to other people?	Yes / Maybe / No
15	What is one thing that could be improved or added to this system?	Open answer (optional)

A.3 Raw Questionnaire Scores

Table A.3 lists the 27 responses without any personal details. C = customer, B = barber, O = owner. Question 11 is shown as it was answered; only the answers from B and O are used in the analysis. Y, M and N are Yes, Maybe and No for question 14.

Table A.3 : Raw Questionnaire Scores

#	Role	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14
1	C	5	5	5	5	5	5	5	5	5	5	5	5	Y
2	C	5	5	5	5	5	5	5	5	5	5	5	5	M
3	B	5	5	5	5	5	5	5	5	5	5	5	5	Y
4	O	5	5	5	5	5	5	5	5	5	5	5	5	Y
5	C	4	5	3	4	4	5	4	4	5	4	4	4	Y
6	C	5	3	4	4	3	3	3	3	4	4	4	3	M
7	C	4	4	4	4	4	4	4	4	4	4	5	5	M
8	C	5	5	5	5	5	5	5	5	5	4	5	5	Y
9	C	5	4	4	5	3	4	4	5	3	4	5	5	Y
10	B	5	5	5	5	5	5	5	5	5	5	5	5	Y
11	C	5	5	5	5	5	5	5	5	5	5	5	4	Y
12	C	4	2	2	3	2	3	2	3	3	5	5	5	Y
13	C	5	4	5	4	5	4	5	4	5	5	5	4	Y
14	C	5	5	5	5	5	5	5	5	5	5	5	5	Y
15	C	5	5	5	5	5	5	5	5	5	5	4	4	Y
16	C	5	5	5	5	5	5	5	5	5	4	4	4	Y
17	C	5	4	4	4	5	4	4	4	4	5	5	5	Y

#	Role	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14
18	C	5	4	3	2	3	4	5	4	3	4	5	3	Y
19	C	5	4	3	5	4	3	5	4	4	4	5	5	Y
20	C	5	5	5	5	5	5	5	5	5	5	5	5	Y
21	C	5	5	5	5	5	5	5	5	5	5	5	5	Y
22	C	5	4	4	5	4	5	4	5	4	4	5	4	Y
23	C	5	4	5	4	5	4	5	4	5	5	5	4	Y
24	C	5	4	5	4	5	4	5	4	5	5	5	5	Y
25	C	5	5	5	5	5	5	5	5	5	4	3	3	M
26	B	4	4	5	4	4	4	4	4	4	3	4	4	Y
27	C	5	5	5	5	4	4	4	4	4	4	4	4	Y

Appendix B – User Manual

B.1 Introduction

This manual is an explanation of the KI'AD booking system. It is divided in a brief getting started section and one section for each type of users (Customers, Barber and the owner). A phone or computer with an ordinary Web browser. Its web address is https://barbershophesmes.site.je.

B.2 Getting Started

1. Type in the web address and click **Make an Appointment**, or click **Log in** at the top of the page.
2. New users select the **Sign up** tab and enter a user name, email address and password (at least 6 characters long) and click **Create account**.
3. Enter your username/email and password. The same page is used by both barbers and the owner, on which is written the accounts the owner handed out.
4. **Forgot password?** (If user forgets his/her password). This will open WhatsApp with a message to the shop. The owner will provide you with a temporary password which will be requested by the system once you log in to select your own password.

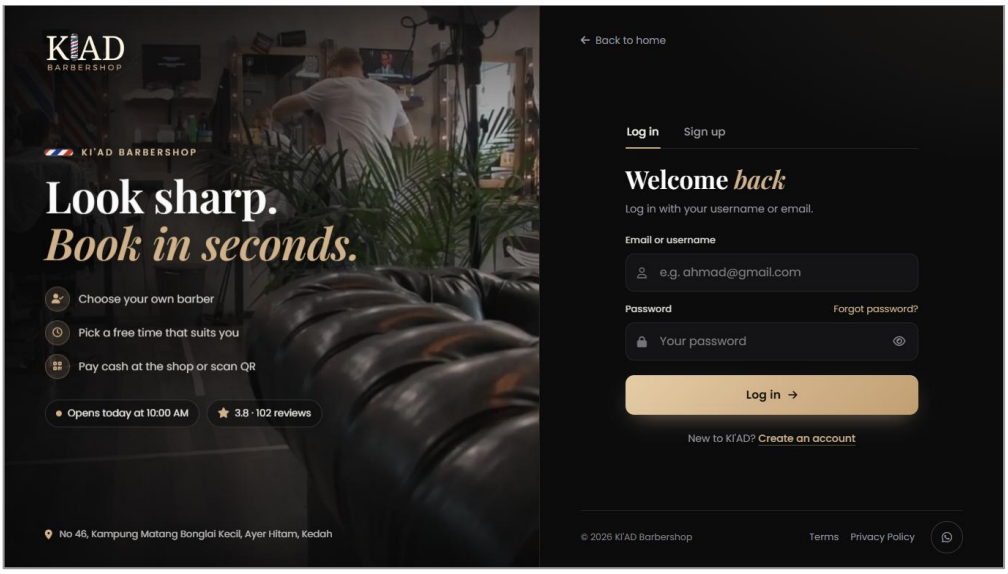


Figure B.1 : Login page

B.3 Customer Guide

Choose services

1. Click on 1. Press **Services** in the menu. If you need to look for something, such as "fade", use the search box.
2. On each service you wish to add, click **Add to cart**.
3. Click on the cart icon to view the list and the total. When you are ready, click **Continue**.

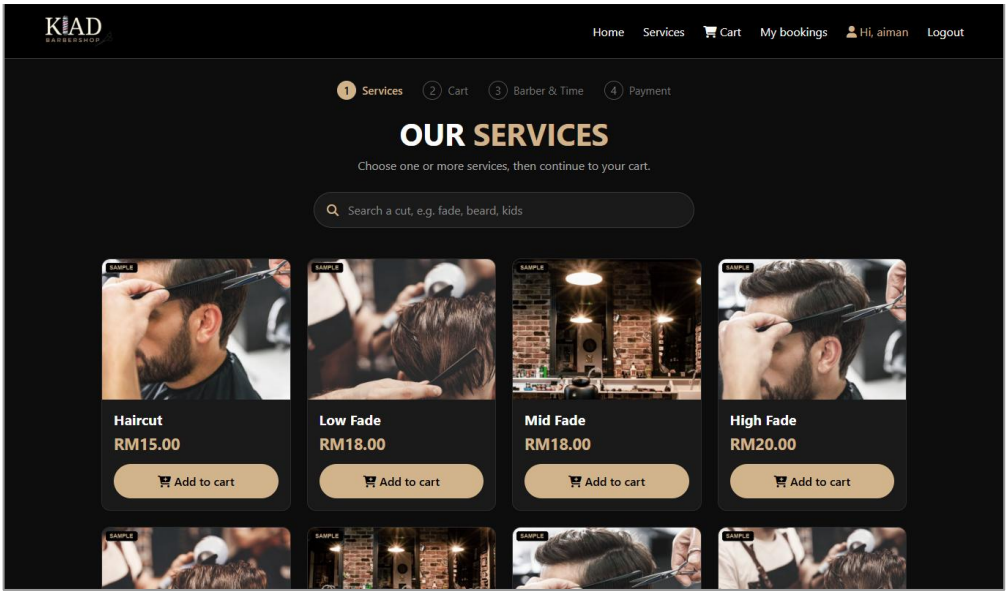


Figure B.2 : Services page

Choose barber, date and time

1. Select **Any barber** or a particular barber.
2. Pick a date. Time that has been taken, is in the past or is on a closed day is not selectable.
3. Choose a free time. You can type a note, for example the style you want.
4. Press **Continue to payment**.

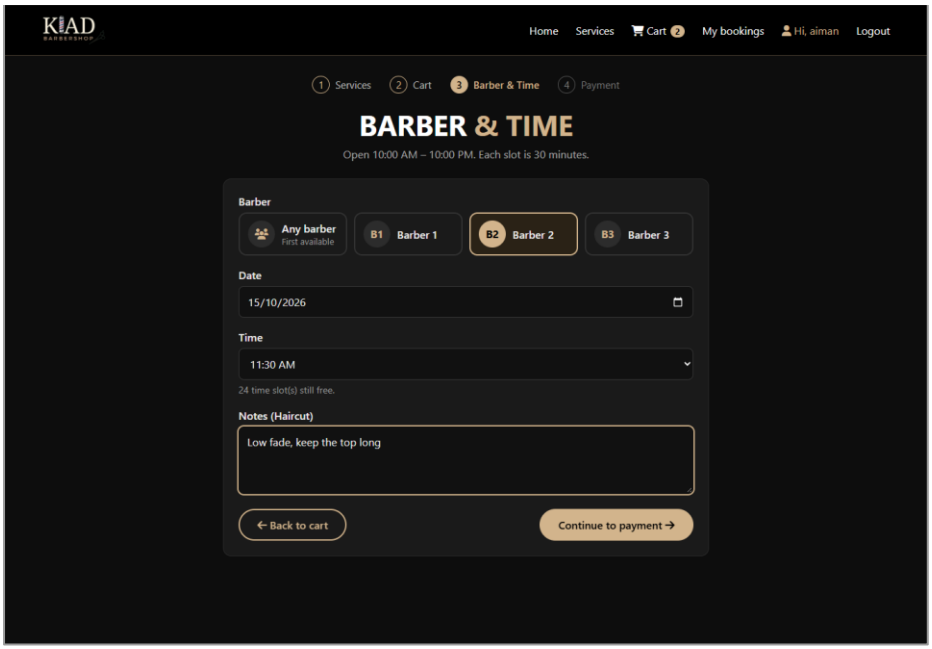


Figure B.3 : Barber and time page

Pay

1. Review Booking Summary. If it's incorrect, you can use the links to switch the services or the barber, date and time.

2. Select **Cash** for payment at the shop or **Online payment** to pay via QR.
3. To pay online, scan the QR with your banking application, pay the total, and upload the receipt (a photo, a screen or a PDF of up to 5 MB).
4. Press **Confirm booking**. You will see your receipt, and you can print it or make a screen shot.

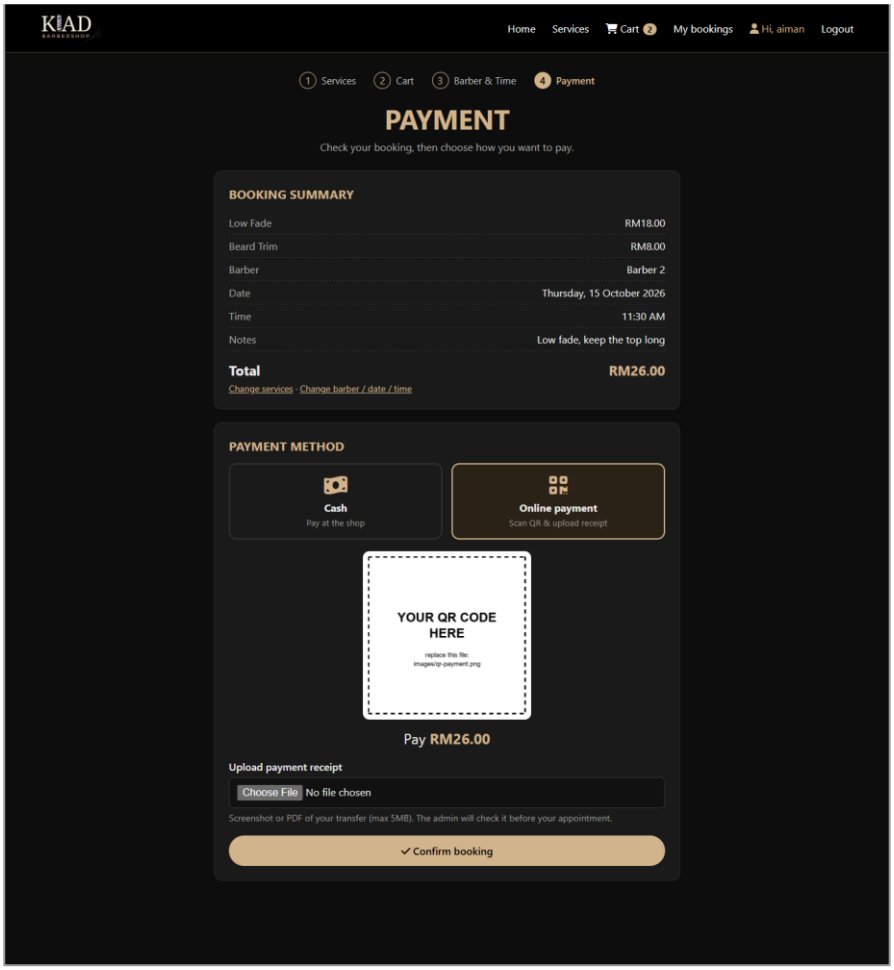


Figure B.4 : Payment page with QR payment

Check your bookings and rate the barber

1. Open **My bookings**. The status is Booked, Waiting for payment check, Payment rejected, Done or Cancelled.
2. Tap **View receipt** to reopen the receipt.
3. Once a haircut has been Done, give 1 to 5 stars and an optional comment. Each booking can only be rated once.
4. Please contact the shop as soon as possible on WhatsApp to cancel or change a booking.

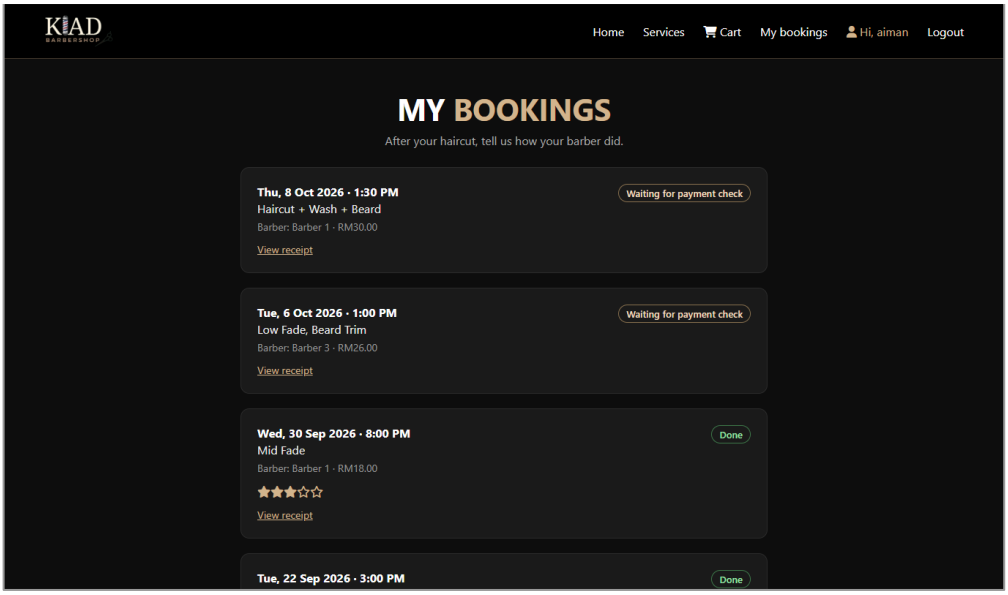


Figure B.5 : My Bookings page

B.4 Barber Guide

1. Log in. You are taken to the STAFF panel.
2. Upon entering the workplace, scan QR code at the counter screen or open **Attendance** and click the scan button. Welcome and indicating if you were on time or late. Repeat scan upon return home.
3. View **Dashboard** for today's bookings, your sales and your rating.
4. Open **Bookings**. If it is a cash booking, click **Cash received** when the monies have been received. If the haircut is completed then press **Done**.
5. Click on **My salary** to view the months when the owner has marked them as paid.

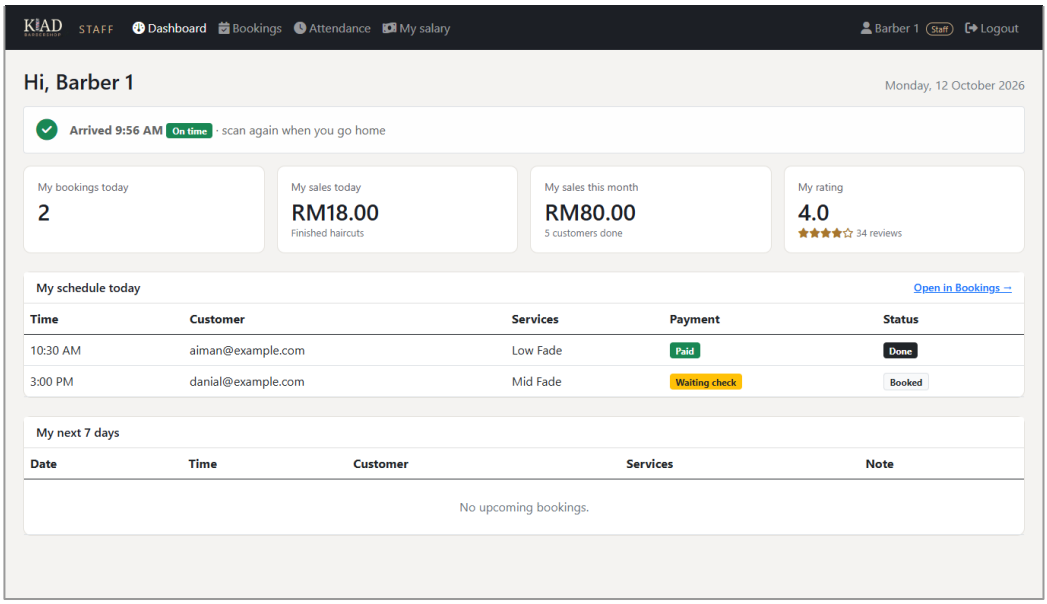


Figure B.6 : Barber dashboard

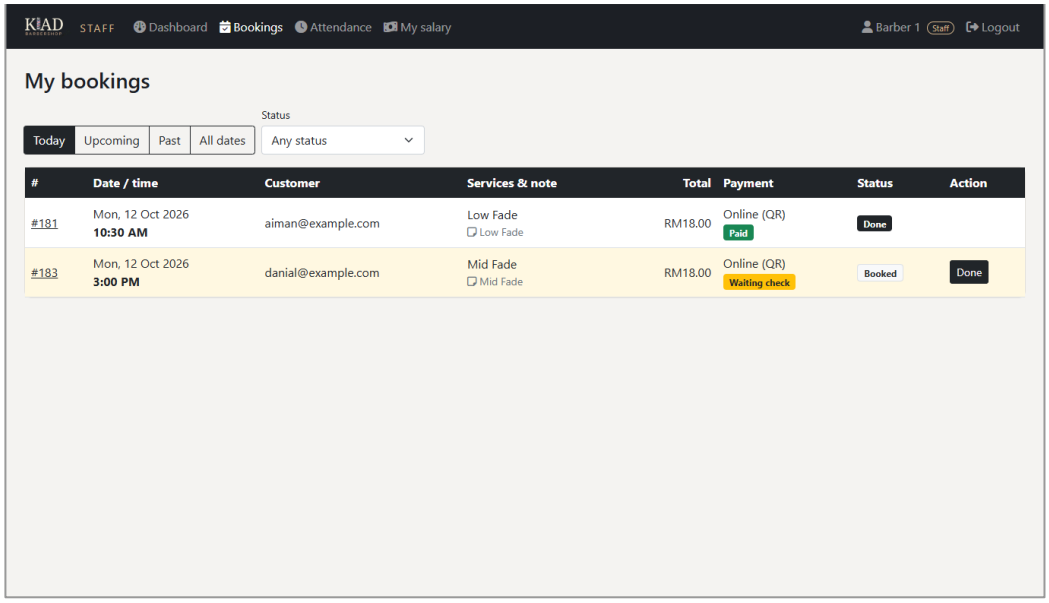


Figure B.7 : Barber bookings page

B.5 Owner Guide

Dashboard and bookings

1. Today's bookings, value booked, money collected this month and payments waiting for a check are all included on the **Dashboard**.
2. Open **Bookings**. Highlighted rows are awaiting a payment check. Open the receipt and then press **Verify payment** / **Reject**.
3. Click the barber drop-down on a row to switch a booking to a different barber or **Cancel** to cancel.

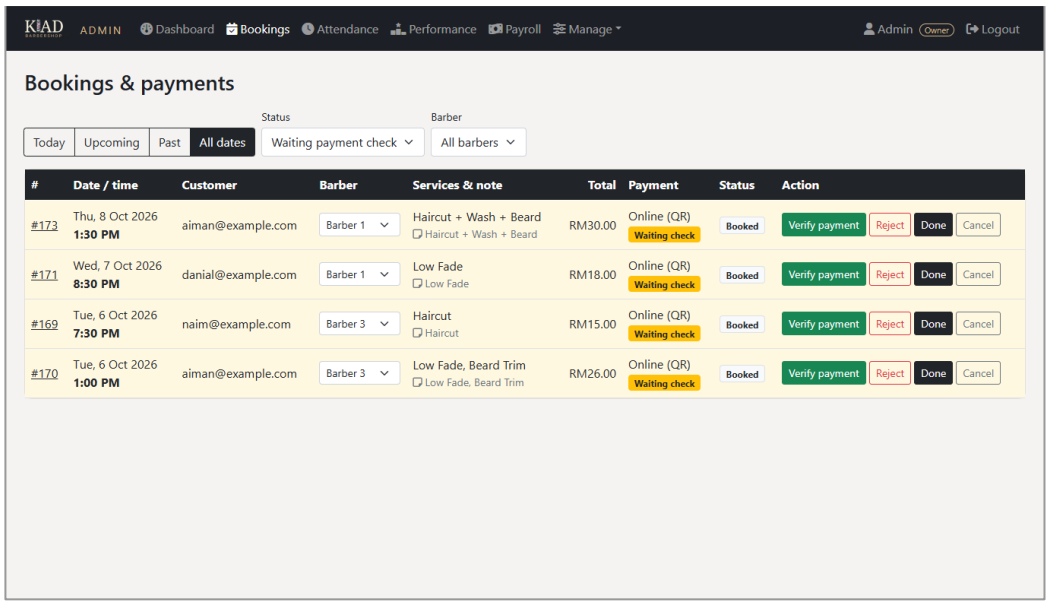


Figure B.8 : Bookings waiting for a payment check

Attendance, performance and payroll

1. Open **Attendance**. Show the counter QR on a tablet or other spare mobile phone by pressing **Open screen**. Hit **New link** should you want to change out the counter link.
2. **Fix** or **Add** next to a name to fix a forgotten scan.
3. Opening Sales & Barber Rating for the selected month for customers served.
4. Open **Payroll**, select the month then enter each salary and **Save**, **Paid**. If a person has a paid salary, it is locked. Only use **Undo** if it's absolutely necessary to change it.

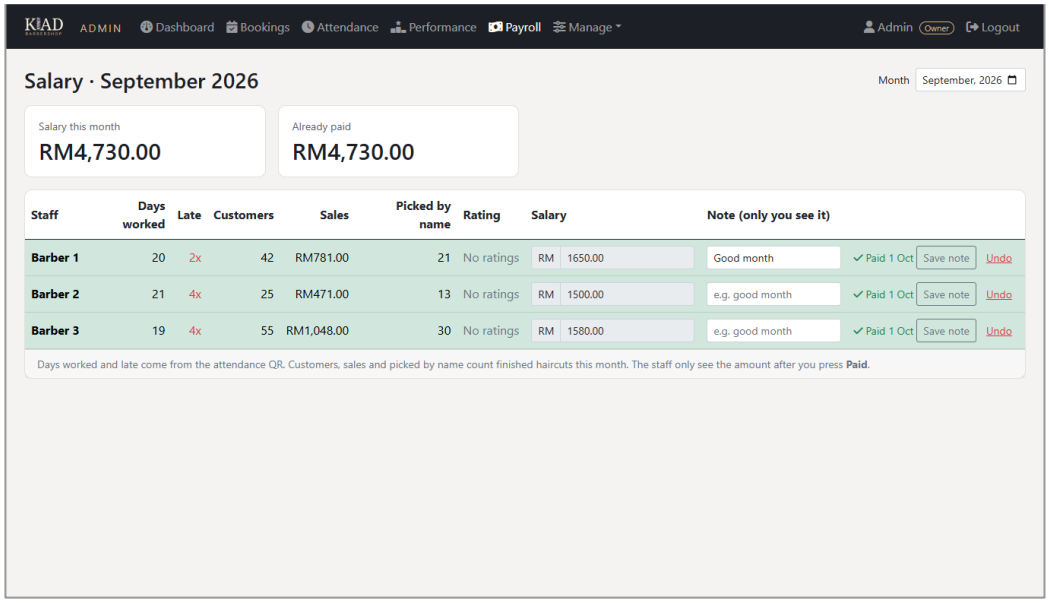


Figure B.9 : Payroll page

Services, staff, customers and settings

1. To add and/or modify a service, the price and photo in **Manage**, open **Services**.
2. Click Open **Staff** to add a barber, select the role, start time and Active status. Have a strong password for each account.
3. Open **Customers** to look for an account and click **Reset password**, if a user forgets it. Confirm that it is them before sending the temporary password.
4. Edit opening hours, closed days/holidays, payment QR, logo, contact information and the terms text in **Settings**.

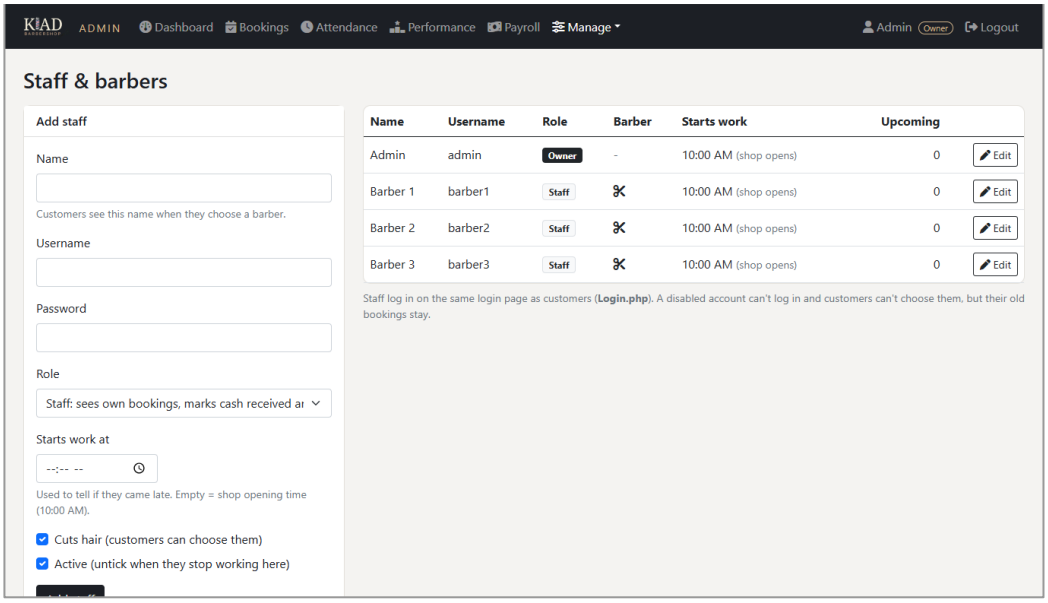


Figure B.10 : Staff and barbers page

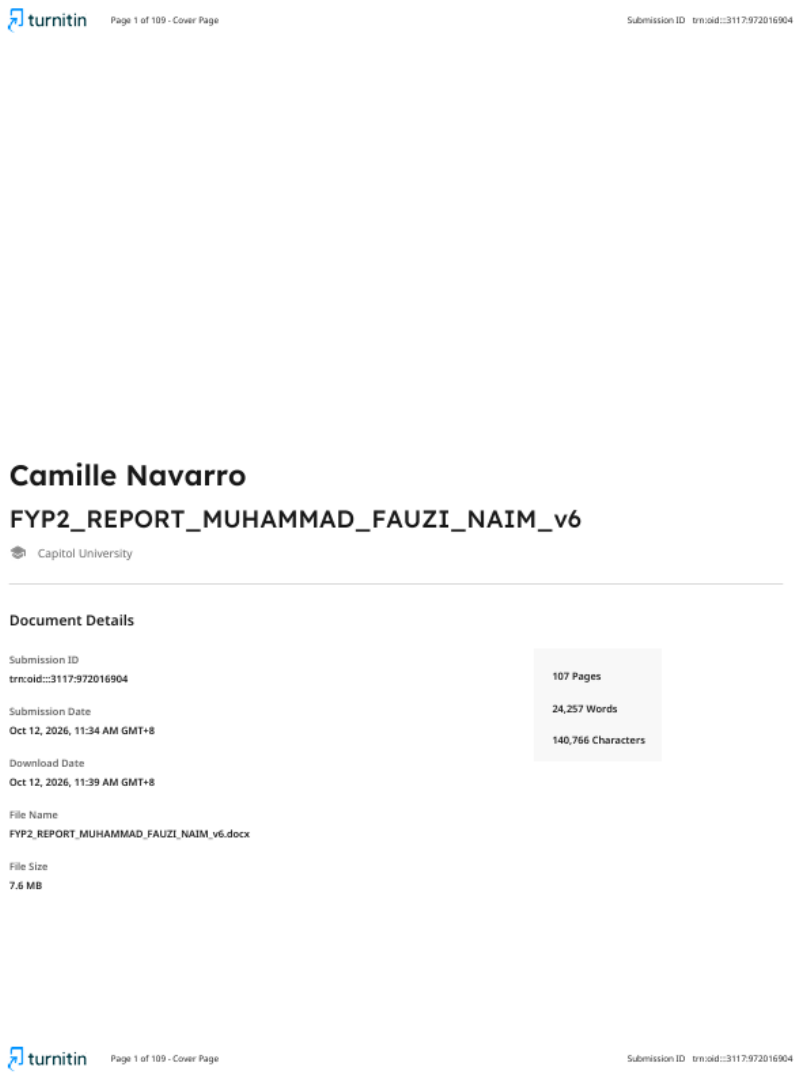
B.6 Troubleshooting

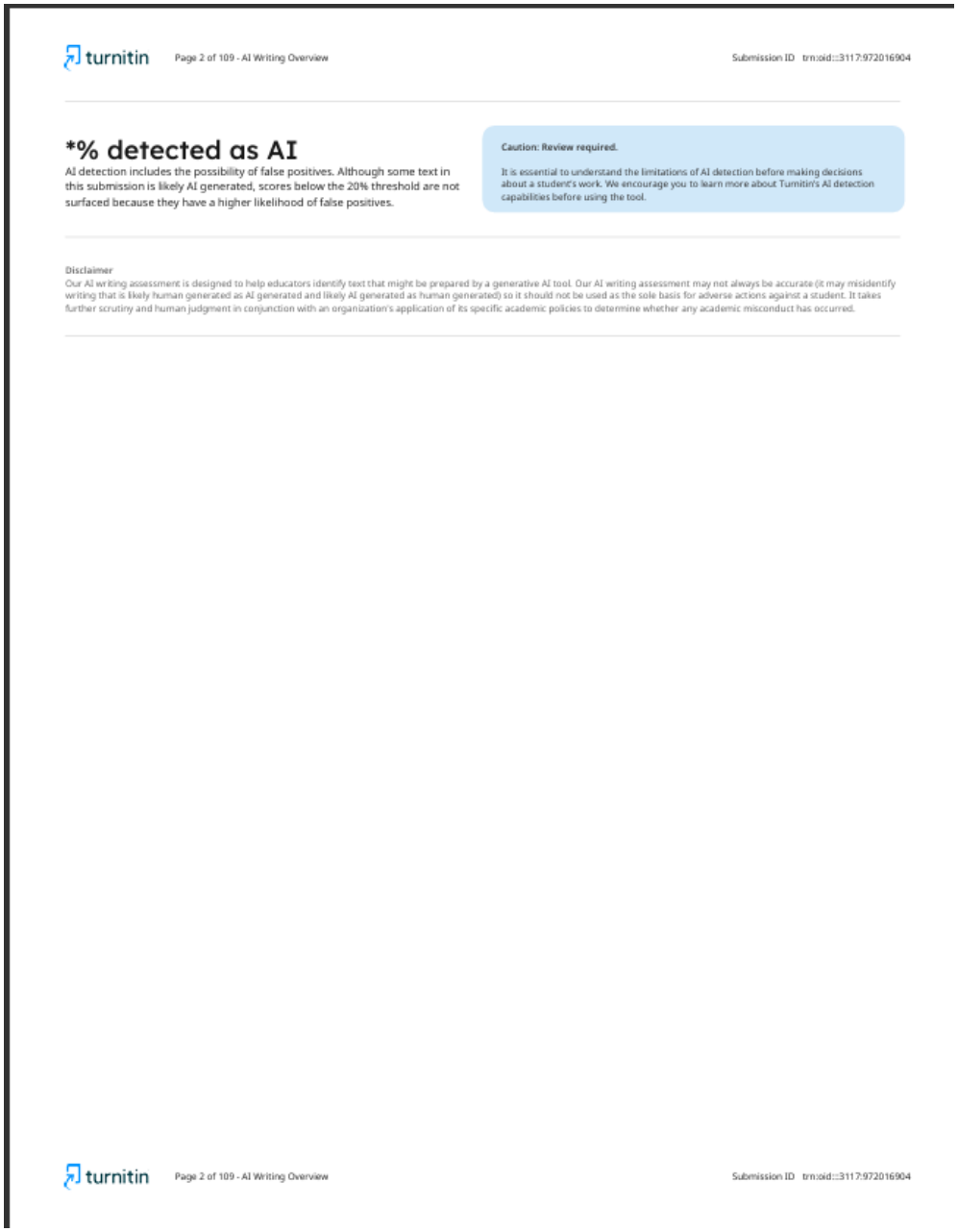
Table B.1 : Troubleshooting

Problem	Likely cause	What to do
"Too many wrong passwords"	Five wrong tries for one account	Wait 15 minutes, or use Forgot password
"This page has expired"	The page was open too long	Go back, refresh the page and try again
A time cannot be chosen	It is taken, has passed, or the shop is closed	Choose another time or date, or try another barber
Receipt upload fails	File is not JPG, PNG, WEBP or PDF, or is over 5 MB	Use a smaller file in an allowed format
"QR code expired" after scanning	The code on the counter changes every minute	Scan the code on the counter screen again
The camera does not open for the scan	The browser does not have camera permission	Allow the camera for the site, or use the phone's camera app to scan
"Only the owner can open that page"	A barber tried an owner-only page	Ask the owner
Cannot log in after a reset	Temporary password typed wrongly	Copy it exactly; it avoids look-alike letters such as 0, O, 1 and l

Appendix C – Turnitin Result

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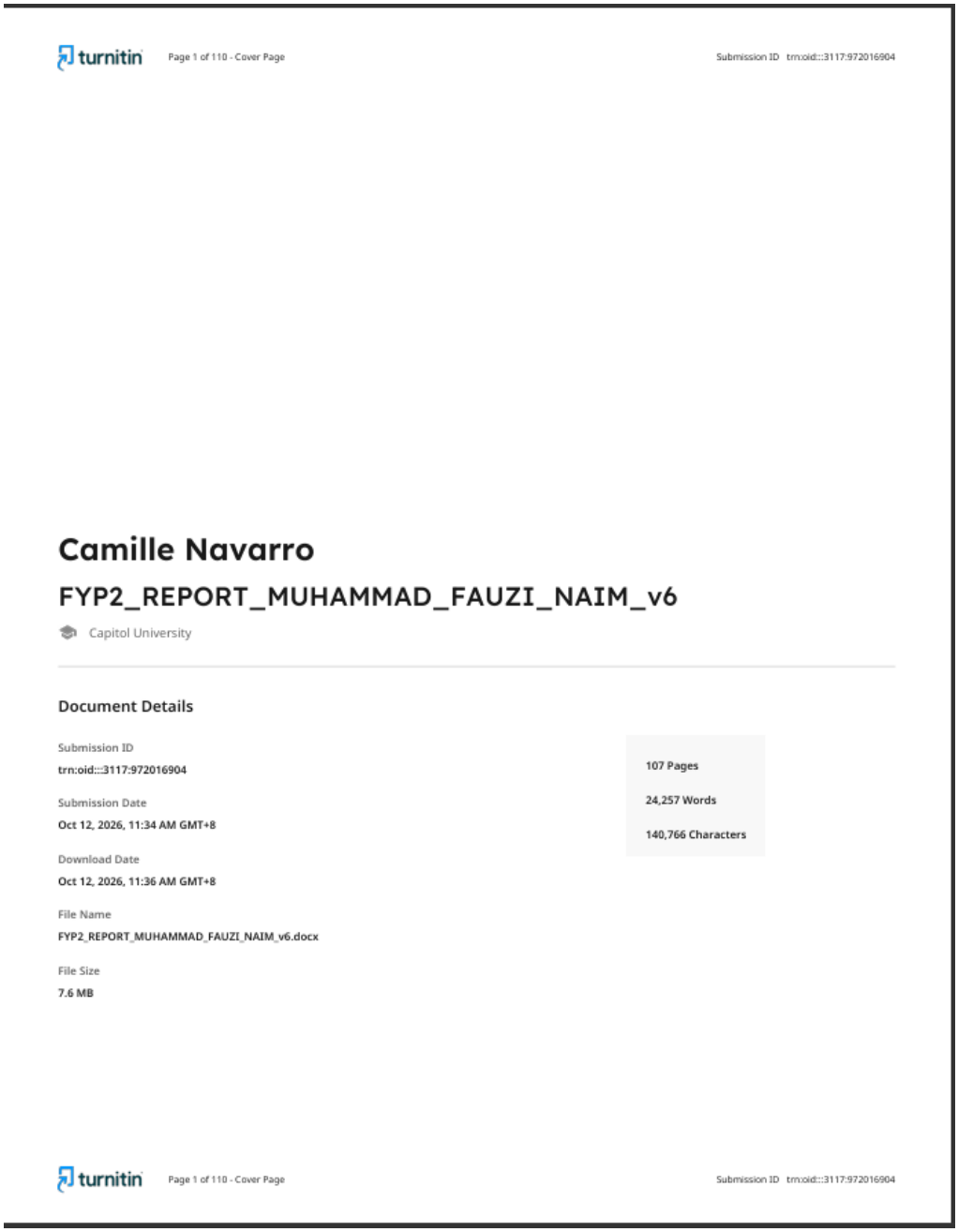
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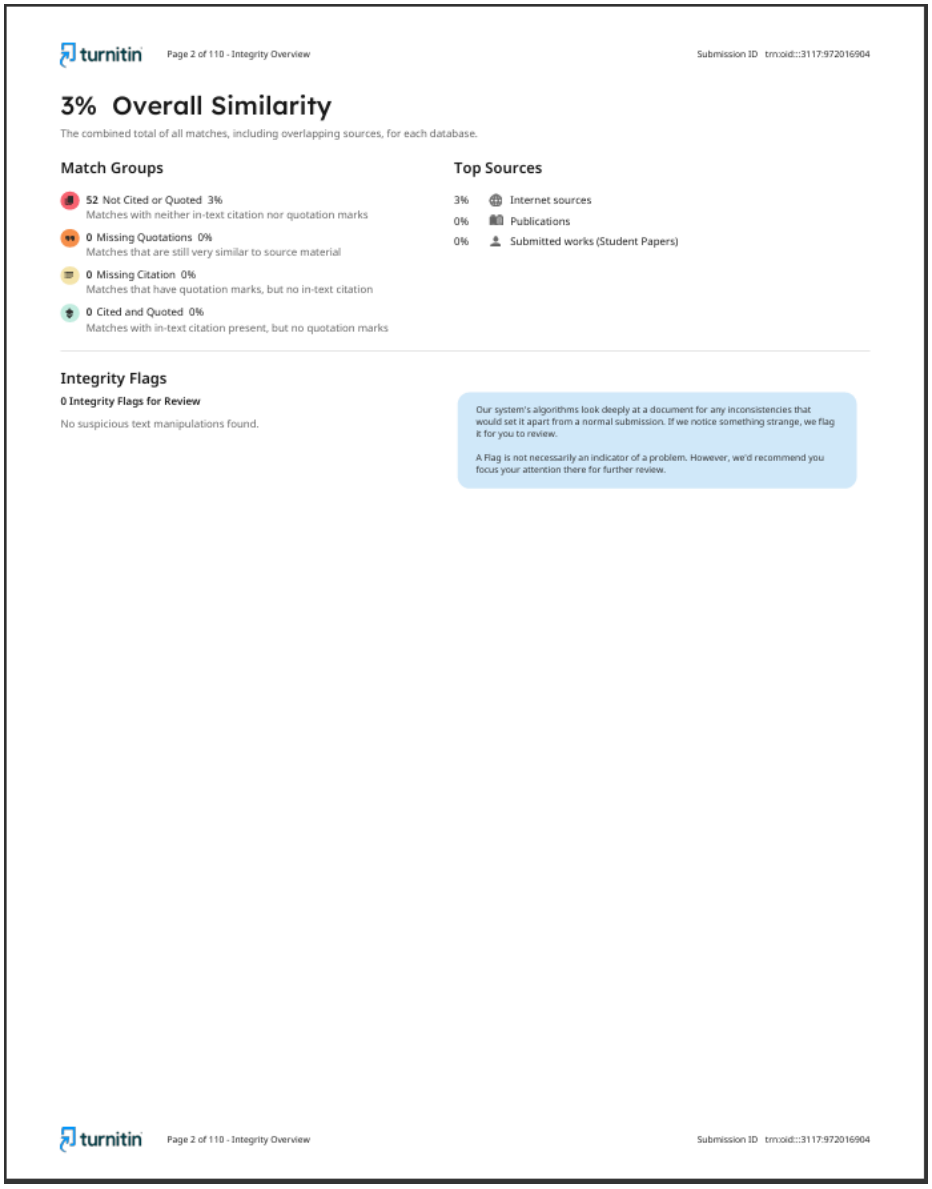
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Appendix D – Deployment Steps

Here are the steps that are taken when the system is put in shared hosting. Further, they are stored in a brief hosting guide, which is available in the project folder.

1. In the admin panel on the development computer, set the actual information of the shop: QR code, photos and prices of products/services, names of the staff, address, telephone number, WhatsApp number, terms text and logo, and change all starter passwords.
2. Export from command line (php tools/export-for-hosting.php). It generates database/kiad_install.sql, which contains all the tables, settings, services and staff logins, but no customers, bookings or other test data. The secret for the attendance QR and the counter link are blank so that the new server can create its own.
3. Setup a MySQL database and user with a strong password in the hosting control panel and grant all privileges to that user and import kiad_install.sql using phpMyAdmin.
4. Zip the project folder (minus the contents of storage/sessions and uploads/receipts) and upload it to the web root then extract it so that home.php and Login.php are directly in the web root.
5. Edit config.php: change the env to production and add the database name, and user/password. When running in production mode, errors are logged in a log file, and all requests are forced to go over HTTPS.
6. Set permission to '755' or '775' for uploads/receipts, images, images/services, storage/sessions, and storage/logs. (If uploads still do not work, set to 775).
7. Turn on the SSL certificate for the domain.
8. Test before announcing – open the home page, log in as the owner, make a test booking with a QR receipt, verify the booking, open the counter QR link on a tablet and scan with a phone, and check the pages on a phone. Delete the test bookings and accounts.
9. Back up database, images and uploads folders on a regular basis. In the case of lost password, it could be reset from the hosting terminal by using php tools/reset-password.php.